A singly exponential construction of matrix multiplication algorithms with exponent approaching 9/4

Abstract

An October 2026 preprint by OpenAI proved that the exponent of complex matrix multiplication is at most 9/4, without specifying algorithms. A subsequent construction gives exact rational algorithms with exponent below $9/4+\varepsilon$ for every rational $\varepsilon > 0$, but the numbers of matrix products in its schemes and the sides of its seeds are doubly exponential in $1/\varepsilon$, necessarily so within that construction. Our construction makes them singly exponential. Its main ingredient, a shifted sum of polynomials, splits a polynomial multiplication tensor times a dot product into equal kernel parts and one quotient part, which a reset turns into shorter polynomial multiplications. The resulting stages, which shorten the polynomial multiplication and enlarge a matrix multiplication tensor, run in sequence, each exactified separately. For rational $0 < \varepsilon \le 1$, an explicit rational scheme computes m products of $B \times B$ matrices from r products of $A \times A$ matrices, with exponent between 9/4 and $9/4+\varepsilon/2$, and with $\log m$, $\log r$ and $\log A$ below $2.7 \cdot 10^{10} \varepsilon^{-9/2} \log^3(4096/\varepsilon)$. It yields rational seeds of side n_0 with $\log n_0 = O(\varepsilon^{-11/2} \log^3(1/\varepsilon))$, and one algorithm for all n with exponent $9/4 + O((\log\log n)^{6/11} (\log n)^{-2/11})$. Within the construction, the exponent exceeds 9/4, and $\log r$ and $\log m$ have the least possible order.

1 Introduction

The exponent ω of matrix multiplication is the infimum of the real numbers a such that two $n \times n$ matrices can be multiplied with $O(n^a)$ arithmetic operations. In a preprint dated 2 October 2026 and released on 6 October, OpenAI [11] proved $\omega \le 9/4$ over the complex numbers, well below the bounds near 2.371 obtained by the laser method [1, 5, 13, 18]. The proof ends with an existence theorem for tensor characters, and specifies neither a size n nor an algorithm for $n \times n$ matrices with fewer than $n^{9/4+\varepsilon}$ multiplications.

The preceding paper [2] constructs, for every rational $\varepsilon > 0$, explicit exact bilinear algorithms with rational coefficients and exponent below $9/4+\varepsilon$. Their sizes are doubly exponential in $1/\varepsilon$, necessarily so within that construction: as $\varepsilon \to 0$, the least source side A (defined below) of a scheme with exponent at most $9/4+\varepsilon$ satisfies $\log\log A = \Theta(1/\varepsilon)$, and its seed sides are doubly exponential too [2, Corollary 11.2]. The second open question of [2, Section 11.5] asks for a finite construction with exponent below $9/4+\varepsilon$ in which $\log A$ is bounded by a polynomial in $1/\varepsilon$. That question was posed for the source side A, but A alone is not a robust measure: composition with the classical algorithm reduces it to 1 (Remark 7.3). The robust measures are the copy counts r_* and m_* and the seed side n_0 (Section 1.1), which are also doubly exponential within [2] (Table 1). Here all three are singly exponential. Our schemes have $\log r_*, \log m_* = O(\varepsilon^{-9/2} \log^3(1/\varepsilon))$, the least possible order within the construction (Theorems 1.1 and 1.4), and seeds with $\log n_0 = O(\varepsilon^{-11/2} \log^3(1/\varepsilon))$ (Theorem 1.2). They also have $\log A = O(\varepsilon^{-9/2} \log^3(1/\varepsilon))$, which answers the question as posed. Sections 3–7 prove Theorems 1.1–1.4 below, with long proofs in Appendices A and B.

1.1 Results

We use the conventions of [2], recalled in Section 2 below; all logarithms are natural. We write $T \succeq T'$ if linear substitutions on each leg of the tensor T transform it into T', M_d for $d \times d$ matrix multiplication, and mT for the direct sum of m copies of T. A *relative scheme* [2, Definition 2.3] is a restriction $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ with $B > A$; it computes m_* products of $B \times B$ matrices from r_* products of $A \times A$ matrices. We call A and B its *source* and *target sides*, r_* and m_* its *copy counts*, and $p = \log(r_*/m_*)/\log(B/A)$ its *pooled exponent*. Applied recursively, with the subproblems of each depth pooled into batches of m_*, it multiplies $n \times n$ matrices with $O(n^p)$ arithmetic operations if $p > 2$, over every field that contains its coefficients [2, Lemma 2.4].

Our schemes use the convolution tensor $C(r,b) = \sum_{\alpha<r}\sum_{\beta<b} x_\alpha y_\beta z_{\alpha+\beta}$, which multiplies polynomials with r and b coefficients, the dot product $B_X(d) = x\sum_{\gamma<d} y_\gamma z_\gamma$, and the product $Q_{r,b}$ of the six copies of $C(r,b)$ with permuted legs; note that $Q_{1,t} \cong \mathsf{M}_{t^2}$. A *stage* $(r,b) \to (c,r)$ of arity d turns a power of $Q_{r,b}$ and auxiliary dot products $B_X(d^N)$ into the same power of $Q_{c,r}$ and a matrix multiplication tensor. A *chain* applies stages in sequence, from a *source pair* (r_1, b_1) to a *terminal pair* $(1, t)$ (Section 1.2).

Theorem 1.1 (Main theorem)**.** *For every rational $0 < \varepsilon \le 1$, an explicit chain of at most $240\,\varepsilon^{-3/2}$ stages, whose parameters are computed from ε by integer arithmetic, gives a relative scheme $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ with rational coefficients such that $9/4 < p < 9/4 + \varepsilon/2$,*

$$\log m_* < 1.8 \cdot 10^{10}\,\varepsilon^{-9/2}\log^3\frac{4096}{\varepsilon}, \qquad \log r_* < 2.7 \cdot 10^{10}\,\varepsilon^{-9/2}\log^3\frac{4096}{\varepsilon},$$

$m_* = B^3$, $r_* < A^{35/4}$, $B \le A^2$ and $\log A < 3 \cdot 10^9\,\varepsilon^{-9/2}\log^3(4096/\varepsilon)$. For $0 < \varepsilon \le \frac{1}{2}$, the scheme with 2ε in place of ε satisfies $9/4 < p < 9/4 + \varepsilon$, $\log m_* < 8.4 \cdot 10^8\,\varepsilon^{-9/2}\log^3(2048/\varepsilon)$, $\log r_* < 1.23 \cdot 10^9\,\varepsilon^{-9/2}\log^3(2048/\varepsilon)$ and $\log A < 1.4 \cdot 10^8\,\varepsilon^{-9/2}\log^3(2048/\varepsilon)$.

Theorem 1.2 (Seeds and one algorithm)**.** *Let $0 < \varepsilon \le 1$ be rational, and let A be the source side of the scheme of Theorem 1.1.*

(a) An explicit integer n_0 with $\log n_0 < 2.1 \cdot 10^{11}\,\varepsilon^{-11/2}\log^3(4096/\varepsilon)$ has the following property: for every $n \ge n_0$, unrolling the recursion of the scheme at side n gives a rational bilinear algorithm for $n \times n$ matrices with fewer than $n^{9/4+\varepsilon}$ multiplications.

(b) The coefficients of the scheme are fractions $\mathfrak{c}/\mathcal{D}$ with a common denominator $\mathcal{D}$, where $\mathcal{D}$ and all $|\mathfrak{c}|$ are below 2^{β_ε} with $\log_2\beta_\varepsilon < 10^4\,\varepsilon^{-3/2}\log^2(4096/\varepsilon)$. If $\varepsilon = u/v$ with positive integers $u, v \le A$, which holds whenever $\varepsilon = 1/v$ (then $v < A$), a generator computes $\mathcal{D}$ and all numerators $\mathfrak{c}$ from u and v with at most A^{46} bit operations.

(c) One algorithm multiplies $n \times n$ matrices over every field of characteristic zero, for every $n \ge 2$, in $n^{9/4+\delta(n)}$ steps, where $\delta(n) = O((\log\log n)^{6/11}(\log n)^{-2/11})$. A step is a step of a fixed multitape Turing machine during the preprocessing, or a field operation.

Theorem 1.1 restates Theorem 4.8 and Corollary 4.9, with Lemma 5.1(a) for the number of stages; the bounds on $\log m_*$ and $\log r_*$ follow from $m_* \le A^6$ and $r_* < A^{35/4}$, as in Remark 7.14. Parts (a), (b) and (c) of Theorem 1.2 are Theorem 5.2, where $\log n_0 \ge (24/\varepsilon)\log A$, Proposition 5.4 and Theorem 5.8. All these sizes are astronomically large, and no algorithm here is shown to use $O(n^{9/4})$ operations. The local maps of Section 3 hold over every field; the relations that use the separator and exactification are proved over $\mathbb{Q}$, and we do not treat positive characteristic.

For a fixed target, much smaller chains suffice. Proposition 6.1, restated next, certifies some with the supplementary script `shifted_chain_certificates.py`, through the *fixed-batch bound*

$\Phi = \log(2r_*/m_*)/\log(B/A) = p + \log 2/\log(B/A)$ of [2, (10.1)], which also bounds the exponent of the recursion without pooling; Theorem 1.1 holds with Φ in place of p (Theorem 4.8).

Theorem 1.3 (Finite examples; computer-assisted)**.** *Explicit chains give relative schemes over* $\mathbb{Q}$ *with* $\Phi < 2.5$, 2.3, 2.26, 2.84063 *and* 2.67283, *whose source sides have* 33,379, 12,371,380, 48,859,824,830, 280 *and* 4,653 *decimal digits, respectively. The last two targets lie about* 10^{-5} *below the floating-point values of* Φ *for the schemes of [2, Table 2] with* $a = 8$ *and* $a = 12$ *(Section 6.2).*

Section 7 asks what the construction cannot do. Its *chain schemes* (Definition 7.1) allow crops and exchanges of the pairs, several blocks with their own widths and powers in each stage, and every *supply* of the auxiliary matrix multiplication tensors, by the source or by tensors that earlier stages produce (*forward recycling*, Lemma 7.2).

Theorem 1.4 (Limits within the construction)**.**

(a) Every chain scheme has $p > 9/4$*, also with compressed auxiliaries of any rank (Definition 7.8).*

(b) Let $0 < \varepsilon < 1/9216$*. Every chain scheme with* $p \le 9/4 + \varepsilon$ *satisfies* $\log r_* > \log m_* > 3.9 \cdot 10^{-6}\, \varepsilon^{-9/2} \log^3(1/\varepsilon)$ *and* $\log(B/A) > 9.8 \cdot 10^{-5}\, \varepsilon^{-4} \log^2(1/\varepsilon)$*. If the source supplies every auxiliary, then also* $\log A > 1.3 \cdot 10^{-6}\, \varepsilon^{-9/2} \log^3(1/\varepsilon)$.

Part (a) is Corollary 7.9, which extends Lemma 4.7(c) by a Hankel flattening. Part (b) is Theorem 7.12. Its bounds on r_*, m_* and $\log(B/A)$ extend to composites and tensor products of chain schemes and classical algorithms, and Theorem 1.1 attains their orders (Remark 7.14). The source side depends on the supply. Forward recycling keeps the r_*, m_* and $p < 9/4 + \varepsilon/2$ of that scheme, changes $\log(B/A)$ by at most $\log 2$, and gives $\log A < 5.32 \cdot 10^6\, \varepsilon^{-3} \log^3(4096/\varepsilon)$ (Corollary 7.13). It lowers $\log n_0$ of the seed only by a factor below 1.34, since the copy counts, not the source side, set the size of the seed (Remark 7.14). With computer assistance, Proposition 7.15 shows that every chain scheme with $p < 3$, also after composition and tensor products, has $m_* > 2^{494}$, and that the least such m_* is attained. None of these limits is a lower bound for ω or for tensor rank.

	Preceding paper		This paper
	size	source	size
least $\log m_*$ and $\log r_*$	$> (6/5)^{9/(2\varepsilon)}$	[2, Prop. 11.1(b)]	$\Theta(\varepsilon^{-9/2} \log^3(1/\varepsilon))$
$\log n_0$ of a seed	$> \frac{1}{2}(6/5)^{9/\varepsilon}$	[2, Cor. 11.2(c)]	$O(\varepsilon^{-11/2} \log^3(1/\varepsilon))$
least $\log A$	$e^{\Theta(1/\varepsilon)}$	[2, Cor. 11.2(b)]	$\Theta(\varepsilon^{-9/2} \log^3(1/\varepsilon))$
$\delta(n)$ of one algorithm	$O(1/\log\log n)$	[2, Rem. 9.8], sketch	$O((\log\log n)^{6/11} (\log n)^{-2/11})$
digits of A, $\Phi \approx 2.6728$	234,905,651,862	[2, Table 2], $a = 12$	4,653
digits of A, $p < 2.26$	$> 9.29 \cdot 10^{34}$	Section 6.2	48,859,824,830

Table 1: The two constructions, each with the source supplying every auxiliary; far from 9/4, other constructions are smaller (Section 6.2). The first three rows concern exponents at most $9/4 + \varepsilon$ as $\varepsilon \to 0$: schemes with $p \le 9/4 + \varepsilon$, and seeds with fewer than $n_0^{9/4+\varepsilon}$ products. Preceding paper: the first and third rows hold for $\varepsilon \le 1/4$, and the second for $\varepsilon \le 1/2$. In the first row, [2, Proposition 11.1(b)] bounds $\log A$ and gives $a \ge 4$ for the target index a; with this, [2, (7.3), (7.6) and (11.2)] and $B > A$ give $\log r_* > \log m_* > 2 \log A$. This paper: Theorems 1.1 and 1.4(b) for the first and third rows (Remark 7.14), Theorem 1.2 for the second and fourth, and Theorem 1.3 for the last two.

1.2 The idea

One stage. Fix $d \geq 2$ and $1 \leq c < b$, and put $h = b - c$. The *shifted sum* $(g_0, \dots, g_{d-1}) \mapsto \sum_{j<d} \xi^{jh} g_j$ maps d polynomials with b coefficients onto those with $b^+ = db - (d-1)c$ coefficients, with kernel the direct sum of $d-1$ copies of the polynomials with c coefficients. As it commutes with multiplication by a polynomial, $C(r,b)B_X(d)$ is block triangular in unimodular bases adapted to the kernel, with a *quotient* part $C(r,b^+)$ and $d-1$ *kernel* parts $C(r,c)$ sharing the first leg (Lemma 3.1). If $(d+2)c \leq db - r + 1$, a crop and a reset cut the quotient part into two *outer* parts $C(r,c)$ and one *middle* part $C(r,c)^{(23)}$ (Lemma 3.3). In the N-th tensor power, prescribing the numbers $(n_{\mathrm{K}}, n_{\mathrm{O}}, n_{\mathrm{mid}})$ of kernel, outer and middle parts on legs 2 and 3 deletes every cross term, since each goes from a part to one of larger weight (the weights are 2, 1, 0, in this order), and leaves $M = N!\,(d-1)^{n_{\mathrm{K}}} 2^{n_{\mathrm{O}}} / (n_{\mathrm{K}}!\, n_{\mathrm{O}}!\, n_{\mathrm{mid}}!)$ sectors sharing the first leg (Lemma 3.4), with $(d+2)^N/(N+1)^2 \leq M < (d+2)^N$ for the largest M (Lemma 4.6). Lü's version [7] of the separator of OpenAI [11] turns $2M-1$ copies of this sum into M independent sectors, each with an extra dot product $B_X(M)$. In the six orientations of the legs, the dot products $B_X(e)$ form M_{e^2}, and the middle parts only permute the orientations, so Proposition 3.5 gives a degeneration over $\mathbb{Q}$, of degree at most $6(M-1)$,

$$(2M-1)^6\, Q_{r,b}^{\otimes N}\, \mathsf{M}_{d^{2N}} \trianglerighteq M^6\, Q_{c,r}^{\otimes N}\, \mathsf{M}_{M^2},$$

where M_e denotes $e \times e$ matrix multiplication and M the sector count. For the largest M, the stage trades an auxiliary of side d^{2N} for a matrix multiplication tensor of side $M^2 \approx (d+2)^{2N}$, at the price of fewer than 64 copies.

Chains, and why $9/4$**.** Interpolation [3], in the rational form of [2, Lemmas 2.2 and 9.1(c)], makes each stage exact at a tensor power k, with $E = 1 + 6k(M-1)$ times as many copies (Corollary 3.6). The power of $Q_{c,r}$ in the target of a stage is the power of Q in the source of the next, so exact stages compose. Unlike [2], a chain needs no balancing of frequencies and no *catalyst*, that is, no factor common to both sides of a relation that each use of the relation consumes and reproduces [2, Proposition 7.1]. A chain, started by evaluating Q_{r_1,b_1} with s^6 products, $s = r_1 + b_1 - 1$, gives $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$ with $A = \prod_i d_i^{2Nk}$ and $B = t^{2Nk} \prod_i M_i^{2k}$ (Proposition 3.8). As N and k grow, with types of the largest M, p approaches the *ideal exponent* $3\log s/(\log t + G)$, where $G = \sum_i \log \frac{d_i+2}{d_i}$ is the *gain*. The least profile F (Section 2.5) satisfies $dF(r,b) \geq (d+2)F(c,r)$ at every stage, $F(1,t) = t$ and $F(r_1,b_1) \leq s^{4/3}$, so $\log t + G \leq \frac{4}{3}\log s$ and $p > 9/4$ (Lemma 4.7). Equality holds on the *arity ramp*, the stages of arities $3j-1$ and $3j$ from (R,R) down to $(1,1)$, with gain $\log F(R,R) \geq \frac{4}{3}\log R$ (Lemma 4.2). Above it, a *bulk* of stages of one arity D multiplies the pair by about the positive root μ of $D\mu^2 - \mu - (D+2) = 0$ at each stage, with ideal exponent at most $9/4 + 27/(32(D+1)^2)$ (Lemmas 4.3 and 4.4). Theorem 1.1 takes $D = \lceil 2\varepsilon^{-1/2} \rceil$, $R = 8D$, about $64D/\varepsilon$ bulk stages and $N = k = O(\varepsilon^{-3/2}\log(1/\varepsilon))$, so that the excess of the ideal exponent over $9/4$ and all losses add up to less than $\varepsilon/2$ (Step 4 of the proof of Theorem 4.8). As A depends only on the arities, N and k, $\log A \leq 2N^2 J \log(24D) = O(\varepsilon^{-9/2}\log^3(1/\varepsilon))$, where J is the number of stages. The exponentially large source pair enters only r_*, through s^{6Nk}, and the copy counts satisfy $m_* \leq A^6$ and $r_* < A^{35/4}$.

1.3 Related work and credits

The bound $\omega \leq 9/4$, the profile axioms and the convolution relations behind them are due to OpenAI [11]. The bound was extended to every field [9, 19] (in Lean), to associative unital rings [10], to algebraically closed fields of every characteristic, with a polynomial separator and the other steps checked in outline [7], and to rectangular products in characteristic zero [17]. The criterion

of Filmus [6] turns finite families of shared-leg block identities with a nonnegative certificate into exponent bounds, and also gives $\omega \le 9/4$ over every field. These works prove that algorithms exist; to our knowledge, apart from [2], no work gives explicit sizes for exponents near $9/4$. An OpenAI preprint [12] proves $\omega < 2.258$ in characteristic zero by a different method.

For $h = 1$ the shifted sum is known: for $d = 2$ it is the map in the proof of [11, Lemma 4.1], and for every d its kernel is the first step $\mathcal{F}^1$ of the determinant filtration [8, Lemma 2.5], [2, Section 4.1] (there for $b \ge d$). For $h \ge 2$ it is a substitution of binary forms (Section 3.1); this is elementary, but we found no source for it. New, to our knowledge, are keeping the kernel whole, as $d-1$ equal parts, the shift $h \ge 2$, the acyclic schedule without a catalyst, the arity ramp, and the explicit bounds proved here. The reset is the partition in the proof of [11, Lemma 4.2] (see also [17, Lemma 2.1(3)]), and our type restriction follows [2, Lemma 4.3]. Shared-leg separation is due to OpenAI [11, Proposition 3.1], with $5M$ source copies; we use the polynomial separator of Lü [7, Proposition 2.1], with $2M-1$ copies and degree at most $M-1$, in the rational form of [2, Lemma 9.1(b)]. The ramp attains the diagonal recursion [11, (5.1)] with equality, and $F(R,R) \ge R^{4/3}$ is [11, Lemma 5.1]. With $N = 1$ and the cross terms removed by a degeneration, a stage gives a shared-leg block identity $C(r,b)B_X(d) \trianglerighteq \sum_i T_i$ with $d+2$ blocks, in the sense of Filmus [6, Theorem 1.1, identity (1.2)], whose criterion gives exponents but no sizes. Relative schemes are a form of relative bilinear complexity [16], and type restrictions are standard in the laser method [4, 16].

2 Preliminaries

We use the conventions of the preceding paper [2] and cite its results by the numbers of its version 5, as in [2, Lemma 2.4]. This section recalls what we use.

2.1 Tensors and restrictions

A *tensor* over a field $\mathbb{F}$ is a trilinear form in three disjoint groups of variables, its *legs*; the variables of legs 1, 2 and 3 are named with the letters x, y and z. We write $T \trianglerighteq T'$, and call T' a *restriction* of T, if linear substitutions on each leg transform T into T': every variable of T is replaced by a linear form in the variables of the same leg of T'. Reading a tensor as a bilinear map, with inputs on legs 1 and 2 and the output read off by leg 3, $T \trianglerighteq T'$ says that T' can be evaluated by linear maps on the inputs, one evaluation of T, and a linear map on the output. We write $T \cong T'$ if a bijection between the variables of each leg transforms T into T'. A restriction or a degeneration (Section 2.2) *holds over* $\mathbb{Q}$ if all its maps have rational coefficients; it then holds over every field of characteristic zero.

The tensor mT is the full direct sum of m copies of T, with independent variables on all three legs. Juxtaposition denotes the tensor product, and $T^{\otimes N}$ a tensor power. Restrictions compose, and they are preserved by direct sums and tensor products, in particular by tensoring with a fixed tensor. Moreover $(mT)T' \cong m(TT')$, and reordering the factors of a tensor product is an isomorphism. For $\pi \in S_3$, the tensor T^π places leg i of T at position $\pi(i)$, so that $(T^{\pi'})^\pi = T^{\pi\pi'}$. In a product of oriented factors, leg i of T, in a factor T^π, lies in leg $\pi(i)$ of the product, its *global* leg. We write $\mathcal{A}_+$ and $\mathcal{A}_-$ for the even and odd permutations and (23) for the transposition of legs 2 and 3 [2, Section 2.1].

Matrix multiplication and convolution tensors. The matrix multiplication tensor of side d is $\mathsf{M}_d = \sum_{\alpha,\beta,\gamma<d} x_{\alpha\beta} y_{\beta\gamma} z_{\gamma\alpha}$, and the dot product $B_X(d) = x \sum_{\gamma<d} y_\gamma z_\gamma$ has a single variable on its first leg. A sum of m products of linear forms equal to T is the same as a restriction $m\,\mathsf{M}_1 \trianglerighteq T$,

since M_1 is the single product xyz. By [2, (2.1) and (2.2)],

$$\mathsf{M}_d\mathsf{M}_{d'} \cong \mathsf{M}_{dd'}, \qquad \bigotimes_{\pi\in S_3} B_X(d)^\pi \cong \mathsf{M}_{d^2}. \tag{2.1}$$

For positive integers r and b, the convolution tensor [2, (2.3)] is

$$C(r,b) = \sum_{\alpha=0}^{r-1}\sum_{\beta=0}^{b-1} x_\alpha y_\beta z_{\alpha+\beta}.$$

As a bilinear map it multiplies a polynomial with r coefficients x_α by a polynomial with b coefficients y_β, and z_γ reads the coefficient of the γ-th power in the product. In particular $C(1,b) = B_X(b)$. Over $\mathbb{Q}$, evaluation at the nodes $1,\dots,r+b-1$ followed by Lagrange interpolation writes $C(r,b)$ as a sum of $r+b-1$ products of linear forms with rational coefficients [2, Lemma 9.1(a)]. The symmetrized tensors [2, (2.5) and (2.7)] are

$$S^\sigma_{r,b} = \bigotimes_{\pi\in\mathcal{A}_\sigma} C(r,b)^\pi \quad (\sigma\in\{+,-\}), \qquad Q_{r,b} = S^+_{r,b}S^-_{r,b} = \bigotimes_{\pi\in S_3} C(r,b)^\pi.$$

By [2, (2.6)], $S^\sigma_{r,b} \cong S^{-\sigma}_{b,r}$ and $S^\sigma_{1,b} \cong \mathsf{M}_b$. With (2.1) and six oriented copies of the rational decomposition, this gives (the first identity is also [2, (2.8)])

$$Q_{r,b} \cong Q_{b,r}, \qquad Q_{1,t} \cong \mathsf{M}_{t^2}, \qquad (r+b-1)^6\,\mathsf{M}_1 \succeq Q_{r,b} \text{ over } \mathbb{Q}. \tag{2.2}$$

2.2 Degenerations and exactification

Let $\theta \geq 0$ be an integer. We write $T \trianglerighteq T'$, a *degeneration of degree at most* θ, if a triple $\mathsf{S}(t)$ of linear maps, one on each leg, with Laurent polynomials in t as entries, transforms T into a polynomial in t of degree at most θ with tensor coefficients and constant term T' [2, Section 2.3]. The entries of $\mathsf{S}(t)$ may contain negative powers of t; the transformed tensor may not. Outside degenerations, t denotes the positive integer in the terminal pair $(1,t)$ of a chain (Section 3.5).

We use two facts. First, [2, Lemma 2.1] (Composition): maps that do not depend on t, applied before a degeneration, or after it if they are defined on all its target variables, keep the degree bound and act on the constant term; applying degenerations to the factors of a tensor product with one common parameter t gives a degeneration of the product whose degree bound is the sum of the degree bounds and whose constant term is the product of the constant terms; and both statements hold for oriented copies T^π, with the maps moved to the permuted legs. Second, [2, Lemma 2.2] (Exactification, after Bini [3]): if $T \trianglerighteq T'$ with degree at most θ, then

$$(k\theta+1)\,T^{\otimes k} \succeq T'^{\otimes k} \qquad (k\geq 1).$$

That lemma uses roots of unity. By [2, Lemma 9.1(c)], the same restriction is given by the maps $\mathsf{S}(\ell)^{\otimes k}$, $1\leq\ell\leq k\theta+1$, with the scalars $(-1)^{\ell-1}\binom{k\theta+1}{\ell}$ applied on one fixed leg; so it holds over $\mathbb{Q}$ when S has rational coefficients.

Lemma 2.1 of [2] does not cover two degenerations applied in sequence: the Laurent maps of the second would act on the terms of positive degree of the first and could create negative powers of t. So the chains of Section 3.5 exactify each stage before its output is used.

2.3 Relative schemes

A *relative scheme* $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$, with positive integers r_*, m_* and $A < B$, is as in Section 1.1 [2, Definition 2.3]: it is given by its three linear maps, and has source side A, target side B and pooled exponent $p = \log(r_*/m_*)/\log(B/A)$. By the pooled recursion [2, Lemma 2.4], $p \geq 2$, and the scheme yields a fixed arithmetic algorithm that multiplies $n \times n$ matrices with $O(n^p)$ operations if $p > 2$, and with $O(n^2 \log n)$ operations if $p = 2$. The algorithm uses only the coefficients of the scheme, so in both cases $\omega \leq p$ over every field that contains them. We also use the *fixed-batch bound* of [2, (10.1)],

$$\Phi = \frac{\log(2r_*/m_*)}{\log(B/A)} = p + \frac{\log 2}{\log(B/A)}, \tag{2.3}$$

which exceeds p. As $p > 0$ gives $r_*/m_* > 1$, and hence $\lceil r_*/m_* \rceil < 2r_*/m_*$, it also bounds the exponent $\log\lceil r_*/m_* \rceil / \log(B/A)$ of the recursion without pooling [2, Section 10.1].

2.4 Sectors, type restrictions and the separator

Following [2, Section 4], a tensor $T = \sum_{g=0}^{M-1} T_g$ with $T_g \in X \otimes Y_g \otimes Z_g$, $Y = \bigoplus_g Y_g$ and $Z = \bigoplus_g Z_g$ is a sum of M *sectors sharing the first leg*. It is not a direct sum: all sectors use the same first-leg variables, so a coordinate change that depends on the sector can be applied on legs 2 and 3 but not yet on leg 1.

Sectors arise from *type restrictions* [2, Section 4.1 and Lemma 4.3], which keep in a tensor power only the coordinates on legs 2 and 3 whose words of labels have a prescribed type; Lemma 3.4 adapts the proof of [2, Lemma 4.3], with weights on classes of labels in place of the filtration index.

The *separator* gives each sector its own first leg. Shared-leg separation is due to OpenAI [11, Proposition 3.1]; we use the polynomial separator of Lü [7, Proposition 2.1], in the rational form [2, Lemma 9.1(b)], which writes it in the coordinates of [2, Proposition 4.6] (see [2, Remark 4.7]). For T as above, it is a degeneration

$$(2M-1)\,T \trianglerighteq \bigoplus_{g=0}^{M-1} T_g\, B_X(M) \qquad \text{over } \mathbb{Q}, \tag{2.4}$$

of degree at most $M-1$, in which each summand on the right has its own copy of X; permuting legs gives the same with any shared leg. The target variables are $X_{\alpha,g'}$, $Y_{g,\beta,e}$ and $Z_{g,\gamma,e'}$ with $g', e, e' < M$, where α, β and γ index coordinates of X, Y_g and Z_g; the constant term consists of the terms with $g' = g$ and $e = e'$. Suppose that coordinate permutations $(\rho_g^X, \rho_g^Y, \rho_g^Z)$ identify each sector T_g with one fixed tensor $\mathcal{V}$. Apply $\rho_{g'}^X$ to the copy $(X_{\alpha,g'})_\alpha$, and ρ_g^Y and ρ_g^Z to the variables $Y_{g,\beta,e}$ and $Z_{g,\gamma,e'}$, for all g, g', e, e'. This *reindexing* [2, (B.6)] is a bijection of all target variables that does not depend on t, so by [2, Lemma 2.1] it can follow the separator; it maps the constant term of (2.4) onto $M\,(\mathcal{V}B_X(M))$.

2.5 Profiles

A *profile* [2, Definition 3.1] is a positive symmetric function P on pairs of positive integers such that, for all positive integers r and b,

$$P(1,b) = b, \qquad 2P(r,b) \geq P(r,b-1) + P(r,b+1) \ \ (b \geq 2), \qquad P(r, 3b+r-1) \geq 3P(r,b).$$

These are the normalization, concavity and tripling axioms [2, (3.1)–(3.3)]. The *least profile* F of [2, (3.4)] is a profile, and every profile P satisfies $P \geq F$ at every pair [2, Theorem 3.2]. Section 4.2 recalls the formula for F, and Lemma 4.7 uses F as a potential.

3 Stages

A *legal stage of arity* d, or simply a *stage*, written $(r,b)\to(c,r)$, is given by integers with

$$r\ge 1,\qquad d\ge 2,\qquad c\ge 1,\qquad (d+2)c\le db-r+1. \tag{3.1}$$

The last condition implies $c<b$, as $dc<(d+2)c\le db-r+1\le db$. The stage has *parent* $C(r,b)$, *child length* c and next pair (c,r). We put $h=b-c\ge 1$ and $b^+=db-(d-1)c=b+(d-1)h$, the *shift* and the *quotient length*. Lemma 3.1 splits $C(r,b)B_X(d)$ block-triangularly into $d-1$ kernel parts $C(r,c)$ and a quotient part $C(r,b^+)$. By Lemma 3.3(c), the last condition of (3.1) says that $b^+\ge 3c+r-1$, so Lemma 3.3 can crop the quotient to $C(r,3c+r-1)$ and reset it into two copies of $C(r,c)$ and one of $C(r,c)^{(23)}$. Lemma 3.4 deletes all cross terms in a tensor power by fixing class counts, and Proposition 3.5, Corollary 3.6 and Proposition 3.8 turn this into exact relations and compose them.

3.1 The shifted sum

Let $\mathbb{F}$ be a field, and let $\mathbb{F}[\xi]_{<\ell}$ be the polynomials in ξ of degree less than ℓ, with basis $\xi^0,\dots,\xi^{\ell-1}$. Fix $d\ge 2$ and $h\ge 1$. For $\ell\ge h+1$ let $W_\ell=(\mathbb{F}[\xi]_{<\ell})^d$, with components $j=0,\dots,d-1$ and monomial basis $\xi^m e_j$, and define the *shifted sum*

$$W_\ell\to\mathbb{F}[\xi]_{<\ell+(d-1)h},\qquad (g_0,\dots,g_{d-1})\longmapsto\sum_{j=0}^{d-1}\xi^{jh}g_j.$$

For $0\le j\le d-2$ and $g\in\mathbb{F}[\xi]_{<\ell-h}$, the *kernel vector* $\kappa_j(g)\in W_\ell$ has component j equal to $-\xi^h g$, component $j+1$ equal to g, and all others zero. For $0\le\gamma<\ell+(d-1)h$, let $j_\ell(\gamma)$ be the least j with $\gamma\le jh+\ell-1$, and let $\lambda_\ell(\gamma)=\xi^{\gamma-j_\ell(\gamma)h}e_{j_\ell(\gamma)}$, the *lift* of ξ^γ. It is a monomial of W_ℓ, the one in the first component that contains the exponent γ: if $j=j_\ell(\gamma)\ge 1$, then $\gamma>(j-1)h+\ell-1\ge jh$.

For integers $r\ge 1$, $d\ge 2$ and $1\le c<b$, with $h=b-c$, the tensor

$$U=C(r,b)\,B_X(d)=\sum_{\alpha=0}^{r-1}\sum_{\beta=0}^{b-1}\sum_{j=0}^{d-1}x_\alpha\,y_{j,\beta}\,z_{j,\alpha+\beta} \tag{3.2}$$

multiplies $f\in\mathbb{F}[\xi]_{<r}$ by $g\in W_b$ componentwise: the variable $z_{j,\gamma}$ reads the coefficient of $\xi^\gamma e_j$ in $fg\in W_{b+r-1}$.

Lemma 3.1 (Shifted sum)**.** *Let $d\ge 2$, $h\ge 1$ and $\ell\ge h+1$.*

(a) The lifts $\lambda_\ell(\gamma)$, $0\le\gamma<\ell+(d-1)h$, and the kernel vectors $\kappa_j(\xi^i)$, $0\le j\le d-2$, $0\le i<\ell-h$, form a basis of W_ℓ, the shifted basis*. Its matrix has entries in $\{0,\pm1\}$ and determinant ±1, and its inverse has entries in $\{0,1\}$: for $0\le m<\ell$ and $0\le j<d$,*

$$\xi^m e_j=\lambda_\ell(jh+m)+\sum_{j'=j_\ell(jh+m)}^{j-1}\kappa_{j'}\big(\xi^{m+(j-1-j')h}\big). \tag{3.3}$$

(b) The shifted sum is surjective, and its kernel is the direct sum of the $d-1$ subspaces $\kappa_j(\mathbb{F}[\xi]_{<\ell-h})$.

(c) Let $r\ge 1$ and $1\le c<b$, with $h=b-c$. Use the shifted basis of W_b on leg 2 of U, the coordinates dual to the shifted basis of W_{b+r-1} on leg 3, and the monomial basis on leg 1. Call

the new coordinates y^{q}_β and $y^{(j)}_i$ (for $\lambda_b(\beta)$ and $\kappa_j(\xi^i)$), and z^{q}_γ and $z^{(j)}_i$ (for $\lambda_{b+r-1}(\gamma)$ and $\kappa_j(\xi^i)$). Then U becomes

$$U' = \sum_{\alpha=0}^{r-1} x_\alpha \left(\sum_{\beta=0}^{b^+-1} y^{\mathrm{q}}_\beta z^{\mathrm{q}}_{\beta+\alpha} + \sum_{j=0}^{d-2} \sum_{i=0}^{c-1} y^{(j)}_i z^{(j)}_{i+\alpha} + \sum_{\beta=0}^{b^+-1} \sum_{j'=j_{b+r-1}(\beta+\alpha)}^{j_b(\beta)-1} y^{\mathrm{q}}_\beta z^{(j')}_{\beta+\alpha-(j'+1)h} \right). \tag{3.4}$$

The first sum is $C(r, b^+)$, the second is $d-1$ copies of $C(r,c)$, all sharing leg 1, and the third, which joins quotient to kernel coordinates, is nonzero if and only if $r \geq 2$. The substitutions are integral with integral inverses, so $U \succeq U' \succeq U$ over every field.

Proof. (a) For each γ, the monomials $m_j = \xi^{\gamma-jh} e_j$ of W_ℓ have $j_0 \leq j \leq j_1$, where $j_0 = j_\ell(\gamma)$ and $j_1 = \min(d-1, \lfloor \gamma/h \rfloor)$; let $W^{(\gamma)}$ be their span. Every monomial $\xi^m e_j$ lies in $W^{(m+jh)}$, so $W_\ell = \bigoplus_\gamma W^{(\gamma)}$. The lift $\lambda_\ell(\gamma)$ is m_{j_0}, and $\kappa_j(\xi^i) = \xi^i e_{j+1} - \xi^{i+h} e_j$ lies in $W^{(\gamma)}$ for $\gamma = i + (j+1)h$, where the conditions $0 \leq i < \ell - h$ and $j \leq d-2$ read $j_0 \leq j < j_1$. So the vectors of (a) in $W^{(\gamma)}$ are m_{j_0} and $m_{j+1} - m_j$, $j_0 \leq j < j_1$. Their matrix in the basis $m_{j_0}, \dots, m_{j_1}$ is unitriangular with entries $0, \pm 1$, and its inverse $m_j = m_{j_0} + \sum_{j'=j_0}^{j-1} (m_{j'+1} - m_{j'})$ is (3.3). Hence (a) holds on every $W^{(\gamma)}$, and so on W_ℓ.

(b) The shifted sum maps $\xi^m e_j$ to ξ^{m+jh}, so it maps the lifts bijectively onto the monomials ξ^γ and is onto; and it maps $\kappa_j(g)$ to $-\xi^{(j+1)h} g + \xi^{(j+1)h} g = 0$. As the lifts and kernel vectors form a basis by (a), the kernel vectors form a basis of the kernel.

(c) As in [2, Section 4.1], substitute $y = \Pi_b y'$ and $z = \Pi^{-\top}_{b+r-1} z'$, where Π_ℓ has the shifted basis as columns; the coefficient of $x_\alpha y'_{\mathbf{u}} z'_{\mathbf{v}}$ is then the coordinate along $\mathbf{v}$ of $\xi^\alpha \mathbf{u}$. First, $\xi^\alpha \kappa_j(\xi^i) = \kappa_j(\xi^{i+\alpha})$ in W_{b+r-1}, since $i + \alpha < c + r - 1 = (b + r - 1) - h$. Second, $\xi^\alpha \lambda_b(\beta) = \xi^{\beta+\alpha-jh} e_j$ with $j = j_b(\beta)$, and by (3.3) for W_{b+r-1} this is $\lambda_{b+r-1}(\beta+\alpha)$ plus $\kappa_{j'}(\xi^{\beta+\alpha-(j'+1)h})$ for $j_{b+r-1}(\beta+\alpha) \leq j' < j_b(\beta)$. This gives (3.4); in the first sum, β runs over $[0, b^+)$ and $\beta + \alpha$ over $[0, b^+ + r - 1)$. If $r = 1$, then $\alpha = 0$ and $j_{b+r-1} = j_b$, so the third sum is empty. If $r \geq 2$, then $\beta = b < b^+$ and $\alpha = 0$ give $j_b(b) = 1$ and $j_{b+r-1}(b) = 0$, so the third sum contains $x_0 y^{\mathrm{q}}_b z^{(0)}_c$. □

Example 3.2. Take $d = 2$, $r = 2$, $b = 3$ and $c = 1$, so that $h = 2$, $b^+ = 5$ and the shifted sum is $(g_0, g_1) \mapsto g_0 + \xi^2 g_1$. The shifted bases are

	lifts $\lambda_\ell(0), \lambda_\ell(1), \dots$	kernel vectors $\kappa_0(\xi^i)$
W_3	$e_0,\ \xi e_0,\ \xi^2 e_0,\ \xi e_1,\ \xi^2 e_1$	$e_1 - \xi^2 e_0$
W_4	$e_0,\ \xi e_0,\ \xi^2 e_0,\ \xi^3 e_0,\ \xi^2 e_1,\ \xi^3 e_1$	$e_1 - \xi^2 e_0,\ \xi e_1 - \xi^3 e_0$

Multiplying a vector of the first row by 1 or ξ gives a vector of the second row in every case but one: $1 \cdot \lambda_3(3) = \xi e_1$ is $\lambda_4(3) + \kappa_0(\xi)$ in W_4. Hence U' consists of $C(2,5)$ in the coordinates $x, y^{\mathrm{q}}, z^{\mathrm{q}}$, of $C(2,1)$ in $x, y^{(0)}, z^{(0)}$, and of the single term $x_0 y^{\mathrm{q}}_3 z^{(0)}_1$.

Binary forms. Identify $\xi^m e_j \in W_\ell$ with $u^{\ell-1-m} v^m\, u'^{d-1-j} v'^j \in V_\ell \otimes V'_d$, where V_ℓ and V'_d are the binary forms with ℓ and d coefficients in u, v and in u', v' of [2, Section 4.1], and identify ξ^γ with $u^{\ell+(d-1)h-1-\gamma} v^\gamma$. Then the shifted sum is the substitution $(u', v') = (u^h, v^h)$, and $\kappa_j(\xi^i) = \Delta_h\, u^{\ell-h-1-i} v^i u'^{d-2-j} v'^j$ with $\Delta_h = u^h v' - v^h u'$. For $h = 1$, the kernel $\Delta_1(V_{\ell-1} \otimes V'_{d-1})$ is the first step $\mathcal{F}^1(V_\ell \otimes V'_d)$ of the determinant filtration of [2, Section 4.1], defined there for $\ell \geq d$. For $b \geq d$, the split of width $d-1$ there refines $\mathcal{F}^1$ further, with quotients carrying $C(r, b + d - 1 - 2j)$, $1 \leq j \leq d-1$; Lemma 3.1 keeps it whole, as $d-1$ equal parts $C(r,c)$.

3.2 Crop and reset

Lemma 3.3 (Crop and reset)**.** *Let $r, c \geq 1$.*

(a) For $\ell \geq 3c + r - 1$, setting $y_\beta = 0$ for $\beta \geq 3c + r - 1$ and $z_\gamma = 0$ for $\gamma \geq 3c + 2r - 2$ restricts $C(r, \ell)$ to $C(r, 3c + r - 1)$.

(b) Label the indices of legs 2 and 3 of $C(r, 3c + r - 1)$ by the intervals

	left	mid	right
β *(leg 2)*	$[0, c - 1]$	$[c, 2c + r - 2]$	$[2c + r - 1, 3c + r - 2]$
γ *(leg 3)*	$[0, c + r - 2]$	$[c + r - 1, 2c + r - 2]$	$[2c + r - 1, 3c + 2r - 3]$

In every term $x_\alpha y_\beta z_{\alpha+\beta}$, either β and $\gamma = \alpha + \beta$ have the same label, or β is mid *and γ is* left *or* right*. The terms with both labels* left *form $C(r, c)$; those with both labels* right *form $C(r, c)$ after subtracting $2c + r - 1$ from β and γ; and those with both labels* mid *form $C(r, c)^{(23)}$ after $\alpha \mapsto r - 1 - \alpha$, $\beta \mapsto \beta - c$ and $\gamma \mapsto \gamma - (c + r - 1)$.*

(c) For $d \geq 2$ and $1 \leq c < b$, $b^+ \geq 3c + r - 1$ if and only if $(d + 2)c \leq db - r + 1$.

Proof. (a) If $\beta < 3c + r - 1$, then $\alpha + \beta \leq 3c + 2r - 3$, so the kept terms are those of $C(r, 3c + r - 1)$. (b) A left β gives $\gamma \leq c + r - 2$, which is left; a right β gives $\gamma \geq \beta$, which is right; and a mid γ gives $\beta = \gamma - \alpha \in [c, 2c + r - 2]$, which is mid. So different labels force a mid β and an outer γ. The left part consists of the terms with $\beta < c$, and the right part is a translate. The mid part consists of the terms with γ mid; with $\alpha' = r - 1 - \alpha$ and $\gamma' = \gamma - (c + r - 1) \in [0, c)$ we get $\beta = c + \alpha' + \gamma'$, so after the substitutions the part is $\sum_{\alpha' < r, \gamma' < c} x_{\alpha'} y_{\alpha'+\gamma'} z_{\gamma'}$, which is $C(r, c)$ with legs 2 and 3 exchanged. (c) $b^+ - (3c + r - 1) = db - (d + 2)c - r + 1$. □

Lemma 3.4 below removes the cross terms of (b), together with the third sum of (3.4).

3.3 The type restriction of a stage

Fix a stage $(r, b) \to (c, r)$ of arity d. Each coordinate on legs 2 and 3 of U' gets a *letter*: $y_i^{(j)}$ and $z_i^{(j)}$ get K_j; y_β^{q} gets its leg-2 label from Lemma 3.3(b) if $\beta < 3c + r - 1$, and cropped otherwise; z_γ^{q} gets its leg-3 label if $\gamma < 3c + 2r - 2$, and cropped otherwise. Since $b^+ \geq 3c + r - 1$, the quotient coordinates that are not cropped carry $C(r, 3c + r - 1)$ and every label occurs. The other letters form three *classes*: *kernel* $(\mathsf{K}_0, \ldots, \mathsf{K}_{d-2})$, of *weight* 2; *outer* (left, right), of weight 1; and *middle* (mid), of weight 0. The *diagonal part* U'_χ of such a letter χ is the sum of the terms of U' with letter χ on both legs. By Lemmas 3.1(c) and 3.3, coordinate permutations of each leg identify U'_{K_j}, U'_{left} and U'_{right} with $C(r, c)$, and U'_{mid} with $C(r, c)^{(23)}$, the last one reversing the first index.

Lemma 3.4 (Stage type restriction)**.** *Let $(r, b) \to (c, r)$ be a stage of arity d, let $N \geq 1$, and let $\tau = (n_{\mathrm{K}}, n_{\mathrm{O}}, n_{\mathrm{mid}})$ be nonnegative integers with sum N, a* type of width N*. In $U'^{\otimes N}$, with product coordinates on legs 2 and 3, the* word *of a coordinate is its sequence of letters, and its* type *is the triple of the numbers of its kernel, outer and middle letters. After the substitutions of Lemma 3.1(c) in every position, setting to zero every coordinate on legs 2 and 3 whose type is not τ gives a restriction of $U^{\otimes N}$, with integer coefficients, to a sum $T = \sum_w T_w$ of*

$$M = \frac{N!}{n_{\mathrm{K}}!\, n_{\mathrm{O}}!\, n_{\mathrm{mid}}!}\, (d - 1)^{n_{\mathrm{K}}}\, 2^{n_{\mathrm{O}}} \tag{3.5}$$

sectors sharing the first leg $X = (\mathbb{F}^r)^{\otimes N}$, one for each word w of type τ (a word that contains cropped *has a type of width less than N). The sector $T_w = U'_{w_1} \otimes \cdots \otimes U'_{w_N}$ consists of the terms with word w on both legs. Moving the middle positions to the end, keeping the order within each group, and reversing the first index at middle positions identifies T_w with*

$$\mathcal{V} = C(r,c)^{\otimes(N-n_{\mathrm{mid}})}\big(C(r,c)^{(23)}\big)^{\otimes n_{\mathrm{mid}}} \tag{3.6}$$

by coordinate permutations of each leg; on the first leg they depend on w.

Proof. One position. Let $x_\alpha y'_{\mathbf{u}} z'_{\mathbf{v}}$ be a term of U' with letters $\chi \neq$ cropped on leg 2 and $\chi' \neq$ cropped on leg 3. We claim that $\chi' = \chi$ or that χ' has larger weight. By (3.4), a kernel coordinate $y_i^{(j)}$ occurs only with $z_{i+\alpha}^{(j)}$. A quotient coordinate, of weight at most 1, occurs either with a kernel coordinate, of weight 2, or in a term $x_\alpha y^{\mathrm{q}}_\beta z^{\mathrm{q}}_{\beta+\alpha}$ of $C(r,b^+)$ with $\beta < 3c + r - 1$, which by Lemma 3.3 has equal labels or goes from mid to an outer label.

The power. Setting coordinates to zero is a restriction, so $U^{\otimes N} \succeq U'^{\otimes N} \succeq T$. The coefficient of a term of $U'^{\otimes N}$ is the product of the coefficients in the N positions. In a surviving term with words w on leg 2 and w' on leg 3, the weight of w'_q is therefore at least that of w_q at every position q. Both words have type τ and hence weight $2n_{\mathrm{K}} + n_{\mathrm{O}}$, so the weights agree at every position, and by the claim $w' = w$. Thus T is the sum of the T_w, which use disjoint coordinates on legs 2 and 3, and $T_w = \bigotimes_q U'_{w_q}$.

Count. A word of type τ is given by the positions of the three classes, a kernel letter at each kernel position and an outer letter at each outer position, which gives (3.5). The identification follows from those of the parts U'_χ. □

In Example 3.2, where (3.1) reads $4 \leq 5$, the crop removes y^{q}_4 and z^{q}_5, and the cross terms are $x_0 y^{\mathrm{q}}_1 z^{\mathrm{q}}_1$ (mid to left), $x_1 y^{\mathrm{q}}_2 z^{\mathrm{q}}_3$ (mid to right) and $x_0 y^{\mathrm{q}}_3 z^{(0)}_1$ (right to K_0). Fixing only the number of middle letters, as in [2, Lemma 4.5], would keep the last one.

3.4 The stage relation

Proposition 3.5 (One stage)**.** *Let $(r,b) \to (c,r)$ be a stage of arity d, let $N \geq 1$, let τ be a type of width N, and let M be given by (3.5). There is a degeneration over $\mathbb{Q}$*

$$(2M-1)^6\, Q_{r,b}^{\otimes N}\, \mathsf{M}_{d^{2N}} \trianglerighteq M^6\, Q_{c,r}^{\otimes N}\, \mathsf{M}_{M^2} \tag{3.7}$$

of degree at most $6(M-1)$. In each orientation $\pi \in S_3$, its maps are the restriction of Lemma 3.4, the separator (2.4) with shared leg $\pi(1)$, and coordinate permutations that do not depend on t.

Proof. Let U be the tensor (3.2); since $B_X(d)^{\otimes N} \cong B_X(d^N)$, $U^{\otimes N} \cong C(r,b)^{\otimes N} B_X(d^N)$. Apply the restriction of Lemma 3.4 to each of $2M-1$ copies of $U^{\otimes N}$, then the separator (2.4), of degree at most $M-1$, and then the reindexing of Section 2.4 built from the identifications $T_w \cong \mathcal{V}$. The first and last maps do not depend on t, and the last is defined on all target variables. By [2, Lemma 2.1], for every $\pi \in S_3$ this gives $(2M-1)(U^{\otimes N})^\pi \trianglerighteq M(\mathcal{V}B_X(M))^\pi$, of degree at most $M-1$, with shared leg $\pi(1)$, and tensoring the six orientations with one parameter t gives

$$(2M-1)^6 \bigotimes_{\pi\in S_3} (U^{\otimes N})^\pi \trianglerighteq M^6 \bigotimes_{\pi\in S_3} (\mathcal{V}B_X(M))^\pi$$

of degree at most $6(M-1)$. All maps are rational. It remains to identify both sides by coordinate permutations within each global leg, which change neither the degree nor the constant term. By (2.1),

$\bigotimes_\pi (U^{\otimes N})^\pi \cong Q_{r,b}^{\otimes N} \bigotimes_\pi B_X(d^N)^\pi \cong Q_{r,b}^{\otimes N} \mathsf{M}_{d^{2N}}$ and $\bigotimes_\pi B_X(M)^\pi \cong \mathsf{M}_{M^2}$. Since $(T^{(23)})^\pi = T^{\pi(23)}$ for every tensor T,

$$\bigotimes_{\pi\in S_3} \mathcal{V}^\pi = \bigotimes_{\pi\in S_3} \big(C(r,c)^\pi\big)^{\otimes(N-n_{\mathrm{mid}})} \big(C(r,c)^{\pi(23)}\big)^{\otimes n_{\mathrm{mid}}},$$

and as π runs through S_3, so does $\pi(23)$. Hence $\bigotimes_\pi \mathcal{V}^\pi \cong Q_{r,c}^{\otimes N} \cong Q_{c,r}^{\otimes N}$ by (2.2). □

Corollary 3.6 (Exact stage)**.** *In the setting of Proposition 3.5, let $k \geq 1$ and $E = 1 + 6k(M-1)$. Then, over $\mathbb{Q}$,*

$$E\,(2M-1)^{6k}\, Q_{r,b}^{\otimes Nk}\, \mathsf{M}_{d^{2Nk}} \succeq M^{6k}\, Q_{c,r}^{\otimes Nk}\, \mathsf{M}_{M^{2k}}. \tag{3.8}$$

Proof. Apply exactification in its rational form [2, Lemmas 2.2 and 9.1(c)] to (3.7), with degree bound $\theta = 6(M-1)$ and $k\theta + 1 = E$, and use $(mT)^{\otimes k} \cong m^k T^{\otimes k}$ and $\mathsf{M}_a^{\otimes k} \cong \mathsf{M}_{a^k}$. □

Remark 3.7 (Both signs). For one class $\mathcal{A}_\sigma$, the proof gives $\bigotimes_{\pi\in\mathcal{A}_\sigma} \mathcal{V}^\pi \cong (S_{r,c}^{\sigma})^{\otimes(N-n_{\mathrm{mid}})} (S_{r,c}^{-\sigma})^{\otimes n_{\mathrm{mid}}}$, since $\pi(23) \in \mathcal{A}_{-\sigma}$. For $c = r$, [2, (4.6)] closes this with $S_{r,r}^{+} \cong S_{r,r}^{-}$. For $r \neq c$ with $r, c \geq 2$, the identities of [2, (2.6)] do not relate $S_{r,c}^{+}$ to $S_{r,c}^{-}$, so we use all six orientations, as in the proof of [11, Lemma 4.2]. This costs nothing in the exponent. A threefold relation of degree $3(M-1)$, if it existed, would have the auxiliaries $\bigotimes_{\pi\in\mathcal{A}_\sigma} B_X(e)^\pi \cong \mathsf{M}_e$ [2, (2.2)] with $e = d^N$ and $e = M$. At power $2k$ it would give the same copy counts, auxiliary sides and number $E = 1 + 2k \cdot 3(M-1)$ of exactification copies as (3.8); only its odd powers, which give smaller sides but no smaller exponent, have no sixfold counterpart (compare [2, Appendix A.4]).

3.5 Chains

A *chain* of $J \geq 1$ stages from the *source pair* (r_1, b_1) to the *terminal pair* $(1, t)$, $t \geq 1$, consists of stages $(r_i, b_i) \to (c_i, r_i)$ of arities d_i, $1 \leq i \leq J$, with $(r_{i+1}, b_{i+1}) = (c_i, r_i)$ for $i < J$ and $(c_J, r_J) = (1, t)$. Its *evaluation size* is $s = r_1 + b_1 - 1$.

Proposition 3.8 (Chain counts)**.** *Fix a chain as above and integers $N, k \geq 1$. At each stage i fix a type of width N, let M_i be its sector count (3.5) with $d = d_i$, and put $E_i = 1 + 6k(M_i - 1)$. Then $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$ over $\mathbb{Q}$, where*

$$\begin{aligned} A &= \prod_{i=1}^{J} d_i^{2Nk}, & B &= t^{2Nk} \prod_{i=1}^{J} M_i^{2k}, \\ r_* &= s^{6Nk} \prod_{i=1}^{J} E_i\,(2M_i - 1)^{6k}, & m_* &= \prod_{i=1}^{J} M_i^{6k}. \end{aligned} \tag{3.9}$$

If $B > A$, this is a relative scheme [2, Definition 2.3], its pooled exponent $p = \log(r_*/m_*)/\log(B/A)$ satisfies $p \geq 2$, and $\omega \leq p$ over every field of characteristic zero [2, Lemma 2.4].

Proof. Put $\mathcal{T}_i = Q_{r_i,b_i}^{\otimes Nk}$ for $1 \leq i \leq J+1$, with $(r_{J+1}, b_{J+1}) = (1, t)$, so that $\mathcal{T}_{J+1} \cong \mathsf{M}_{t^{2Nk}}$ by (2.2) and (2.1). Put $\rho_i = E_i(2M_i - 1)^{6k}$ and $\rho_i' = M_i^{6k}$. Since $Q_{c_i,r_i} = Q_{r_{i+1},b_{i+1}}$, Corollary 3.6 reads $\rho_i \mathcal{T}_i \mathsf{M}_{d_i^{2Nk}} \succeq \rho_i' \mathcal{T}_{i+1} \mathsf{M}_{M_i^{2k}}$. With $A_{\geq j} = \prod_{i\geq j} d_i^{2Nk}$ and $B_{<j} = \prod_{i<j} M_i^{2k}$, we show by induction on $j = 1, \ldots, J+1$ that

$$\Big(\prod_{i=1}^{J} \rho_i\Big) \mathcal{T}_1 \mathsf{M}_A \succeq \Big(\prod_{i<j} \rho_i'\Big) \Big(\prod_{i\geq j} \rho_i\Big) \mathcal{T}_j\, \mathsf{M}_{A_{\geq j} B_{<j}}.$$

For $j = 1$ the two sides are equal. For the step, write $\mathsf{M}_{A_{\geq j}B_{<j}} \cong \mathsf{M}_{d_j^{2Nk}}\mathsf{M}_{A_{\geq j+1}B_{<j}}$. The right side is a direct sum of copies of $\rho_j\mathcal{T}_j\mathsf{M}_{d_j^{2Nk}}\mathsf{M}_{A_{\geq j+1}B_{<j}}$. Applying Corollary 3.6 for stage j, tensored with $\mathsf{M}_{A_{\geq j+1}B_{<j}}$, to each copy turns it into $\rho'_j\mathcal{T}_{j+1}\mathsf{M}_{A_{\geq j+1}B_{<j+1}}$, which is the claim for $j+1$. For $j = J+1$, the right side is isomorphic to $m_*\mathsf{M}_B$. Finally, (2.2) gives $s^{6Nk}\mathsf{M}_1 \succeq \mathcal{T}_1$ over $\mathbb{Q}$, so $r_*\mathsf{M}_A \succeq (\prod_i \rho_i)\mathcal{T}_1\mathsf{M}_A$, and composing gives $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$. The rest is Section 2.3. □

Remark 3.9 (Stage-dependent N and k). The proof needs only that the Q-power in the target of each stage is the Q-power in the source of the next. So stage i may use its own $N_i, k_i \geq 1$, a type of width N_i and $E_i = 1 + 6k_i(M_i - 1)$, provided $N_i k_i = N_1 k_1$ for all i. The same induction then gives $A = \prod_i d_i^{2N_1k_1}$, $B = t^{2N_1k_1}\prod_i M_i^{2k_i}$, $r_* = s^{6N_1k_1}\prod_i E_i(2M_i-1)^{6k_i}$ and $m_* = \prod_i M_i^{6k_i}$.

Remark 3.10 (Fields). Lemmas 3.1, 3.3 and 3.4 use integer matrices with integer inverses and coordinate selections, so they hold over every field. Proposition 3.5, Corollary 3.6 and Proposition 3.8 are proved over $\mathbb{Q}$, hence hold over every field of characteristic zero; their coefficients come from the interpolation of [2, Lemma 9.1] at the nodes $1, \dots, s$ and, at each stage i, $1, \dots, M_i$, $1, \dots, 2M_i - 1$ and $1, \dots, E_i$. We do not treat positive characteristic (Section 8).

4 Chains and the main theorem

This section chooses the chain and its parameters and proves the main theorem (Theorem 4.8). Next to the terminal pair, an *arity ramp* climbs the diagonal (Section 4.2); above it, a *bulk* of stages of one arity D makes the source pair grow geometrically (Section 4.3). Section 4.4 shows that the chain schemes of Section 4.1 have $p > 9/4$.

4.1 Chain schemes

Fix a chain (Section 3.5) with source pair (r_1, b_1), evaluation size $s = r_1 + b_1 - 1$ and terminal pair $(1, t)$; we call $G = \sum_{i=1}^{J} \log \frac{d_i+2}{d_i}$ its *gain*. For integers $N, k \geq 1$ and, at each stage i, a type of width N with sector count M_i, Proposition 3.8 gives a restriction $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ over $\mathbb{Q}$ with the counts (3.9). If $B > A$, it is a relative scheme, which we call a *chain scheme* (Definition 7.1 extends the term to crops, exchanges, several blocks per stage and other supplies). Dividing the numerator and the denominator of its pooled exponent $p = \log(r_*/m_*)/\log(B/A)$ by $2Nk$ gives

$$p = \frac{3\log s + C_{\mathrm{sep}} + C_{\mathrm{ex}}}{\Delta}, \qquad \Delta = \frac{\log(B/A)}{2Nk} = \log t + \sum_i \Big(\frac{\log M_i}{N} - \log d_i\Big), \tag{4.1}$$

where $C_{\mathrm{sep}} = \frac{3}{N}\sum_i \log(2 - 1/M_i)$ is the *separator charge* and $C_{\mathrm{ex}} = \frac{1}{2Nk}\sum_i \log E_i$ is the *exactification charge*. The fixed-batch bound (2.3) is

$$\Phi = \frac{\log(2r_*/m_*)}{\log(B/A)} = p + \frac{C_{\mathrm{pool}}}{\Delta}, \qquad C_{\mathrm{pool}} = \frac{\log 2}{2Nk}. \tag{4.2}$$

For large N and k the charges are small, and for types with the largest sector counts Δ is close to $\log t + G$ (Lemma 4.6). So the best a chain can give is its *ideal exponent* $3\log s/(\log t + G)$; Lemma 4.7 shows that p always exceeds it.

Lemma 4.1 (Smallest parent)**.** *Let $d \geq 2$, and let (c, r) be a pair of positive integers. Put*

$$b = \Big\lceil \frac{(d+2)c + r - 1}{d} \Big\rceil, \qquad e = db - (d+2)c - r + 1.$$

Then $0 \leq e \leq d - 1$, the stage $(r, b) \to (c, r)$ of arity d is legal, and b is the least integer with this property.

We call (r, b) the *smallest parent* of (c, r) for arity d, and e its *slack*.

Proof. The bounds on e restate the definition of the ceiling. For the stage $(r, b) \to (c, r)$, the last condition of (3.1) reads $db \geq (d+2)c + r - 1$, that is, $e \geq 0$, and the ceiling is the least integer b with this property. □

4.2 The ramp

Recall that the least profile of [2, (3.4)] is $F(m, n) = \varsigma_{\min(m,n)}(\max(m, n) + \frac{\min(m,n)-1}{2})$, where $\varsigma_m = \prod_{\ell=1}^{m-1}(1 + \frac{1}{3\ell})$ (written s_m in [2]), so that $F(R, R) = \varsigma_R(3R-1)/2$.

Lemma 4.2 (Zero-slack ramp)**.** *Let $R \geq 2$. Start at the pair $(1, 1)$ and, for $j = 1, \ldots, R-1$ in turn, take the smallest parent for arity $3j-1$ and then the smallest parent for arity $3j$. All slacks are zero, and the pairs are*

$$(j, j) \xleftarrow{3j-1} (j, j+1) \xleftarrow{3j} (j+1, j+1) \qquad (1 \leq j \leq R-1),$$

where each arrow points from a parent to its child and carries the arity. Read forwards, this is a chain of $2(R-1)$ legal stages from (R, R) to the terminal pair $(1, 1)$, with largest arity $3R-3$. Its gain is

$$G = \sum_{j=1}^{R-1} \log \frac{(3j+1)(3j+2)}{(3j-1)\,3j} = \log F(R, R), \qquad \text{and} \qquad F(R, R) \geq R^{4/3}.$$

Proof. For arity $3j-1$ and the pair (j, j), Lemma 4.1 gives $b = \lceil ((3j+1)j + j - 1)/(3j-1) \rceil = \lceil (3j-1)(j+1)/(3j-1) \rceil = j+1$ with slack zero. For arity $3j$ and the pair $(j, j+1)$, it gives $b = \lceil ((3j+2)j+j)/(3j) \rceil = j+1$ with slack zero. The two stages contribute the factors $(3j+1)/(3j-1)$ and $(3j+2)/(3j)$ to $\exp(G)$. Since $3j+2 = 3(j+1)-1$, the product of the factors $(3j+2)/(3j-1)$ over $j < R$ telescopes to $(3R-1)/2$, and the product of the factors $(3j+1)/(3j)$ is ς_R. Hence $\exp(G) = F(R, R)$. Finally, $(1 + \frac{1}{3\ell})^3 \geq 1 + \frac{1}{\ell}$ gives $\varsigma_R^3 \geq \prod_{\ell=1}^{R-1}(1 + \frac{1}{\ell}) = R$, and $(3R-1)/2 \geq R$. □

Each pair of ramp stages multiplies $\exp(G)$ by $F(j+1, j+1)/F(j, j)$, and $F(j+1, j+1) - F(j, j) = \varsigma_j + \varsigma_{j+1}$. For F, the quantity H_a in the proof of [11, Lemma 5.1] equals ς_a, so the ramp attains the diagonal recursion [11, (5.1)] with equality, with integer pairs and no rounding. Used alone, the ramp is a chain with $s = 2R-1$, and its ideal exponent $3\log(2R-1)/\log F(R, R)$ is the profile exponent of [2, Section 3]. It exceeds $9/4$ by less than $\frac{9}{4}\log 2/\log R$, since $F(R, R) \geq R^{4/3}$ (Lemma 4.2), and, as the proof of [2, Proposition 11.1(b)] shows, by at least $\frac{9}{4}\log\frac{6}{5}/\log\frac{3R-1}{2}$. So, by Lemma 4.7(c), a chain scheme on the ramp alone with $p \leq 9/4 + \varepsilon$ needs R and the number of stages exponential in $1/\varepsilon$. The bulk is what makes $\log A$ polynomial in $1/\varepsilon$.

4.3 The bulk

Lemma 4.3 (Bulk potential)**.** *Let $D \geq 2$, let μ be the positive root of $D\mu^2 - \mu - (D+2) = 0$, and put $\eta = \mu - 1/D$. Then $1 + 1/D < \mu < 5/4 + 1/D$, so $1 < \eta < 5/4$, and $\mu\eta = (D+2)/D$. For a pair (r, b) put $V(r, b) = \eta(r - \frac{1}{3}) + (b - \frac{1}{3})$.*

(a) If (r, b) is the smallest parent of (c, r) for arity D, with slack e, then $V(r, b) = \mu V(c, r) + e/D$.

(b) Let $R \geq 1$ and $L \geq 0$, and let the pair (r', b') arise from (R, R) by L smallest-parent steps of arity D. Then $r' + b' - 1 < \mu^L(\frac{9}{4}R + D)$.

Proof. Put $f(u) = Du^2 - u - (D+2)$. Since $f(0) < 0$ and f is a convex quadratic, f has exactly one positive root, $\mu = (1+\sqrt{4D^2+8D+1})/(2D)$. As $f(1+\frac{1}{D}) = -1$ and $f(\frac{5}{4}+\frac{1}{D}) = (9D-12)/16 > 0$, it lies strictly between $1+\frac{1}{D}$ and $\frac{5}{4}+\frac{1}{D}$. The equation $D\mu^2 = \mu+D+2$ gives $\mu\eta = \mu^2 - \mu/D = (D+2)/D$.

(a) By the definition of the slack, $b = ((D+2)c + r - 1 + e)/D$. In

$$V(r,b) - \mu V(c,r) = \eta(r - \tfrac{1}{3}) + \frac{(D+2)c + r - 1 + e}{D} - \tfrac{1}{3} - \mu\eta(c - \tfrac{1}{3}) - \mu(r - \tfrac{1}{3}),$$

the coefficient of c is $(D+2)/D - \mu\eta = 0$, and that of r is $\eta + 1/D - \mu = 0$. The constant term is $\frac{\mu-\eta}{3} + \frac{\mu\eta-1}{3} + \frac{e-1}{D} = \frac{1}{3D} + \frac{2}{3D} + \frac{e-1}{D} = \frac{e}{D}$.

(b) Let V_ℓ be the value of V after ℓ steps, and $e_\ell \le D-1$ the slack of step ℓ. By (a),

$$V_L = \mu^L V_0 + \sum_{\ell=1}^{L} \mu^{L-\ell}\frac{e_\ell}{D} \le \mu^L V_0 + \sum_{\ell=0}^{L-1} \mu^\ell \le \mu^L\Big(V_0 + \frac{1}{\mu-1}\Big).$$

Here $V_0 = (1+\eta)(R - \frac{1}{3}) < \frac{9}{4}R$, and $\mu - 1 > 1/D$ gives $1/(\mu-1) < D$. Finally, $\eta > 1$ and $r' \ge 1$ give $V_L = \eta(r' - \frac{1}{3}) + b' - \frac{1}{3} > r' + b' - 1$. □

The potential V comes from the map $(c,r) \mapsto (r, ((D+2)c + r - 1)/D)$, the smallest parent without the ceiling: the map fixes $(\frac{1}{3}, \frac{1}{3})$, and $(\eta, 1)$ is a left eigenvector of its linear part, for the Perron root μ.

Each bulk stage adds $g_D = \log\frac{D+2}{D}$ to the gain and multiplies s by about μ: in the setting of Lemma 4.3(b), part (a) and $e \ge 0$ also give $V(r',b') \ge \mu^L V(R,R)$, and $V(r,b) < \frac{5}{4}r + b \le \frac{5}{2}(r+b-1)$ for every pair of positive integers, so $\log(r'+b'-1) = L\log\mu + O(\log(R+D))$. For a long bulk, $3\log s/G$ therefore tends to $\omega_D = 3\log\mu/g_D$, the ideal exponent of the bulk.

Lemma 4.4 (Ideal gap)**.** *Let $D \ge 2$, let μ be as in Lemma 4.3, and put $y = g_D/2$. Then* $\sinh y = 1/\sqrt{D(D+2)}$,

$$\log\mu = y + \operatorname{arsinh}(\tfrac{1}{2}\sinh y), \qquad \textit{and} \qquad \frac{9}{4} \le \omega_D \le \frac{9}{4} + \frac{27}{32(D+1)^2}.$$

Proof. Since $e^y = \sqrt{(D+2)/D}$, we have $\sinh y = \frac{1}{2}(e^y - e^{-y}) = 1/\sqrt{D(D+2)}$. With $z = \frac{1}{2}\sinh y$,

$$z + \sqrt{1+z^2} = \frac{1+\sqrt{4D^2+8D+1}}{2\sqrt{D(D+2)}}, \qquad \text{so} \qquad e^y(z+\sqrt{1+z^2}) = \frac{1+\sqrt{4D^2+8D+1}}{2D} = \mu.$$

This is the identity, since $\operatorname{arsinh} z = \log(z + \sqrt{1+z^2})$. For the lower bound, $\operatorname{arsinh}(\frac{1}{2}\sinh y) \ge \frac{y}{2}$ is equivalent to $\frac{1}{2}\sinh y \ge \sinh\frac{y}{2}$, that is, to $\sinh\frac{y}{2}\cosh\frac{y}{2} \ge \sinh\frac{y}{2}$. Hence $3\log\mu \ge \frac{9}{2}y = \frac{9}{4}g_D$.

For the upper bound, $\frac{D+2}{D} = (1+\frac{1}{D+1})/(1-\frac{1}{D+1})$ gives $g_D = 2\operatorname{artanh}\frac{1}{D+1} \ge \frac{2}{D+1}$. Using $\operatorname{arsinh} z \le z$,

$$\omega_D = \frac{3}{2} + \frac{3\operatorname{arsinh}(\frac{1}{2}\sinh y)}{g_D} \le \frac{3}{2} + \frac{3\sinh y}{2g_D} \le \frac{3}{2} + \frac{3}{4}\cdot\frac{D+1}{\sqrt{D(D+2)}}.$$

Finally, $D(D+2) = (D+1)^2 - 1$ lies between $\frac{8}{9}(D+1)^2$ and $(D+1)^2$, as $D \ge 2$. Hence $\frac{D+1}{\sqrt{D(D+2)}} \le \frac{(D+1)^2}{D(D+2)} = 1 + \frac{1}{D(D+2)} \le 1 + \frac{9}{8(D+1)^2}$. □

The expansion $\operatorname{arsinh}(\frac{1}{2}\sinh y) = \frac{y}{2} + \frac{y^3}{16} + O(y^5)$ gives $\omega_D - \frac{9}{4} = \frac{3}{32D^2} + O(D^{-3})$, about a ninth of the upper bound.

Lemma 4.5 (Defect)**.** *Let $D \geq 2$, $R = 8D$ and $L \geq 0$. Starting at $(1,1)$, build the ramp of Lemma 4.2 up to (R,R), and then take L further smallest parents of arity D. Read forwards, this is a chain with terminal pair $(1,1)$ and $J = 16D - 2 + L$ stages: L bulk stages of arity D, then the $2R-2$ ramp stages. All its arities satisfy $2 \leq d_i \leq 24D-3$. Its gain is $G = \log F(R,R) + Lg_D$, and*

$$3\log s - \tfrac{9}{4}G < 3\log\tfrac{19}{8} + Lg_D(\omega_D - \tfrac{9}{4}) \leq 3 + \frac{27L}{16D(D+1)^2}. \tag{4.3}$$

We call this chain the *ramp-and-bulk chain* for D and L.

Proof. The stages, arities and gain follow from Lemmas 4.1 and 4.2. Since $\frac{9}{4}R + D = \frac{19}{8}R$, Lemma 4.3(b) gives $3\log s < 3\log\frac{19R}{8} + 3L\log\mu$, and Lemma 4.2 gives $\frac{9}{4}\log F(R,R) \geq 3\log R$. Subtracting proves the first inequality, since $3\log\mu - \frac{9}{4}g_D = g_D(\omega_D - \frac{9}{4})$. For the second, $\log\frac{19}{8} < 1$, and $g_D(\omega_D - \frac{9}{4}) \leq \frac{2}{D}\cdot\frac{27}{32(D+1)^2}$ by Lemma 4.4 and $g_D = \log(1+\frac{2}{D}) \leq \frac{2}{D}$. □

4.4 Types, and the least profile as a potential

For $d \geq 2$ and $N \geq 1$, a type $(n_{\mathrm{K}}, n_{\mathrm{O}}, n_{\mathrm{mid}})$ of width N, that is, with $n_{\mathrm{K}} + n_{\mathrm{O}} + n_{\mathrm{mid}} = N$, has the sector count $N!\,(d-1)^{n_{\mathrm{K}}}2^{n_{\mathrm{O}}}/(n_{\mathrm{K}}!\,n_{\mathrm{O}}!\,n_{\mathrm{mid}}!)$ (Lemma 3.4). Let $M(N,d)$ be the largest sector count over all types of width N.

Lemma 4.6 (Types)**.** *For $d \geq 2$ and $N \geq 1$, every sector count is less than $(d+2)^N$, and $(d+2)^N/(N+1)^2 \leq M(N,d) < (d+2)^N$.*

Proof. By the multinomial theorem, the sector counts of all types add up to $((d-1)+2+1)^N = (d+2)^N$. There are $\binom{N+2}{2} \geq 3$ types, and every count is positive, so each count is less than the sum. The largest count is at least the average, and $\binom{N+2}{2} = \frac{(N+2)(N+1)}{2} \leq (N+1)^2$. □

Lemma 4.7 (The least profile is a potential)**.** *Let F be the least profile of [2, (3.4)].*

(a) Every legal stage $(r,b) \to (c,r)$ of arity d satisfies $d\,F(r,b) \geq (d+2)\,F(c,r)$, with equality on every stage of the ramp of Lemma 4.2.

(b) Every chain (Section 3.5) satisfies $\log t + G \leq \log F(r_1,b_1) \leq \frac{4}{3}\log s$.

(c) Every chain scheme of Section 4.1, and every relative scheme given by the variant of Remark 3.9, satisfies $p > 3\log s/(\log t + G) \geq 9/4$.

Proof. (a) Put $b^+ = db - (d-1)c$ and $h = b - c$, so that $c < b < b^+$ and $b^+ - b = (d-1)h$. By [2, Theorem 3.2], F is a profile: it satisfies the concavity axiom [2, (3.2)], and the tripling axiom [2, (3.3)] at every column. By concavity, the increments $F(r,b'+1) - F(r,b')$ of row r do not increase in b'. So each of the h increments from column c to b is at least each of the $(d-1)h$ increments from b to b^+, and $(d-1)(F(r,b) - F(r,c)) \geq F(r,b^+) - F(r,b)$, that is,

$$d\,F(r,b) \geq (d-1)F(r,c) + F(r,b^+).$$

By [2, (3.5)], the increments of row r from column $r-1$ on equal $\varsigma_r > 0$. As they do not increase, the row is increasing. By Lemma 3.3(c), $b^+ \geq 3c + r - 1$, so $F(r,b^+) \geq F(r,3c+r-1) \geq 3F(r,c)$ by tripling. Hence $dF(r,b) \geq (d+2)F(r,c)$, and $F(r,c) = F(c,r)$ by symmetry.

On the ramp, [2, (3.4)] gives $F(j,j) = \varsigma_j\frac{3j-1}{2}$, $F(j,j+1) = \varsigma_j\frac{3j+1}{2}$ and $F(j+1,j+1) = \varsigma_{j+1}\frac{3j+2}{2} = \varsigma_j\frac{(3j+1)(3j+2)}{6j}$. Hence $(3j-1)F(j,j+1) = (3j+1)F(j,j)$ and $3j\,F(j+1,j+1) = (3j+2)F(j,j+1)$: these are the equality cases for the stages of arities $3j-1$ and $3j$.

(b) Applying (a) to every stage and multiplying gives $F(r_1, b_1) \geq \exp(G) F(c_J, r_J) = t \exp(G)$, by the normalization [2, (3.1)]. The second inequality is $F(m, n) \leq (m + n - 1)^{4/3}$ [2, (3.6)].

(c) By (3.9), $r_*/m_* = s^{6Nk} \prod_i ((2M_i - 1)/M_i)^{6k} E_i$, and every factor other than s^{6Nk} is at least one, so $\log(r_*/m_*) \geq 6Nk \log s$. By Lemma 4.6, $M_i < (d_i + 2)^N$, so $\log(B/A) < 2Nk(\log t + G)$, since $J \geq 1$. As $B > A$ and $s \geq b_1 > c_1 \geq 1$, we get $p \geq 6Nk \log s / \log(B/A) > 3 \log s/(\log t + G)$. Since $G > 0$, (b) gives $3 \log s/(\log t + G) \geq 9/4$. The same argument, with $N_1 k_1$ in place of Nk, applies to the variant of Remark 3.9, with stage-dependent N_i and k_i and $N_i k_i = N_1 k_1$, since there $M_i^{2k_i} < (d_i + 2)^{2N_i k_i} = (d_i + 2)^{2N_1 k_1}$. □

Lemma 4.7 is the analogue for chain schemes of the scalar limit in [2, Remark 3.3]. Section 7 shows more: $p > 9/4$ persists if the auxiliary of each stage is replaced by a compressed auxiliary of any rank, and for every supply (Definition 7.8, Corollary 7.9). On a ramp stage, $c \geq r - 1$, $b^+ = 3c + r - 1$, and row r of F is affine from column $r - 1$ on, so every inequality in the proof of (a) is an equality.

4.5 The main theorem

Theorem 4.8 (Main theorem). *Let $0 < \varepsilon \leq 1$ be rational. Put*

$$D = \lceil \sqrt{4/\varepsilon} \rceil, \qquad R = 8D, \qquad L = \lceil 64D/\varepsilon \rceil,$$

let H be the number of binary digits of $\lceil 128D/\varepsilon \rceil$, and put $N = k = \lceil 256DH/\varepsilon \rceil$. Take the ramp-and-bulk chain for D and L (Lemma 4.5), the width N and the power k, and at stage i a type of width N with the largest sector count $M_i = M(N, d_i)$. Then the restriction $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$ of Proposition 3.8 for these data has rational coefficients and $B > A$, so it is a relative scheme over $\mathbb{Q}$. Its pooled exponent p, its fixed-batch bound Φ and its counts satisfy

$$\frac{9}{4} < p < \Phi < \frac{9}{4} + \frac{719}{1988}\,\varepsilon < \frac{9}{4} + \frac{\varepsilon}{2}, \qquad \log A < 3 \cdot 10^9\, \varepsilon^{-9/2} \log^3 \frac{4096}{\varepsilon},$$
$$B \leq A^2, \qquad m_* = B^3 \leq A^6, \qquad r_* < A^{35/4}.$$

All parameters are computed from ε by integer arithmetic; for example, if $\varepsilon = u/v$ with positive integers u and v, then D is the least integer with $D^2 u \geq 4v$.

Proof. Here $t = 1$, so by (4.1) and (4.2), $\Phi = (3 \log s + C_{\mathrm{sep}} + C_{\mathrm{ex}} + C_{\mathrm{pool}})/\Delta$ with $\Delta = \sum_i (\frac{1}{N} \log M_i - \log d_i)$, once we know $\Delta > 0$. Put $x = 128D/\varepsilon$, $a_N = 2 \log(N + 1)/N$ and $\Lambda = \log(4096/\varepsilon)$.

Step 0: parameter facts. Since $\varepsilon \leq 1$, we have $D \geq 2$, $D^2 \varepsilon \geq 4$ and $x \geq 256$. From $2^{H-1} \leq \lceil x \rceil < 2^H$ we get $H > \log_2 x \geq \log x$, $H \leq 1 + \log_2(x + 1) \leq x$ and, as $\lceil x \rceil \geq 2^8$, $H \geq 9$. Next, $N \geq 2xH$, so $N = k \geq 4608$, and $N + 1 \leq 2xH + 2 \leq 2x^2 + 2 \leq x^3$. Also $\log(6k) < k$. Since $L \geq 64D$, $J = 16D - 2 + L \leq \frac{5}{4} L$. Every arity satisfies $d_i \leq 24D - 3$, so $\log d_i < \log(d_i + 2) < \log(24D) \leq \log x < H$.

Step 1: the denominator, and $B > A$. By Lemma 4.6, $\frac{1}{N} \log M_i \geq \log(d_i + 2) - a_N$, so $\Delta \geq G - J a_N$. By Step 0, $a_N \leq 6 \log x/N < 6H/(2xH) = 3/x$, that is, $a_N < 3\varepsilon/(128D)$. Since $\log(1 + u) \geq u/(1 + u)$, we have $g_D \geq 2/(D + 2)$. As $\log F(R, R) \geq 0$,

$$G \geq L g_D \geq \frac{2L}{D + 2} \geq \frac{128D}{(D + 2)\varepsilon} \geq \frac{64}{\varepsilon}, \qquad \frac{J}{G} \leq \frac{5L}{4} \cdot \frac{D + 2}{2L} \leq \frac{5D}{4}.$$

Hence $J a_N/G \leq \frac{15}{512} \varepsilon < 1$, and $\Delta \geq G - J a_N > 0$. So $\log(B/A) = 2Nk\Delta > 0$, that is, $B > A$.

Step 2: the charges. Since $2 - 1/M_i < 2$, $C_{\mathrm{sep}} \leq 3J \log 2/N$. Since $E_i \leq 6kM_i \leq 6k(d_i + 2)^N$, $C_{\mathrm{ex}} \leq J \log(6k)/(2Nk) + \frac{1}{2k} \sum_i \log(d_i + 2)$.

Step 3: reduction. Put $X = 3\log s + C_{\mathrm{sep}} + C_{\mathrm{ex}} + C_{\mathrm{pool}} - \frac{9}{4}\Delta$, so that $\Phi - \frac{9}{4} = X/\Delta$. By Lemma 4.7(c), $p > 9/4$, that is, $X > C_{\mathrm{pool}} > 0$. With $\Delta \geq G - Ja_N$ and $-\frac{9}{4}\Delta \leq -\frac{9}{4}G + \frac{9}{4}Ja_N$ from Step 1,

$$\Phi - \frac{9}{4} = \frac{X}{\Delta} \leq \frac{X}{G - Ja_N} \leq \frac{1}{1 - Ja_N/G}\Big(\frac{3\log s - \frac{9}{4}G}{G} + \frac{C_{\mathrm{sep}} + C_{\mathrm{ex}} + C_{\mathrm{pool}}}{G} + \frac{9}{4}\cdot\frac{Ja_N}{G}\Big).$$

Step 4: the charges, divided by G. We bound the six terms of the bracket: (i) the geometric defect, (ii) the type loss, (iii) the separator copies, (iv) and (v) the two parts of the exactification charge, and (vi) the fixed-batch ceiling.

(i) By (4.3), $3\log s - \frac{9}{4}G < 3 + \frac{27L}{16D(D+1)^2}$. Here $3/G \leq \frac{3}{64}\varepsilon$, and $G \geq 2L/(D+2)$ gives $\frac{27L}{16D(D+1)^2 G} \leq \frac{27(D+2)}{32D(D+1)^2} < \frac{27}{32D^2} \leq \frac{27}{128}\varepsilon$, using $D(D+2) < (D+1)^2$ and $D^2\varepsilon \geq 4$. Together, $\frac{3}{64} + \frac{27}{128} = \frac{33}{128}$.

(ii) $\frac{9}{4}\cdot\frac{Ja_N}{G} \leq \frac{9}{4}\cdot\frac{15}{512}\varepsilon$, by Step 1.

(iii) $\frac{3J\log 2}{NG} \leq \frac{15D\log 2}{4N} \leq \frac{15\log 2}{1024H}\varepsilon < \frac{15}{1024}\varepsilon$, by $J/G \leq 5D/4$ and $N \geq 256DH/\varepsilon$.

(iv) $\frac{J\log(6k)}{2NkG} \leq \frac{J}{2NG} \leq \frac{5D}{8N} \leq \frac{5}{2048H}\varepsilon \leq \frac{5}{2048}\varepsilon$.

(v) $\frac{1}{2kG}\sum_i \log(d_i+2) \leq \frac{JH}{2kG} \leq \frac{5DH}{8k} \leq \frac{5}{2048}\varepsilon$.

(vi) $\frac{C_{\mathrm{pool}}}{G} = \frac{\log 2}{2NkG} \leq \frac{\log 2}{128Nk}\varepsilon < \frac{1}{128}\varepsilon$, by $G \geq 64/\varepsilon$.

The bounds add up to $\frac{528+135+30+5+5+16}{2048}\varepsilon = \frac{719}{2048}\varepsilon$, and $1 - Ja_N/G \geq \frac{497}{512}$. Hence $\Phi - \frac{9}{4} < \frac{719}{2048}\cdot\frac{512}{497}\varepsilon = \frac{719}{1988}\varepsilon < \frac{\varepsilon}{2}$. With $9/4 < p$ (Lemma 4.7(c)) and $\Phi = p + C_{\mathrm{pool}}/\Delta > p$, this proves the exponent bounds.

Step 5: the source side. We have $D \leq 2\varepsilon^{-1/2} + 1 \leq 3\varepsilon^{-1/2}$, so $J \leq 16D + 64D/\varepsilon - 1 \leq 240\,\varepsilon^{-3/2}$ and $x \leq 384\,\varepsilon^{-3/2}$. Since $H \leq 1 + \log_2(2x) \leq 1 + \frac{3}{2}\log(2x)$ and $\log(2x) \geq \log 512 > 2$, we get $H \leq 2\log(2x) \leq 2\log 768 + 3\log\frac{1}{\varepsilon} \leq 3\Lambda$, using $768^2 < 4096^3$. Hence $N = k \leq 256DH/\varepsilon + 1 \leq 2304\,\varepsilon^{-3/2}\Lambda + 1 \leq 2305\,\varepsilon^{-3/2}\Lambda$. Finally, $\log d_i < \log(24D) \leq \log(72\varepsilon^{-1/2}) \leq \Lambda$. Therefore

$$\log A = 2Nk\sum_i \log d_i < 2N^2 J\Lambda \leq 2\cdot 2305^2\cdot 240\,\varepsilon^{-9/2}\Lambda^3 = 2{,}550{,}252{,}000\,\varepsilon^{-9/2}\Lambda^3 < 3\cdot 10^9\,\varepsilon^{-9/2}\Lambda^3.$$

Step 6: the other counts. Since $d_i \geq 2$, we have $d_i + 2 \leq d_i^2$, so $M_i < (d_i+2)^N \leq d_i^{2N}$ by Lemma 4.6, and $B = \prod_i M_i^{2k} \leq A^2$. Since $t = 1$, (3.9) gives $m_* = B^3 \leq A^6$. By the definition of p, $r_* = m_*(B/A)^p$. As $1 < B/A \leq A$ and $p < \frac{9}{4} + \frac{\varepsilon}{2} \leq \frac{11}{4}$, we get $r_* < B^3(B/A)^{11/4} \leq A^{35/4}$. The coefficients are rational by Proposition 3.8. □

With the separator of [2, Proposition 4.6] in place of Lü's, $E_i = 1 + 6k(M_i - 1)^2$ doubles bound (v) of Step 4, so the proof gives $\frac{724}{1988}\varepsilon$ in place of $\frac{719}{1988}\varepsilon$, with coefficients in a cyclotomic field instead of $\mathbb{Q}$. For the bulk alone, started at $(1,1)$, Lemma 4.3(b) with $R = 1$ puts $3\log(D + \frac{9}{4})$ in place of $3\log\frac{19}{8}$ in (4.3); the same argument would then need L larger by a factor of order $\log\frac{1}{\varepsilon}$ and would give only $\log A = O(\varepsilon^{-9/2}\log^4\frac{1}{\varepsilon})$.

Corollary 4.9. *Let $0 < \varepsilon \leq \frac{1}{2}$ be rational. The scheme of Theorem 4.8 with 2ε in place of ε is a relative scheme over $\mathbb{Q}$ with $\frac{9}{4} < p < \Phi < \frac{9}{4} + \varepsilon$, $\log A < 1.4\cdot 10^8\,\varepsilon^{-9/2}\log^3(2048/\varepsilon)$, $B \leq A^2$, $m_* \leq A^6$ and $r_* < A^{35/4}$.*

Proof. Theorem 4.8 applies to $2\varepsilon \le 1$. It gives $\Phi < \frac{9}{4} + \frac{719}{994}\varepsilon$ and $\log A < 3 \cdot 10^9 (2\varepsilon)^{-9/2} \log^3(2048/\varepsilon)$. Finally, $3 \cdot 10^9 \cdot 2^{-9/2} < 1.4 \cdot 10^8$, because $9 \cdot 10^{18} < 1.96 \cdot 10^{16} \cdot 2^9$. □

Remark 4.10 (Φ versus p, and the bound $\frac{9}{4} + \frac{\varepsilon}{2}$). By (2.3), Φ also bounds the exponent of the recursion without pooling. Here $\Phi - p = \log 2 / \log(B/A)$, and Steps 0 and 1 give $\log(B/A) \ge 2Nk(G - Ja_N) \ge 2 \cdot 4608^2 \cdot \frac{497}{512} \cdot \frac{64}{\varepsilon}$, so $\Phi - p < 10^{-9}\varepsilon$. The theorem keeps half of ε in reserve because Section 5 unrolls the pooled recursion into one seed, as in [2, Theorem 9.2]; the factor lost in unrolling is absorbed into the remaining $\varepsilon/2$. For an algorithm with exponent below $\frac{9}{4} + \varepsilon$ alone, Corollary 4.9 suffices.

5 Seeds and one algorithm for all sizes

This section turns the relative scheme of Theorem 4.8 into a rational seed (Section 5.2), bounds its coefficients and the cost of generating them (Section 5.3), and gives one algorithm for all sizes (Section 5.4).

Setting. Throughout, $\varepsilon \in (0, 1]$ is rational, $\Lambda = \log(4096/\varepsilon)$, and $\mathcal{S}_\varepsilon$ is the relative scheme $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$ of Theorem 4.8, with the parameters D, R, L, H and $N = k$ defined there. Its chain has $J = 2(R-1) + L$ stages, from the source pair (r_1, b_1) to the terminal pair $(1, 1)$, with arities $2 \le d_i < 24D$. At stage i, $\mathcal{S}_\varepsilon$ uses the first type, in the lexicographic order of $(n_\mathrm{K}, n_\mathrm{O})$, with the largest sector count M_i; so $(d_i + 2)^N/(N+1)^2 \le M_i < (d_i + 2)^N$ (Lemma 4.6). The counts are those of (3.9) for $t = 1$, and s and E_i are as in Proposition 3.8; the three maps are rational, and $9/4 < p < 9/4 + \varepsilon/2$.

The pooled recursion. We use the pooled recursion of [2, Lemma 2.4] in its notation, except that we write $\bar{n}_\ell$ for the side at depth ℓ, because n_0 denotes the side of a seed here. Thus $\rho = A/B$, $n_\infty = AB/(B - A)$, and $n_* = \lceil n_\infty \rceil + 1$ is the leaf threshold. At side n, the recursion starts from $\nu_0 = 1$ product of side $\bar{n}_0 = n$ and sets $\nu_{\ell+1} = r_* \lceil \nu_\ell / m_* \rceil$ and $\bar{n}_{\ell+1} = A \lceil \bar{n}_\ell / B \rceil$; the last depth $\ell_{\max}$ is the first ℓ with $\bar{n}_\ell \le n_*$, and its products are computed classically. Step 3 of the proof of [2, Lemma 2.4] shows that, for every $n \ge 1$ and $0 \le \ell \le \ell_{\max}$,

$$\nu_\ell \le \Big(1 + \frac{r_* m_*}{r_* - m_*}\Big)\Big(\frac{r_*}{m_*}\Big)^\ell, \qquad \rho^\ell n \le \bar{n}_\ell < n_\infty + \rho^\ell n, \qquad \rho < \rho^{\ell_{\max}} n \le n_*. \tag{5.1}$$

5.1 Resources

Lemma 5.1 (Resources)**.** *For every rational $\varepsilon \in (0, 1]$, the scheme $\mathcal{S}_\varepsilon$ satisfies:*

(a) $N = k \ge 4608$, $J \le 240\,\varepsilon^{-3/2}$, $N \le 2305\,\varepsilon^{-3/2}\Lambda$, *and* $2NkJ \le \log_2 A \le 2NkJ \log_2(24D)$; *in particular* $\log A \le 2{,}550{,}252{,}000\,\varepsilon^{-9/2}\Lambda^3$;

(b) $m_* = B^3$ *and* $\log(B/A) \ge 124Nk$; *in particular* $2A \le B \le A^2$;

(c) $4 \le (B/A)^2 \le r_*/m_* \le (B/A)^{11/4}$ *and* $r_* < A^{35/4}$;

(d) $A < n_* \le 2A + 1$;

(e) $A^{12} \le \Xi$ *and* $\Xi + 1 \le A^{27}$, *where* $\Xi = r_*(1 + r_* m_*)\, n_*^3$.

Proof. (a) Steps 0 and 5 of the proof of Theorem 4.8 give $N = k \geq 4608$, $J \leq 240\,\varepsilon^{-3/2}$, $N \leq 2305\,\varepsilon^{-3/2}\Lambda$ and the bound on $\log A$. Since $2 \leq d_i < 24D$, the number $\log_2 A = 2Nk\sum_i \log_2 d_i$ lies between $2NkJ$ and $2NkJ\log_2(24D)$.

(b) Theorem 4.8 gives $m_* = B^3$ and $B \leq A^2$. Let $G = \sum_i \log((d_i+2)/d_i)$ be the gain of the chain. Step 1 of the proof of Theorem 4.8 gives

$$\log(B/A) \geq \frac{497}{512}\cdot 2NkG, \qquad G \geq \frac{64}{\varepsilon}, \qquad J \leq \frac{5D}{4}\,G, \tag{5.2}$$

so $\log(B/A) \geq 2Nk\cdot\frac{64}{\varepsilon}\cdot\frac{497}{512} \geq 124Nk$, and in particular $B \geq 2A$.

(c) By Step 1 of the proof of [2, Lemma 2.4], $p \geq 2$, so $r_*/m_* = (B/A)^p \geq (B/A)^2 \geq 4$. Also $p < 9/4 + \varepsilon/2 \leq 11/4$, and $r_* < A^{35/4}$ by Theorem 4.8.

(d) Since $B \geq 2A$, we have $A < AB/(B-A) = A/(1-A/B) \leq 2A$, so $A+2 \leq n_* \leq 2A+1$.

(e) By (b)–(d), $r_* \geq 4m_*$, $m_* = B^3 > A^3$ and $n_* > A$, so $\Xi \geq r_*^2 m_* n_*^3 > A^{12}$. Also $1 + r_*m_* \leq 2r_*m_*$ and $n_* \leq 3A$, so $\Xi \leq 54\,r_*^2 m_* A^3 \leq 54A^{35/2+6+3}$. Hence $\Xi + 1 \leq 55A^{53/2} \leq A^{27}$, because $A \geq 2^{2NkJ} > 55^2$. □

5.2 An explicit seed

The next theorem is the analogue of [2, Theorem 9.2] for $\mathcal{S}_\varepsilon$, and its proof follows the proof given there.

Theorem 5.2 (An explicit seed)**.** *Let $\varepsilon \in (0,1]$ be rational, let n_* and Ξ be as in Lemma 5.1, and put*

$$n_0 = (\Xi+1)^{\lceil 2/\varepsilon\rceil}.$$

For every $n \geq n_0$, unrolling the pooled recursion of $\mathcal{S}_\varepsilon$ at side n, with the zero padding of [2, Lemma 2.4] and classical multiplication at the last depth, gives a decomposition of M_n into at most $\Xi n^p < n^{9/4+\varepsilon}$ rank-one tensors with rational coefficients. In particular, $n = n_0$ gives a rational seed of side n_0 with fewer than $n_0^{9/4+\varepsilon}$ products, and

$$\frac{24}{\varepsilon}\log A \;\leq\; \log n_0 \;\leq\; 27\left\lceil\frac{2}{\varepsilon}\right\rceil\log A \;<\; 2.1\cdot 10^{11}\,\varepsilon^{-11/2}\log^3\frac{4096}{\varepsilon}.$$

Proof. Run the pooled recursion at side n. By (5.1) and $r_* - m_* \geq 1$, we have $\nu_{\ell_{\max}} \leq (1 + r_*m_*)(r_*/m_*)^{\ell_{\max}}$. Since $r_*/m_* = \rho^{-p}$ and $\rho^{\ell_{\max}} n > \rho$,

$$\left(\frac{r_*}{m_*}\right)^{\ell_{\max}} = (\rho^{\ell_{\max}})^{-p} < \left(\frac{n}{\rho}\right)^p = \frac{r_*}{m_*}\,n^p,$$

and therefore $\nu_{\ell_{\max}} < r_*(1 + r_*m_*)\,n^p$, as $m_* \geq 1$. Every product at the last depth has side $\bar{n}_{\ell_{\max}} \leq n_*$, and classical multiplication computes it with at most n_*^3 scalar products. Each of them multiplies an entry of a left input by an entry of a right input, which by Step 5 of the proof of [2, Lemma 2.4] are linear forms in the entries of the two original $n \times n$ matrices. All later operations (the output maps and the removal of padding) are fixed linear maps, so every entry of the computed product is a fixed linear combination of these scalar products. As in the proof of [2, Corollary 8.7], the scalar products therefore give a decomposition of M_n into at most $\nu_{\ell_{\max}} n_*^3 < \Xi n^p$ rank-one tensors, one for each scalar product. Its coefficients are rational because the maps of $\mathcal{S}_\varepsilon$ are rational. Finally, $n \geq n_0$ gives $n^{\varepsilon/2} \geq (\Xi+1)^{\lceil 2/\varepsilon\rceil\varepsilon/2} \geq \Xi + 1 > \Xi$, and $p < 9/4 + \varepsilon/2$ gives $\Xi n^p < n^{9/4+\varepsilon}$.

For the size, $\log n_0 = \lceil 2/\varepsilon\rceil\log(\Xi+1)$, and Lemma 5.1(e) gives $(2/\varepsilon)\cdot 12\log A \leq \log n_0 \leq 27\lceil 2/\varepsilon\rceil\log A$. Since $\varepsilon \leq 1$, we have $\lceil 2/\varepsilon\rceil \leq 3/\varepsilon$, and Lemma 5.1(a) gives $\log n_0 \leq 81\cdot 2{,}550{,}252{,}000\,\varepsilon^{-11/2}\Lambda^3 < 2.1\cdot 10^{11}\,\varepsilon^{-11/2}\Lambda^3$. □

Block recursion with the seed is covered by [2, Lemma 9.4 and Propositions 9.5–9.6].

Remark 5.3 (The factor $1/\varepsilon$). The factor of order $1/\varepsilon$ between $\log n_0$ and $\log A$ is necessary for the procedure of Theorem 5.2 when $\varepsilon \le 3/4$, as in [2, Remark 8.8]: if the procedure, run at side n, executes fewer than $n^{9/4+\varepsilon}$ scalar products at its last depth, counting those in batch slots filled with zero matrices, then $\log n > (\frac{13}{4}\log A - \log 27)/\varepsilon$. Indeed, $\ell_{\max} = 0$ would give $n^3 \ge n^{9/4+\varepsilon}$ products, so $\ell_{\max} \ge 1$. Then $\nu_1 = r_*$ and $\nu_{\ell+1} \ge (r_*/m_*)\nu_\ell$, so $\nu_{\ell_{\max}} \ge m_*\rho^{-p\ell_{\max}} \ge m_*(n/n_*)^p$ by (5.1). Every product at the last depth has side $A\lceil \bar{n}_{\ell_{\max}-1}/B\rceil \ge A$, hence needs at least A^3 scalar products. With $m_* = B^3 \ge A^3$, $n_* \le 3A$ and $p \le 3$, the procedure executes at least $n^p A^{6-p}/27$ products, and $9/4 < p < 21/8$ (Lemma 4.7 and Theorem 4.8) gives $\varepsilon \log n > (9/4 + \varepsilon - p)\log n > (6-p)\log A - \log 27 > \frac{13}{4}\log A - \log 27$. This does not bound the rank after rank-one terms with a zero factor are deleted.

5.3 Coefficients and generation

Appendix B writes each of the three maps of $\mathcal{S}_\varepsilon$ as a product of explicit matrices and describes a simple generator that multiplies them out. We count bit operations of schoolbook integer arithmetic; an addition, subtraction, multiplication, comparison or division with remainder of integers with at most $b \ge 1$ bits costs at most $16b^2$ of them.

Proposition 5.4 (Coefficients and generation)**.** *Let $\varepsilon \in (0,1]$ be rational, and put*

$$\mathcal{D} = ((s-1)!)^{2Nk} \prod_{i=1}^{J} \left((M_i-1)!\,(2M_i-2)!\right)^{2k} (E_i!)^{2k(M_i-1)}, \qquad \beta_\varepsilon = (24D)^{3N}.$$

(a) Every coefficient of the three maps of $\mathcal{S}_\varepsilon$ is $\mathfrak{c}/\mathcal{D}$ with an integer $\mathfrak{c}$, $|\mathfrak{c}| < 2^{\beta_\varepsilon}$, and $\mathcal{D} < 2^{\beta_\varepsilon}$.

(b) $\beta_\varepsilon \le A^{1/k}$, and $\log_2 \beta_\varepsilon < 10^4\,\varepsilon^{-3/2}\Lambda^2$.

(c) Let $\varepsilon = u/v$ with positive integers $u, v \le A$; this holds whenever $\varepsilon = 1/v$, since $v < N < A$. The generator of Appendix B.4 computes $\mathcal{D}$ and the integers $\mathfrak{c}$ of all three maps from u and v with at most A^{46} bit operations, and every integer that it forms has absolute value below 2^{β_ε}.

Proof. Parts (a) and (c) are proved in Appendices B.3 and B.4. For (b), $\log_2 A^{1/k} = 2N\sum_i \log_2 d_i \ge 2NJ$, and $2J \ge 2L \ge 128D \ge 3\log_2(24D)$, so $\beta_\varepsilon \le A^{1/k}$. Also $\log_2 \beta_\varepsilon = 3N\log_2(24D) \le 3\cdot 2305\,\varepsilon^{-3/2}\Lambda\cdot\Lambda/\log 2 < 10^4\,\varepsilon^{-3/2}\Lambda^2$, by Lemma 5.1(a) and $\log(24D) \le \Lambda$ (Step 5 of the proof of Theorem 4.8). □

The power 46 in (c) comes from dense products of matrices of side at most $r_*A^2B^2$; the generator exploits no sparsity.

Remark 5.5 (Precision of one batch). Apply $\mathcal{S}_\varepsilon$ once to m_* pairs of $B\times B$ matrices X_j, Y_j, and multiply the r_* pairs of $A\times A$ matrices that it forms classically. Evaluate every linear combination by recursive summation in binary floating point with round-to-nearest and unit roundoff $\mathsf{u} \le 2^{-4\beta_\varepsilon}$, without underflow or overflow. Then every entry of every computed product X_jY_j has error at most $2^{4\beta_\varepsilon}\mathsf{u}\max_j\|X_j\|_{\max}\max_j\|Y_j\|_{\max}$. The proof follows that of [2, Proposition 9.6], for one level with classical leaves of side A. A bilinear monomial carries at most $\chi = 2m_*B^2 + r_*A^2 + A + 3$ rounding factors, and $\chi \le 4A^{43/4}$ by Lemma 5.1(b),(c). An output entry combines at most r_*A^2 entries of the $A\times A$ products, each a sum of A products of two linear combinations, and the leg matrices have entries and absolute row sums below 2^{β_ε} (Appendix B.3). So the exact monomials of one entry have absolute sum at most $\Upsilon = r_*A^3 2^{3\beta_\varepsilon}$ times $\max_j\|X_j\|_{\max}\max_j\|Y_j\|_{\max}$. As

$2\chi\Upsilon \le 8A^{45/2}2^{3\beta_\varepsilon} < 2^{4\beta_\varepsilon}$, we have $\chi\mathsf{u} < 1/2$, and the error is at most $\Upsilon\chi\mathsf{u}/(1-\chi\mathsf{u}) \le 2^{4\beta_\varepsilon}\mathsf{u}$ times that product. We do not analyze the precision or the integer bit complexity of the pooled recursion, in which the inputs of a batch are themselves computed.

5.4 One algorithm for all sizes

We first make the constants of the pooled recursion explicit.

Lemma 5.6 (Pooled recursion with explicit constants)**.** *Let $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ be a relative scheme with pooled exponent $p \le 3$, $B \ge 2A$ and $m_* \le B^3$. Store its three maps as dense matrices over a field, and run the pooled recursion of [2, Lemma 2.4] with classical multiplication at the last depth. For every $n \ge 1$, the number of additions, subtractions and multiplications in the field, including multiplications by coefficients of the scheme, is at most*

$$C_\nu n^p\Big[2\frac{r_*}{m_*}n_*^3 + 12m_*B^2\Big(\frac{4}{3}\frac{r_*}{m_*}n_\infty^2 + \ell_{\max}\Big(\frac{B}{A}\Big)^{p-2}\Big)\Big] \le (304 + 28\log_2 n)\,B^{11}n^p,$$

where $C_\nu = 1 + r_*m_*/(r_* - m_*)$ and $\ell_{\max}$ is the last depth at side n.

Proof. Let $\ell < \ell_{\max}$ and $c_\ell = \lceil \bar{n}_\ell/B \rceil$. One batch at depth ℓ is an instance of $m_*\mathsf{M}_B$ whose entries are blocks of side c_ℓ. Each of the r_*A^2 left blocks of the created products is a combination of at most m_*B^2 blocks, and so is each right block; each of the m_*B^2 output blocks is a combination of at most r_*A^2 blocks of the products. So a batch costs fewer than $6r_*m_*A^2B^2c_\ell^2$ operations. Since $\nu_{\ell+1} = r_*\lceil \nu_\ell/m_* \rceil$ and $\bar{n}_{\ell+1} = Ac_\ell$, depth ℓ costs at most $\lceil \nu_\ell/m_* \rceil \cdot 6r_*m_*A^2B^2c_\ell^2 = 6m_*B^2\nu_{\ell+1}\bar{n}_{\ell+1}^2$. The last depth costs at most $2\nu_{\ell_{\max}}n_*^3$.

By (5.1), $\nu_\ell \le C_\nu(r_*/m_*)^\ell$ and $\bar{n}_\ell^2 \le 2n_\infty^2 + 2\rho^{2\ell}n^2$. Since $r_*/m_* = (B/A)^p \ge 4$ and $\rho^{\ell_{\max}}n > \rho$,

$$\sum_{\ell=1}^{\ell_{\max}}\Big(\frac{r_*}{m_*}\Big)^\ell \le \frac{4}{3}\Big(\frac{r_*}{m_*}\Big)^{\ell_{\max}} < \frac{4}{3}\,\frac{r_*}{m_*}\,n^p.$$

The terms $\rho^{(2-p)\ell}n^2 = (r_*/m_*)^\ell\rho^{2\ell}n^2$ do not decrease with ℓ, and the last one is $(\rho^{\ell_{\max}}n)^{2-p}n^p \le \rho^{2-p}n^p$, so their sum over $1 \le \ell \le \ell_{\max}$ is at most $\ell_{\max}(B/A)^{p-2}n^p$. Adding the leaves, which cost at most $2C_\nu(r_*/m_*)^{\ell_{\max}}n_*^3 < 2C_\nu(r_*/m_*)n^pn_*^3$, gives the first bound.

For the second, $r_* \ge 4m_*$ gives $C_\nu \le 1 + \frac{4}{3}m_* \le \frac{7}{3}B^3$. Also $r_*/m_* \le (B/A)^3$, $n_* \le 2A+1 \le 3A$, $n_\infty \le 2A$ and $(B/A)^{p-2} \le B/A$. Since $\rho \le 1/2$ and $\rho^{\ell_{\max}-1}n > 1$, we have $\ell_{\max} \le \log_2 n + 1$. The three terms are then at most $126B^6$, $\frac{448}{3}B^{11}/A$ and $28\ell_{\max}B^9/A$, and their sum is at most $(126 + 150 + 28(\log_2 n + 1))B^{11}$. □

The lemma counts field operations, as [2, Corollary 9.7] does. Padding, cutting and reassembling at depth ℓ move at most $3\lceil \nu_\ell/m_* \rceil m_*(Bc_\ell)^2$ entries, at most half the operations counted for that depth.

For an integer $v \ge 1$, let D_v, L_v, H_v, N_v and J_v be D, L, H, N and J for $\varepsilon = 1/v$, and put

$$\mathsf{a}(v) = 2N_v^2J_v \cdot (\text{number of binary digits of } 24D_v).$$

By Lemma 5.1(a), $\log_2 A \le \mathsf{a}(v)$ for the scheme $\mathcal{S}_{1/v}$. Each of D_v, L_v, H_v, N_v and J_v is nondecreasing in v, so a is nondecreasing, and $\mathsf{a}(1) = 2 \cdot 4608^2 \cdot 158 \cdot 6 = 40{,}259{,}026{,}944$.

Definition 5.7 (Algorithm $\mathcal{U}$)**.** The input is an integer $n \ge 1$ and two $n \times n$ matrices over a field of characteristic zero.

1. Compute $\kappa = \lfloor \log_2 n \rfloor$. If $3000\,\mathsf{a}(1) > \kappa$, multiply the matrices classically and stop. Otherwise compute $v(n) = \max\{v \geq 1 : 3000\, v\, \mathsf{a}(v) \leq \kappa\}$ by binary search over $1 \leq v \leq \kappa$.

2. Run the generator of Appendix B.4 for $\varepsilon = 1/v(n)$. It outputs $\mathcal{D}$ and the integer numerators of the three maps of $\mathcal{S}_{1/v(n)}$.

3. Convert $\mathcal{D}$ and every numerator to the field by Horner's rule in base 2, invert $\mathcal{D}$, and multiply every numerator by $\mathcal{D}^{-1}$.

4. Multiply the matrices by the pooled recursion of $\mathcal{S}_{1/v(n)}$ with classical leaves, as in Lemma 5.6.

Theorem 5.8 (One algorithm for all sizes)**.** *Count steps as in [2, Corollary 9.7]: a step is one step of a fixed multitape Turing machine during the preprocessing (Steps 1 and 2 of Algorithm $\mathcal{U}$, and the reading of the generator's output in Step 3), or one addition, subtraction, multiplication or division in the field on matrix entries or scheme coefficients. There is a constant c_0, depending only on the fixed machine, with the following properties. Algorithm $\mathcal{U}$ multiplies two $n \times n$ matrices over a field of characteristic zero. If $\lfloor \log_2 n \rfloor \geq 3000\,\mathsf{a}(1) \approx 1.21 \cdot 10^{14}$, it uses at most*

$$(c_0 + 28 \log_2 n)\, n^{9/4+0.51/v(n)}$$

steps; otherwise it uses at most $2n^3 + c_0 \log_2(n+1)$ steps. If $\kappa = \lfloor \log_2 n \rfloor \geq 2 \cdot 10^{18}$, then

$$\frac{0.51}{v(n)} \leq 244\, \frac{\big(\log(4096\kappa)\big)^{6/11}}{\kappa^{2/11}}. \tag{5.3}$$

Hence, for every $n \geq 2$, Algorithm $\mathcal{U}$ uses $n^{9/4+\delta(n)}$ steps with $\delta(n) = O\big((\log\log n)^{6/11}(\log n)^{-2/11}\big)$.

Proof. Correctness follows from [2, Lemma 2.4], since the maps computed in Steps 2 and 3 are those of the relative scheme $\mathcal{S}_{1/v(n)}$, whose denominator $\mathcal{D}$ is invertible in characteristic zero. Step 1 takes $O(\log n)$ machine steps: a is nondecreasing, and each value $\mathsf{a}(v)$ with $v \leq \kappa$ is computed from integers with $O(\log \kappa)$ bits. If $3000\,\mathsf{a}(1) > \kappa$, classical multiplication adds fewer than $2n^3$ field operations.

Otherwise let $v = v(n)$, and let A, B, r_*, m_*, p be the counts and the pooled exponent of $\mathcal{S}_{1/v}$. Then $\log_2 A \leq \mathsf{a}(v) \leq \kappa/(3000v)$, so $A \leq n^{1/(3000v)}$.

Step 2. By Proposition 5.4(c), the generator performs at most A^{46} bit operations, and it stores at most that many bits. A fixed multitape Turing machine executes it, and reads its output in Step 3, in at most $c_{\mathrm{TM}} A^{92}$ steps, for a constant c_{TM}: it keeps the stored integers on one tape, brings the operands of each integer operation to its work tapes by one scan of that tape, and performs schoolbook arithmetic there in a constant times the counted bit operations. Since $92/3000 < 0.031$, this is at most $c_{\mathrm{TM}}\, n^{0.031/v}$ steps.

Step 3. The three maps have at most $3r_*A^2m_*B^2 \leq 3A^{83/4}$ coefficients, by Lemma 5.1(b),(c). Each numerator, and $\mathcal{D}$, has at most $\beta_\varepsilon \leq A^{1/k}$ bits (Proposition 5.4), so Horner's rule takes fewer than $2A^{1/k}$ field operations for each. With the inversion and the final multiplications, Step 3 uses at most $A^{22} \leq n$ field operations.

Step 4. Lemma 5.1 gives the hypotheses of Lemma 5.6, and $B^{11} \leq A^{22}$. By Theorem 4.8, $p < 9/4 + 1/(2v)$. Hence Step 4 uses at most $(304 + 28\log_2 n)\, A^{22} n^p \leq (304 + 28\log_2 n)\, n^{9/4+(1500+22)/(3000v)}$ field operations, and $1522/3000 < 0.51$.

Since $0.031/v \leq 9/4$, the four steps together use at most $(c_0 + 28\log_2 n)\, n^{9/4+0.51/v}$ steps, where c_0 is $c_{\mathrm{TM}} + 305$ plus the constant of Step 1.

For the rate, apply Lemma 5.1(a) with $\varepsilon = 1/v$ and $\Lambda(v) = \log(4096v)$: $N_v \le 2305\, v^{3/2}\Lambda(v)$ and $J_v \le 240\, v^{3/2}$. Since $D_v \le 2\sqrt{v}+1 \le 3\sqrt{v}$, the number of binary digits of $24D_v$ is at most $\log_2(144\sqrt{v}) \le \Lambda(v)$. Hence $\mathsf{a}(v) \le 2 \cdot 2305^2 \cdot 240\, v^{9/2}\Lambda(v)^3 < 4 \cdot 10^9\, v^{9/2}\Lambda(v)^3$. Put

$$y = \Big(\frac{\kappa}{1.2 \cdot 10^{13}\,\Lambda(\kappa)^3}\Big)^{2/11}.$$

Then $y \le \kappa^{2/11} \le \kappa$. Every integer v with $1 \le v \le y$ satisfies $\Lambda(v) \le \Lambda(\kappa)$, hence $3000\, v\, \mathsf{a}(v) < 1.2 \cdot 10^{13}\, v^{11/2}\Lambda(v)^3 \le \kappa$, so $v(n) \ge \lfloor y \rfloor$. The function $\kappa/\Lambda(\kappa)^3$ increases for $\kappa \ge 1$, and $1.2 \cdot 10^{13}\,\Lambda(2 \cdot 10^{18})^3 < 1.6 \cdot 10^{18}$. So $\kappa \ge 2 \cdot 10^{18}$ gives $y \ge 1$, hence $v(n) \ge y/2$, and

$$\frac{0.51}{v(n)} \le \frac{1.02}{y} = 1.02\,(1.2 \cdot 10^{13})^{2/11}\,\frac{\Lambda(\kappa)^{6/11}}{\kappa^{2/11}} < 244\,\frac{\Lambda(\kappa)^{6/11}}{\kappa^{2/11}},$$

because $(1.2 \cdot 10^{13})^{2/11} < 239$. Finally, the factor $c_0 + 28\log_2 n$ adds $O(\log\log n/\log n)$ to the exponent, which is of smaller order. □

Remark 5.9 (Comparison with the preceding paper). For a suitable implementation of the algorithm of [2, Corollary 9.7], Remark 9.8 there sketches a bound of $O(n^{9/4+547200/\log_2\log_2 n})$ steps (its (9.4)), an exponent below 3 only when $\log_2\log_2 n > 729{,}600$, and sketches why that argument gives no better order: the seed for $\varepsilon = 1/v$ has a side doubly exponential in v. Theorem 5.8 improves the rate to $(\log n)^{-2/11}$, up to a factor $(\log\log n)^{6/11}$. It uses that $\log A$ is polynomial in $1/\varepsilon$ and that the preprocessing costs $A^{O(1)}$ (Proposition 5.4). With a preprocessing cost $2^{O((\log A)^c)}$ for a constant $c > 1$, the same argument would still give a power of $1/\log n$, with a smaller exponent. A cost bound exponential in A would destroy the rate; the bit-length bound of [2, Theorem 9.3] is already above 2^A for the A of that paper. The exponent $2/11$ balances the excess $1/(2v)$ of the pooled exponent against $\log(A^{22})/\log n$ with $\log A = O(v^{9/2}\log^3 v)$; we do not know whether it can be improved.

Remark 5.10 (The seed route). Following the proof of [2, Corollary 9.7] instead, with the seed of Theorem 5.2 and block recursion, costs at most $7n_0^{17/2}n^{9/4+\varepsilon}$ operations [2, Lemma 9.4], with $\log n_0 = O(\varepsilon^{-11/2}\Lambda^3)$. Balancing gives only $\delta(n) = O((\log\log n)^{6/13}(\log n)^{-2/13})$, because of the factor $1/\varepsilon$ between $\log n_0$ and $\log A$, which Remark 5.3 shows to be necessary for this seed.

6 Finite examples

For a fixed target exponent, chain schemes much smaller than those of Theorem 4.8 suffice. This section lists certified examples and compares them with the examples of [2, Section 10]. No other result depends on this section.

6.1 Certified witnesses

A *witness* is given by a target Φ_0, a terminal pair $(1, t)$, arities $d_1, \dots, d_J \ge 2$ and integers $N, k \ge 1$. Its chain is built backwards from $(1, t)$ by smallest parents (Lemma 4.1), which fixes the source pair and makes every stage legal. Stage i uses a type of width N with the largest sector count $M(N, d_i)$ (Section 4.4). Proposition 3.8 gives a restriction $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ over $\mathbb{Q}$ with the counts (3.9), and we certify $\Phi < \Phi_0$ for its fixed-batch bound Φ (4.2), so that also $p < \Phi_0$. For example, the first row of Table 2 is the single stage $(128, 66) \to (1, 128)$ of arity 2, legal since $4 \le 2 \cdot 66 - 128 + 1$.

Φ_0	J	arities	t	N	k	$\Phi <$	$\Phi - p$	digits of A	digits of B
3	1	2	128	24	2	2.999745128438	$1.34 \cdot 10^{-3}$	29	255
2.77429	1	2	5	215	10	2.774233389609	$7.08 \cdot 10^{-5}$	1 295	5 547
2.7	5	2	3	147	8	2.699881286935	$6.71 \cdot 10^{-5}$	3 541	8 025
2.5	15	2	3	264	14	2.499903515815	$8.39 \cdot 10^{-6}$	33 379	69 247
2.3	125	2, ramp 3	1	1 999	80	2.299993369880	$2.54 \cdot 10^{-8}$	12 371 380	24 208 459
2.26	1 264	4, ramp 11	1	28 336	1 118	2.259999981185	$2.16 \cdot 10^{-11}$	48 859 824 830	62 827 651 353
2.84063	1	2	5	93	5	2.840553213477	$3.31 \cdot 10^{-4}$	280	1 190
2.67283	7	2	3	138	8	2.672186713007	$5.51 \cdot 10^{-5}$	4 653	10 114

Table 2: Certified chain schemes (Proposition 6.1), built backwards from $(1, t)$ by smallest parents. Column J: the number of stages. Column "arities": the arity d of every stage, or "d, ramp R" for $J - 2(R-1)$ stages of arity d followed by the $2(R-1)$ stages of Lemma 4.2 from (R, R) to $(1, 1)$, of arities $3j$ and $3j-1$ for $j = R-1, \dots, 1$, in this order from the source. Column $\Phi <$: certified upper bound for Φ. Column $\Phi - p$: the value of $\log 2 / \log(B/A)$, rounded to three significant digits. The last two rows are the comparison points of Section 6.2.

Proposition 6.1 (Witnesses)**.** *For each row of Table 2, the restriction defined above is a relative scheme over $\mathbb{Q}$, its fixed-batch bound Φ is less than the entry in the column $\Phi <$, hence less than Φ_0, and $\Phi - p$ and the numbers of decimal digits of A and B are as stated.*

Proof. The proof is computer-assisted, by the supplementary script `shifted_chain_certificates.py`, which reproduces its report `shifted-chain-certificates.json` byte for byte. The script checks every stage against (3.1), and the maximality of each type $\tau = (\tau_1, \tau_2, \tau_3) = (n_{\mathrm{K}}, n_{\mathrm{O}}, n_{\mathrm{mid}})$ by the inequalities $w_i \tau_j \le w_j(\tau_i + 1)$ for $i \ne j$ with $\tau_j > 0$, where $w = (d-1, 2, 1)$, which say that moving one letter from class j to class i does not increase the sector count; these characterize the maximizers, because the logarithm of (3.5) is $\log N!$ plus a sum of concave functions $\tau_i \mapsto \tau_i \log w_i - \log \tau_i!$. It forms the integers d_i, M_i, $2M_i - 1$, E_i, s and t exactly, encloses their logarithms in intervals with dyadic endpoints (an artanh series with directed rounding and a tail bound), and evaluates Φ, p, $\Phi - p$, $\log_{10} A$ and $\log_{10} B$ in exact interval arithmetic. If the enclosure of $\log_{10} A$ or $\log_{10} B$ contains an integer, the digit count is decided exactly. Printed bounds are rounded outwards. □

The source pairs are $(128, 66)$, $(5, 4)$, $(18, 27)$, $(3098, 5205)$, a pair of 28-digit integers, a pair of 166-digit integers, $(5, 4)$ and $(49, 78)$; the report lists them with the types and intervals for p. The rows are selected, not optimized; we do not claim that any of them is the smallest chain scheme below its target. The 2.3 and 2.26 rows have the ramp-and-bulk shape of Theorem 4.8, but much smaller parameters than that theorem uses. The column $\Phi - p$ is below $1.4 \cdot 10^{-3}$ in every row and below 10^{-5} for the targets 2.5, 2.3 and 2.26.

6.2 Comparison with the preceding paper

Table 2 of [2] bounds the same quantity Φ [2, (10.1)] for three schemes of the construction there, with target indices $a = 4$, 8 and 12 in the notation of [2]. For the last two it gives the certified upper bounds 2.848599165 and 2.675820359, the floating-point values 2.8406393 and 2.6728401 of Φ, and source sides with 11 877 489 885 and 234 905 651 862 digits. The last two rows of Table 2 have targets about 10^{-5} below these floating-point values. They are the one-stage chain $(5, 4) \to (1, 5)$ of arity 2, with $A = 2^{930}$, and a chain of seven stages of arity 2 from $(49, 78)$ to $(1, 3)$, with $A = 2^{15456}$. Their source sides have 280 and 4 653 digits, fewer by factors of about $4.2 \cdot 10^7$ and $5.0 \cdot 10^7$, and

even their target sides, with 1 190 and 10 114 digits, have fewer digits than the source sides there. Pooling lowers the Φ of those schemes by less than 10^{-7} [2, Section 10.2], so the targets are also below their pooled exponents, provided the floating-point values are accurate to 10^{-6}.

Near 9/4 the gap is much larger. By [2, Corollary 11.2(a)], every scheme of the construction there with $p < 3$ has $p > \frac{9}{4} + 9\log(6/5)/(2\log(2\log A))$. For $p < 2.26$ this forces $2\log A > (6/5)^{450}$, so A has more than $(6/5)^{450}/(2\log 10) > 9.29 \cdot 10^{34}$ digits, against 48 859 824 830 for the 2.26 row. For $p < 2.3$ it forces only $2\log A > (6/5)^{90}$, that is, more than $2.9 \cdot 10^{6}$ digits, so it does not separate the two constructions there.

These comparisons concern the schemes as constructed, with the source supplying every auxiliary. Forward recycling and composition with other relative schemes, for example with the classical algorithm $n^3\mathsf{M}_1 \succeq \mathsf{M}_n$ of source side 1, change A and B (Section 7.1).

Far from 9/4, chain schemes are not the smallest schemes known. Strassen's algorithm [15] is the relative scheme $7\mathsf{M}_1 \succeq \mathsf{M}_2$, with $A = 1$, $B = 2$ and $p = \log_2 7 < 2.808$, below the targets 3 and 2.84063. Schwartz and Zwecher [14] report an algorithm with exponent 2.773203 whose base case has side below 1000, below the target 2.77429. The advantage of chain schemes lies near 9/4, where, to our knowledge, the only other explicit sizes are those of [2], which grow doubly exponentially in $1/\varepsilon$ [2, Corollary 11.2(b)].

7 Limits within the construction

This section asks what the construction of Sections 3 and 4 cannot do. It fixes the class of schemes (Section 7.1), shows that $p > 9/4$ even with compressed auxiliaries (Section 7.2), bounds the copy counts and $\log(B/A)$ from below as $p \to 9/4$ (Section 7.3) and the number of target copies for $p < 3$ (Section 7.4), and lists what the class leaves out (Section 7.5). None of these statements is a lower bound for ω or for tensor rank.

7.1 Chain data and supplies

Definition 7.1 (Chain data and supplies)**.** *Chain data* consist of the following.

(a) Integers $J, W \geq 1$, and pairs (r_i, b_i) of positive integers for $1 \leq i \leq J+1$.

(b) For each $i \leq J$, an arity $d_i \geq 2$ and a child length $c_i \geq 1$ such that, for one ordering (r, b) of the entries of (r_i, b_i), the stage $(r, b) \to (c_i, r)$ of arity d_i is legal (3.1) and, after exchanging r_{i+1} and b_{i+1} if necessary, $r_{i+1} \leq c_i$ and $b_{i+1} \leq r$.

(c) For each $i \leq J$, finitely many *blocks* j, each with a power $k_{ij} \geq 1$ and a type τ_{ij} of width $N_{ij} \geq 1$ (Lemma 3.4), such that $\sum_j N_{ij}k_{ij} = W$.

(d) A *terminal* t with $1 \leq t \leq \max(r_{J+1}, b_{J+1})$.

Put $s = r_1 + b_1 - 1$. Let M_{ij} be the sector count (3.5) of τ_{ij} at arity d_i, $E_{ij} = 1 + 6k_{ij}(M_{ij} - 1)$, $U_i = d_i^{2W}$ and $V_i = \prod_j M_{ij}^{2k_{ij}}$, and put

$$\bar{A} = \prod_{i=1}^{J} U_i, \quad \bar{B} = t^{2W} \prod_{i=1}^{J} V_i, \quad r_* = s^{6W} \prod_{i,j} E_{ij}(2M_{ij} - 1)^{6k_{ij}}, \quad m_* = \prod_{i,j} M_{ij}^{6k_{ij}}. \tag{7.1}$$

A *supply* for chain data consists of positive integers $A = \mathfrak{b}_0$ and $f_1, \ldots, f_J$ such that $U_i f_i \leq \mathfrak{b}_{i-1}$ for every i, where $\mathfrak{b}_i = f_i V_i$. Its *source side* is A, and its *target side* is $B = t^{2W}\mathfrak{b}_J$.

- The *source supply* has $A = \bar{A}$ and $f_i = \prod_{l>i} U_l \prod_{l<i} V_l$.
- The *recycling supply* with source side A has $f_i = \lfloor \mathfrak{b}_{i-1}/U_i \rfloor$. It is a supply if every $f_i \geq 1$.

By Lemma 7.2, chain data and a supply give a restriction $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$ over $\mathbb{Q}$. If $B > A$, it is a relative scheme, which we call a *chain scheme*.

Block j of stage i applies the exact relation of Corollary 3.6 with $(N, k, \tau) = (N_{ij}, k_{ij}, \tau_{ij})$ to its own $N_{ij}k_{ij}$ of the W factors of $Q^{\otimes W}_{r_i,b_i}$. So stage i consumes an auxiliary of side U_i and produces one of side V_i, and $\bar{A}$ is the total side consumed. Before stage i the scheme holds one matrix multiplication factor $\mathsf{M}_{\mathfrak{b}_{i-1}}$, the *bank*; stage i cuts $\mathsf{M}_{U_i}\mathsf{M}_{f_i}$ out of it, turns M_{U_i} into M_{V_i}, and keeps $\mathsf{M}_{V_i}\mathsf{M}_{f_i} \cong \mathsf{M}_{\mathfrak{b}_i}$. Under the source supply, $U_i f_i = \mathfrak{b}_{i-1}$ for every i, and $B = \bar{B}$. Under the recycling supply each stage takes what it can from the factors that earlier stages produced.

The class fixes the stage relation of Corollary 3.6 for every block (uncompressed auxiliaries, Lü's separator with $2M-1$ copies, separate exactification of each block), the evaluation of the source with s^{6W} products, and the crop of the last pair to $(1, t)$. The arities, children, crops, exchanges, blocks, types, widths, powers, the terminal and the supply are free. The restrictions of Proposition 3.8 and Remark 3.9 are given by chain data with one block per stage and the source supply, without crops or exchanges; so the chain schemes of Section 4 are chain schemes in the sense of Definition 7.1.

Lemma 7.2 (Supplies and forward recycling)**.** *Let chain data and a supply be given, with source side A and target side B. Then $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$ over $\mathbb{Q}$. Moreover:*

(a) $B/A \leq \bar{B}/\bar{A}$, with equality for the source supply;

(b) if $B > A$, then this restriction and the one given by the source supply are chain schemes, and the pooled exponent of the first is at least that of the second;

(c) if $A \geq 4J \max_{1\leq i\leq J} U_i \prod_{l<i}(U_l/V_l)$, then the recycling supply with source side A is a supply, $\mathfrak{b}_{i-1} \geq 2JU_i$ for every i, and $B/A \geq \bar{B}/(2\bar{A})$.

The proof (Appendix A.1) is an induction over the stages. The bank only carries factors forward, so this is not a catalyst, and every auxiliary keeps its full side. It works because every stage is exactified before its output is used (Section 2.2). The same induction can also fund the blocks of one stage one after another, since they act on disjoint factors of Q. This keeps r_*, m_* and $B/A \leq \bar{B}/\bar{A}$, so the results of Section 7.3 on r_*, m_* and $\log(B/A)$ hold for it with the same proofs, as they use only Lemma 7.2(a) and (b).

Remark 7.3 (Composition). The step of this induction composes any two relative schemes. Let $r^{(l)}\mathsf{M}_{A^{(l)}} \succeq m^{(l)}\mathsf{M}_{B^{(l)}}$, $l = 1, 2$, be relative schemes with $B^{(1)} \geq A^{(2)}$, and put $f = \lfloor B^{(1)}/A^{(2)} \rfloor$. Take $r^{(2)}$ copies of the first, restrict each $\mathsf{M}_{B^{(1)}}$ to its principal block $\mathsf{M}_{A^{(2)}f} \cong \mathsf{M}_{A^{(2)}}\mathsf{M}_f$, and apply the second, tensored with the identity on M_f. This gives $r^{(1)}r^{(2)}\mathsf{M}_{A^{(1)}} \succeq m^{(1)}m^{(2)}\mathsf{M}_{B^{(2)}f}$. Since $B^{(2)}f/A^{(1)} \leq (B^{(1)}/A^{(1)})(B^{(2)}/A^{(2)})$ and $r^{(l)}/m^{(l)} > 1$ (Section 2.3), the composite, if $B^{(2)}f > A^{(1)}$, has pooled exponent at least the smaller of the two.

The classical algorithm is the relative scheme $n^3\mathsf{M}_1 \succeq \mathsf{M}_n$, with source side 1 and pooled exponent 3. Let $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ be a relative scheme, and let $n \geq 2AB/(B-A)$. Put the classical algorithm first, and then apply the scheme j times, each time to the factor produced by the previous step. The composite has source side 1, copy counts $n^3 r_*^j$ and m_*^j, and target side $\mathfrak{b}_j$, where $\mathfrak{b}_0 = n$ and $\mathfrak{b}_{l+1} = \lfloor \mathfrak{b}_l/A \rfloor B$. Every step can be applied, and $\log \mathfrak{b}_j \geq \log n + j \log(B/A) - 2$ (Appendix A.1). So the pooled exponent of the composite tends to that of the scheme as $j \to \infty$, and with Theorem 4.8, for every $\delta > 0$ some composite with source side 1 has pooled exponent below $9/4 + \delta$.

So the source side alone does not measure the size of an algorithm. A lower bound on A, such as Theorem 7.12(c) below, concerns chain schemes with the source supply. The copy counts r_* and m_*, and hence r_*/m_*, multiply under composition and tensor products.

7.2 The exponent limit

Scalar arguments. Lemma 4.7 is a scalar argument, the analogue of [2, Remark 3.3]: it bounds the gain $\log \frac{d+2}{d}$ of each stage by the drop of $\log F$ along the stage. A compressed auxiliary changes the gain of a stage, so the rest of this subsection works with tensors.

Sector tensors. Let $r, c, N \geq 1$, and write $[r] = \{0, \dots, r-1\}$. The *middle child* is

$$C^{\mathrm{mid}}(r,c) = \sum_{\alpha=0}^{r-1} \sum_{\gamma=0}^{c-1} x_\alpha \, y_{r-1-\alpha+\gamma} \, z_\gamma,$$

with legs of sizes r, $r+c-1$ and c. Reversing the first index turns it into $C(r,c)^{(23)}$; we keep the first index unreversed, because the bound below needs the first leg fixed. Let $X = (\mathbb{F}^r)^{\otimes N}$, with coordinates x_α indexed by words $\alpha \in [r]^N$. An $(r, c; N)$ *sector tensor* is a sum $T = \sum_{g=1}^{M} T_g$ of M sectors sharing the first leg X (Section 2.4) such that, for each g, invertible linear maps on Y_g and Z_g, together with the identity on X, carry T_g to a product $L_{g,1} \otimes \cdots \otimes L_{g,N}$; here each $L_{g,\ell}$ is $C(r,c)$ or $C^{\mathrm{mid}}(r,c)$, and letter ℓ of α is its first index. Let $n_{\mathrm{mid}}(g)$ be the number of factors $C^{\mathrm{mid}}(r,c)$. For tensors S and T with the same first leg, we write $S \succeq_1 T$ if linear substitutions on legs 2 and 3, together with the identity on leg 1, transform S into T.

Let $(r,b) \to (c,r)$ be a legal stage of arity d, and let T be the tensor of Lemma 3.4 for a type $\tau = (n_{\mathrm{K}}, n_{\mathrm{O}}, n_{\mathrm{mid}})$. Then T is an $(r, c; N)$ sector tensor with M sectors given by (3.5), and $n_{\mathrm{mid}}(g) = n_{\mathrm{mid}}$ for all g: its sectors are $T_w = \bigotimes_\ell U'_{w_\ell}$, and by Lemmas 3.1(c) and 3.3(b), renaming the coordinates of legs 2 and 3 alone identifies U'_χ with $C(r,c)$ for kernel and outer letters and with $C^{\mathrm{mid}}(r,c)$ for the middle letter. The substitutions of Lemma 3.1(c) and the zeroing of Lemma 3.4 act on legs 2 and 3 only, so $C(r,b)^{\otimes N} B_X(d^N) \succeq_1 T$.

The Hankel lift. Let $u, v \geq 1$ with $u + v = r + 1$. For a tensor T with first leg X, write $T[\alpha, \eta, \zeta]$ for its coefficient at $x_\alpha y_\eta z_\zeta$. Its *Hankel lift* $\mathcal{L}_u(T)$ is the matrix with rows indexed by pairs (ζ, a), where ζ is a coordinate of leg 3 and $a \in [u]^N$, columns indexed by pairs (η, e), where η is a coordinate of leg 2 and $e \in [v]^N$, and entries

$$\mathcal{L}_u(T)_{(\zeta,a),(\eta,e)} = T[a+e, \eta, \zeta],$$

where $a+e \in [r]^N$ is the word with letters $a_\ell + e_\ell$. Equivalently, $\mathcal{L}_u(T)$ is an ordinary flattening of the tensor obtained from T by pulling its first leg back along the multiplication map $(\mathbb{F}^u \otimes \mathbb{F}^v)^{\otimes N} \to X$ of $C(u,v)^{\otimes N}$; this is a catalecticant (Hankel) flattening. Three facts follow from the definition.

(i) If $S \succeq_1 T$, with matrices B and Γ of the substitutions on legs 2 and 3, then $\mathcal{L}_u(T) = (\Gamma^\top \otimes I)\, \mathcal{L}_u(S)\, (\mathsf{B} \otimes I)$, so $\operatorname{rank} \mathcal{L}_u(T) \leq \operatorname{rank} \mathcal{L}_u(S)$.

(ii) For tensors with first legs $(\mathbb{F}^r)^{\otimes N}$ and $(\mathbb{F}^r)^{\otimes N'}$, $\mathcal{L}_u(TT')$ is $\mathcal{L}_u(T) \otimes \mathcal{L}_u(T')$ up to permutations of rows and columns.

(iii) $\mathcal{L}_u(C(r,b)^{\otimes N} B_X(q))$ is $\mathcal{L}_u(C(r,b)^{\otimes N}) \otimes I_q$ up to permutations, since the coefficient of $B_X(q)$ at $y_\eta z_\zeta$ is $[\eta = \zeta]$.

Over every field, the rank of a Kronecker product is the product of the ranks.

Lemma 7.4 (Local ranks)**.** *For $N = 1$ and over every field,* $\operatorname{rank} \mathcal{L}_u(C(r,b)) = b + v - 1$ *and* $\operatorname{rank} \mathcal{L}_u(C^{\mathrm{mid}}(r,c)) = c + u - 1$.

The proof (Appendix A.2) factors each lift through the sum of two indices and exhibits an identity submatrix of full size.

Theorem 7.5 (Hankel flattening bound)**.** *Let T be an $(r, c; N)$ sector tensor with M sectors over a field $\mathbb{F}$, and let $b, q \geq 1$. If $C(r,b)^{\otimes N} B_X(q) \succeq_1 T$, then for all integers $u, v \geq 1$ with $u + v = r + 1$,*

$$q\,(b+v-1)^N \;\geq\; \sum_{g=1}^{M} (c+v-1)^{N-n_{\mathrm{mid}}(g)} (c+u-1)^{n_{\mathrm{mid}}(g)}. \tag{7.2}$$

Proof. By (ii), (iii) and Lemma 7.4, the left side is the rank of $\mathcal{L}_u(C(r,b)^{\otimes N} B_X(q))$. Since $T_g \in X \otimes Y_g \otimes Z_g$, the matrix $\mathcal{L}_u(T)$ is block diagonal, with the block $\mathcal{L}_u(T_g)$ on the rows $Z_g \times [u]^N$ and the columns $Y_g \times [v]^N$, so its rank is $\sum_g \operatorname{rank} \mathcal{L}_u(T_g)$. By (i), applied to the invertible maps and to their inverses, (ii) and Lemma 7.4, the rank of $\mathcal{L}_u(T_g)$ is the g-th term on the right. Now apply (i) to $C(r,b)^{\otimes N} B_X(q) \succeq_1 T$. □

Write $\varphi_r(y) = (r + 2y - 1)^2 - 1$ for real $y \geq 1$, and define the potential $\Psi(r,b) = r^2 + 4rb + b^2 - 2r - 2b - \frac{1}{6}$ for real $r, b \geq 1$.

Lemma 7.6 (Gain per factor)**.** *In Theorem 7.5, if $b \geq c$, then* $\frac{1}{N} \log(M/q) \leq \frac{1}{2} \log(\varphi_r(b)/\varphi_r(c))$.

Lemma 7.7 (Potential)**.** *For real $r, b \geq 1$:*

(a) Ψ is symmetric and increasing in each argument, and $\Psi(r,b) \geq \Psi(1,1) = \frac{11}{6}$;

(b) if $b > c \geq 1$, then $\frac{1}{2} \log(\varphi_r(b)/\varphi_r(c)) < \frac{2}{3} \log(\Psi(r,b)/\Psi(r,c))$;

(c) $\Psi(r,b) \leq \frac{11}{6}(r + b - 1)^2$ if $r + b \geq 3$, and $\Psi(r,b) \geq \frac{11}{6} b^{3/2}$.

The proofs are in Appendix A.2. For Lemma 7.6, one takes $u = v$ if r is odd, and combines the two nearly balanced choices of (u, v) by the Cauchy–Schwarz inequality if r is even. Part (b) of Lemma 7.7 says that $\frac{2}{3} \log \Psi(r,y) - \frac{1}{2} \log \varphi_r(y)$ increases in y.

Definition 7.8 (Compressed auxiliaries)**.** Let $(r,b) \to (c,r)$ be a legal stage of arity d, let $N \geq 1$, let τ be a type of width N, and let T be the tensor of Lemma 3.4. A *compressed auxiliary of rank q* for these data is a restriction $C(r,b)^{\otimes N} B_X(q) \succeq_1 T$, where $q \geq 1$. *Compressed chain data* are chain data together with, for every block (i,j), a compressed auxiliary of some rank q_{ij}, used in all six orientations, for the stage of Definition 7.1(b) and (N_{ij}, τ_{ij}); in their counts (7.1), $U_i = \prod_j q_{ij}^{2k_{ij}}$.

The stage itself has $q = d^N$. Using a compressed auxiliary in every orientation of the proof of Proposition 3.5 and then Corollary 3.6 gives, since $\bigotimes_\pi B_X(q)^\pi \cong \mathsf{M}_{q^2}$,

$$E\,(2M-1)^{6k}\, Q_{r,b}^{\otimes Nk}\, \mathsf{M}_{q^{2k}} \succeq M^{6k}\, Q_{c,r}^{\otimes Nk}\, \mathsf{M}_{M^{2k}}$$

over every field of characteristic zero that contains the coefficients of the compressed auxiliary. So Lemma 7.2 and its proof hold for compressed chain data, with the new U_i.

Corollary 7.9 (Compression limit)**.** *Let compressed chain data and a supply give a relative scheme $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$. Then its pooled exponent satisfies $p > 9/4$.*

Proof. By Lemma 7.2(b), we may assume the source supply, so $A = \bar{A}$ and $B = \bar{B}$. Put $w_{ij} = N_{ij}k_{ij}/W$, so that $\sum_j w_{ij} = 1$. Then

$$\frac{\log(B/A)}{2W} = \log t + \sum_{i,j} w_{ij} h_{ij}, \qquad h_{ij} = \frac{1}{N_{ij}} \log \frac{M_{ij}}{q_{ij}}.$$

Fix i, and let $(r, b) \to (c_i, r)$ be the stage of Definition 7.1(b); it has $c_i < b$. By Theorem 7.5 and Lemmas 7.6 and 7.7(b), applied to each block, and by the symmetry of Ψ, $\sum_j w_{ij} h_{ij} < \frac{2}{3} \log(\Psi(r_i, b_i)/\Psi(c_i, r))$. By Lemma 7.7(a), $\Psi(r_{i+1}, b_{i+1}) \le \Psi(c_i, r)$ and $\Psi(1, t) \le \Psi(r_{J+1}, b_{J+1})$. So the sum telescopes, and Lemma 7.7(c), with $r_1 + b_1 \ge 3$, gives

$$\frac{\log(B/A)}{2W} < \log t + \frac{2}{3} \log \frac{\Psi(r_1, b_1)}{\Psi(1, t)} \le \log t + \frac{2}{3} \log \frac{s^2}{t^{3/2}} = \frac{4}{3} \log s.$$

By the proof of Lemma 7.2(b), $\log(r_*/m_*) \ge 6W \log s > 0$. Hence $p = \log(r_*/m_*)/\log(B/A) > 6/(8/3) = 9/4$. □

For $q_{ij} = d_i^{N_{ij}}$, the corollary gives the conclusion $p > 9/4$ of Lemma 4.7(c) for every chain scheme, with any supply, but not the intermediate bound $p > 3 \log s/(\log t + G)$. One rank in all six orientations is no loss of generality for the barrier: if orientation π has a compressed auxiliary of rank q_π, then $B_X(\max_\pi q_\pi) \succeq B_X(q_\pi)$ (set the extra coordinates to zero) gives one of rank $\max_\pi q_\pi$ in every orientation, as the square auxiliary M_{q^2} requires, and (7.2) holds for each q_π separately.

Remark 7.10 (Ordinary sectors only). Middle sectors are what make exponents below 3 possible. For $u = 1$, $\mathcal{L}_1(T)$ is the flattening of T with rows indexed by leg 3, whose rank does not increase under any restriction, and if no sector has a middle factor, then (7.2) reads $M(r + c - 1)^N \le q(r + b - 1)^N$. So let compressed chain data have only types with $n_{\mathrm{mid}} = 0$, and put $s_i = r_i + b_i - 1$, so that $s_1 = s$. In the notation of the proof above, every block of stage i has $h_{ij} \le \log(s_i/(r + c_i - 1)) \le \log(s_i/s_{i+1})$. These bounds telescope, and $t \le s_{J+1}$ gives $\log(B/A) \le 2W \log s$. Hence $p \ge 3$ for every supply.

Remark 7.11 (The first leg, and sharpness). For $u = 1$ and $u = r$, the lift is a flattening of T along leg 3 or leg 2. The lifts with $2 \le u \le r - 1$ use the Hankel structure of the shared first leg, so they need $\succeq_1$. Crops of the first leg to a prefix are harmless: they turn $C(r, b)$ into $C(r', b)$ with unused coordinates, and Ψ increases in r. General first-leg maps are not covered. Theorem 7.5 also holds if T is the constant term of a degeneration with the identity on leg 1: the minors of size $q(b + v - 1)^N + 1$ of the lift of the transformed tensor vanish as polynomials in the parameter, hence at 0. So removing cross terms by powers of a parameter does not escape the bound. In the homogeneous limit, Lemma 7.6 is tight on the efficient ray $r = \lambda b$, $c = \lambda^2 b$ with $(d+2)\lambda^2 + \lambda - d = 0$, where $(r + 2b)/(r + 2c) = (d + 2)/d$, the gain per factor of an uncompressed stage. The potential step of Lemma 7.7(b) is tight only to first order as $\lambda \to 1$. For compressed auxiliaries this suggests, but we do not prove, that 9/4 can be approached only with growing arity; for chain schemes, which have uncompressed auxiliaries, Theorem 7.12 proves it.

7.3 Resource bounds

Theorem 7.12 (Resource lower bound within the construction)**.** *Let $0 < \varepsilon < 1/9216$, and let a chain scheme $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ have pooled exponent at most $9/4 + \varepsilon$. Then, with $\bar{A}$ as in (7.1):*

(a) $\log r_* > \log m_* \ge 3 \log \bar{A} > 3.9 \cdot 10^{-6}\, \varepsilon^{-9/2} \log^3(1/\varepsilon)$;

(b) $\log(B/A) > 9.8 \cdot 10^{-5}\, \varepsilon^{-4} \log^2(1/\varepsilon)$;

(c) under the source supply, $\log A = \log \bar{A} > 1.3 \cdot 10^{-6}\, \varepsilon^{-9/2} \log^3(1/\varepsilon)$.

Moreover, some block (i,j) *has* $d_i > (96\varepsilon)^{-1/2}$, $N_{ij} > 9d_i \log(1/\varepsilon)/(16\varepsilon)$ *and* $k_{ij} > d_i \log d_i/(4\varepsilon)$, *and* $W > 3\log^2(1/\varepsilon)/(8192\,\varepsilon^3)$.

The proof is in Appendix A.5. In (a), $\bar{A}$ is the auxiliary side consumed by the stages; the supplied source side A may be smaller (Corollary 7.13) or larger, and (a) bounds r_* and m_* in every case.

The bounds $3.9 \cdot 10^{-6}\, \varepsilon^{-9/2} \log^3(1/\varepsilon)$ on $\log r_*$ and $\log m_*$ in (a), and the bound (b), also hold for every relative scheme built from chain schemes and classical algorithms by composition (Remark 7.3) and tensor products. Indeed, under both operations m_* and r_*/m_* are the products of those of the parts, B/A is at most the product, and every part has $r_*/m_* > 1$. So if the result has $\log(r_*/m_*) \le (9/4+\varepsilon)\log(B/A)$, the mediant inequality gives a part with $\log(r_*/m_*) \le (9/4+\varepsilon)\log(B/A)$, where $9/4+\varepsilon < 3$. This part is a chain scheme, and its m_* and r_*/m_*, hence also its r_*, bound those of the result from below. For (b), the result has $\log(B/A) \ge \log(r_*/m_*)/(9/4+\varepsilon)$, and the proof of (b) bounds this quotient from below for that part.

Corollary 7.13 (Recycling the main construction)**.** *Let* $0 < \varepsilon \le 1$ *be rational, and take the chain, type width, power and types of Theorem 4.8. The recycling supply with source side* $A' = 4JD^{2Nk}$ *is a supply. It gives a relative scheme* $r_*\mathsf{M}_{A'} \succeq m_*\mathsf{M}_{B'}$ *over* $\mathbb{Q}$ *with the same* r_* *and* m_* *as Theorem 4.8, whose pooled exponent* p' *satisfies* $p \le p' < p + 10^{-9}\varepsilon$, *and so* $9/4 < p' < 9/4 + \varepsilon/2$. *Moreover,*

$$\log A' < 5.32 \cdot 10^6\, \varepsilon^{-3} \log^3 \frac{4096}{\varepsilon}, \qquad \log B' < 1.5 \cdot 10^9\, \varepsilon^{-4} \log^2 \frac{4096}{\varepsilon}.$$

The proof is in Appendix A.6. For example, on the chain of Theorem 4.8 at $\varepsilon = 1$, the source side $A = \prod_i d_i^{2Nk}$ has 3,250,941,263 decimal digits and the recycled source side $A' = 4JD^{2Nk}$ has 12,783,943 (supplementary script `shifted_chains_check_recycling.py`, Part 5).

Remark 7.14 (Orders). Within the construction, as $\varepsilon \to 0$:

- *Copy counts.* For every chain scheme, and for the composites above, $p \le 9/4+\varepsilon$ with $\varepsilon < 1/9216$ forces $\log m_*$ and $\log r_*$ above $3.9 \cdot 10^{-6}\varepsilon^{-9/2}\log^3(1/\varepsilon)$. For rational $\varepsilon \le \frac{1}{2}$, Corollary 4.9 gives $p < 9/4+\varepsilon$ with $m_* \le A^6$ and $r_* < A^{35/4}$, so $\log m_* < 8.4 \cdot 10^8 \varepsilon^{-9/2}\log^3(2048/\varepsilon)$ and $\log r_* < 1.23 \cdot 10^9 \varepsilon^{-9/2}\log^3(2048/\varepsilon)$. So the least values of $\log m_*$ and $\log r_*$ have order $\varepsilon^{-9/2}\log^3(1/\varepsilon)$.

- *Ratio.* The least $\log(B/A)$ has order $\varepsilon^{-4}\log^2(1/\varepsilon)$, by Theorem 7.12(b) and by the scheme of Corollary 4.9, for which the proof of Corollary 7.13, with 2ε in place of ε, gives $\log(B/A) < 1.43 \cdot 10^9 (2\varepsilon)^{-4}\log^2(2048/\varepsilon) < 9 \cdot 10^7 \varepsilon^{-4}\log^2(2048/\varepsilon)$. Recycling keeps this order, since $B/(2A) \le B'/A' \le B/A$ in the proof of Corollary 7.13.

- *Source side.* Under the source supply, the least $\log A$ has order $\varepsilon^{-9/2}\log^3(1/\varepsilon)$, by Theorem 7.12(c) and Corollary 4.9. Otherwise the counts do not determine it: Corollary 7.13 gives $\log A' = O(\varepsilon^{-3}\log^3(1/\varepsilon))$, and composition with the classical algorithm gives source side 1. Part (c) needs its hypothesis: for rational $\varepsilon \le 2^{-26}$, the scheme of Corollary 7.13 has pooled exponent below $9/4+\varepsilon/2$ and a source side below the bound of (c) with $\varepsilon/2$ in place of ε, since the ratio of the two bounds, $\frac{5.32\cdot 10^{12}}{1.3}\, 2^{-9/2}\varepsilon^{3/2}(\log(4096/\varepsilon)/\log(2/\varepsilon))^3$, increases with ε and is below 0.92 at $\varepsilon = 2^{-26}$.

- *Seeds.* The seed of Theorem 5.2 depends on the scheme through $\Xi = r_*(1+r_*m_*)n_*^3$, and $r_*^2 m_* \le \Xi \le 54 r_*^2 m_* A^3$, where $r_*^2 m_* \ge m_*^3 = B^9 \ge (2A)^9$ (Lemma 5.1). So the seed built in the same way from the scheme of Corollary 7.13 has $\log n_0$ smaller by a factor less than $\frac{4}{3} + \frac{\log 55}{9\log A} < 1.34$: the copy counts, not the source side, set the size of the seed.

7.4 Small batches

Unlike the source side, the number m_* of target copies does not depend on the supply, and it multiplies under composition. It is the batch size of the pooled recursion [2, Lemma 2.4], and the maps of a relative scheme have at least $m_* B^2$ nonzero coefficients on each leg (see the proof of Proposition 7.15).

Proposition 7.15 (Small batches; computer-assisted)**.** *Let* $M_0 = \frac{23!}{6!\,11!\,6!}\,2^{11} = 2\,558\,607\,507\,456$*; then* $2^{494} < M_0^{12} < 2^{495}$*.*

(a) Every chain scheme with the source supply and $p < 3$ *has* $m_* \geq M_0^{12}$*.*

(b) So does every relative scheme with $p < 3$ *obtained from such chain schemes and the classical restrictions* $n^3\mathsf{M}_1 \succeq \mathsf{M}_n$ *by repeated use of these operations: (i) replacing* $r\mathsf{M}_A \succeq m\mathsf{M}_B$ *by some* $r\mathsf{M}_{A'} \succeq m\mathsf{M}_{B'}$ *with* $B'/A' \leq B/A$*, as other supplies do (Lemma 7.2); (ii) composition as in Remark 7.3, applied to restrictions* $r^{(l)}\mathsf{M}_{A^{(l)}} \succeq m^{(l)}\mathsf{M}_{B^{(l)}}$*,* $l = 1, 2$*, with* $B^{(1)} \geq A^{(2)}$*; and (iii) tensor products. The maps of such a scheme have more than* 2^{496} *nonzero coefficients on each leg.*

(c) The bound in (a) is attained: the one-stage chain $(361, 182) \to (1, 361)$ *of arity* 2 *with* $N = 23$*,* $k = 2$ *and type* $(6, 11, 6)$ *has* $m_* = M_0^{12}$ *and* $p < 2.9999985$*.*

The proof is in Appendix A.7; it checks two finite lists of cases with the supplementary script `shifted_chains_check_small_sources.py`. The bound uses the charges in (7.1) and uncompressed auxiliaries. It says nothing about schemes outside the construction, such as Strassen's, with $m_* = 1$, and it concerns only the operations listed in (b), since keeping some of the target copies is a restriction too.

7.5 Scope

The results above cover the class of Definition 7.1, compressed auxiliaries for the exponent, and supplies and composition for r_*, m_* and $\log(B/A)$. For compressed chain data we have no resource bound. The bounds concern counts only, so they do not depend on the field.

The proofs extend to two variants of the class.

- *Other separators and exactifiers.* The proofs of Corollary 7.9 and Theorem 7.12 use the separator and the exactifier of a block (i, j) only through their numbers of copies. Corollary 7.9 needs only that the separator has at least M_{ij} copies, and Theorem 7.12 needs this for the exactifier as well. For the tensor T of Lemma 3.4, with M sectors, every separator with output (2.4) has at least M copies. Indeed, maps on legs 2 and 3 identify each sector with a product of N factors $C(r, c)$ or $C^{\mathrm{mid}}(r, c)$ (Section 7.2), each of rank r along leg 1. So the flattening of $\bigoplus_g T_g B_X(M)$ along leg 1 has rank Mr^N. That of cT has rank at most cr^N, and degenerations do not raise flattening ranks, so $cT \trianglerighteq \bigoplus_g T_g B_X(M)$ needs $c \geq M$. Proposition 7.15 uses the exact counts $2M_{ij} - 1$ and E_{ij}.

- *Joint exactification of the blocks of one stage* keeps Theorem 7.12 except for the bound on k_{ij} (Remark A.6).

The class charges s^{6W} products for the source. Since the flattening of $Q_{r,b}$ along leg 1 has rank $(rb(r + b - 1))^2$, every evaluation of $Q_{r_1,b_1}^{\otimes W}$ needs at least $(r_1 b_1 s)^{2W}$ products, and with such a source the proof of Corollary 7.9 gives only $p > \frac{3}{4}(1 + \log(r_1 b_1)/\log s)$, which may be below $9/4$ for a finite chain.

Outside the class are:

(i) exactification of several stages together, and exactifiers with fewer than M_{ij} copies;

(ii) auxiliaries shared between orientations or stages, or cut from the Q-factors;

(iii) blocks of one stage with different children, factors that skip a stage, and states that split and merge;

(iv) other local maps, such as the determinant splits of [2, Section 4], children longer than their parents (for which Lemma 7.7(b) fails), stages of arity 1, and different types in different orientations;

(v) other terminal restrictions, and combinations with other schemes beyond Remark 7.3.

8 Open questions

1. *Smaller finite sizes.* For the chain schemes of Definition 7.1 with $p \le 9/4 + \varepsilon$, the least copy counts have logarithms of order $\varepsilon^{-9/2} \log^3(1/\varepsilon)$ (Remark 7.14), but the constants of the two bounds are far apart. Do compressed auxiliaries (Definition 7.8) that attain (7.2) exist at large widths, and what sizes do they give? What is the least source side over all supplies? Can an evaluation of the source with fewer than s^{6W} products (Section 7.5) reduce the sizes?

2. *Coupled auxiliaries.* The limits of Section 7 assume an independent auxiliary for every block, stage and orientation, and separate exactification of every stage. Can auxiliaries shared between orientations or stages, or joint exactification of several stages, change the orders of Theorem 7.12?

3. *Seeds without the factor* $1/\varepsilon$. The seed of Theorem 5.2 has $\log n_0 \ge (24/\varepsilon) \log A$, and for $\varepsilon \le 3/4$ the unrolling procedure needs a factor of this order (Remark 5.3). Is there a rational seed with exponent below $9/4 + \varepsilon$ and $\log n_0 = O(\varepsilon^{-9/2} \log^3(1/\varepsilon))$? It would improve the rate of the seed route of Remark 5.10. Independently, can the rate $2/11$ of Theorem 5.8 be improved?

4. *Bit complexity and precision.* Block recursion with a fixed rational seed is covered by [2, Lemma 9.4 and Propositions 9.5–9.6], but Remark 5.5 bounds the precision of one application of the scheme only. What are the precision and the integer bit complexity of the pooled recursion, which Algorithm $\mathcal{U}$ of Section 5.4 uses?

5. *Positive characteristic.* The local maps of Section 3 hold over every field. The separator and exactification are charged here by rational interpolation, and we prove our results in characteristic zero only; the common denominator $\mathcal{D}$ of Proposition 5.4 is divisible by $(s-1)!$. Over every field, $\omega \le 9/4$ is known [6, 9, 19], but without sizes. Is there a construction over every field, for example over $\mathbb{F}_2$, whose copy counts have logarithms polynomial in $1/\varepsilon$?

6. *Practical sizes.* Every chain scheme with $p < 3$ has more than 2^{494} target copies (Proposition 7.15), and far from $9/4$ other constructions are much smaller (Section 6.2). Which exponents can shifted stages, alone or combined with other schemes, reach with sizes that can be executed in practice?

A Proofs of the limits

This appendix contains the proofs deferred from Section 7. For $d \ge 2$, let μ_d be the positive root of $d\mu^2 - \mu - (d+2) = 0$, as in Lemma 4.3, and put $g_d = \log \frac{d+2}{d}$ and $\omega_d = 3 \log \mu_d / g_d$. Then $\mu_d > 1$ (Lemma 4.3), and $\lambda_d = 1/\mu_d$ is the positive root of $f_d(y) = (d+2)y^2 + y - d$. The function f_d increases on $[0, \infty)$, as $f_d'(y) = 2(d+2)y + 1 > 0$ there. We write $\kappa_0 = 21/20$ and

$$\hat{\Psi}(r,b) = \Psi(r,b) + \tfrac{5}{6} = (r - \tfrac{1}{3})^2 + 4(r - \tfrac{1}{3})(b - \tfrac{1}{3}) + (b - \tfrac{1}{3})^2.$$

Like Ψ, the function $\hat{\Psi}$ is symmetric and increasing in each argument for real $r, b \ge 1$ (Lemma 7.7(a)).

A.1 Supplies

Proof of Lemma 7.2. Stage relations. Fix i, and let $(r,b) \to (c_i, r)$ be the legal stage of Definition 7.1(b). For each block j, Corollary 3.6 with $(N,k,\tau) = (N_{ij}, k_{ij}, \tau_{ij})$ gives

$$E_{ij}(2M_{ij}-1)^{6k_{ij}} Q_{r,b}^{\otimes N_{ij}k_{ij}} \mathsf{M}_{d_i^{2N_{ij}k_{ij}}} \succeq M_{ij}^{6k_{ij}} Q_{c_i,r}^{\otimes N_{ij}k_{ij}} \mathsf{M}_{M_{ij}^{2k_{ij}}}.$$

Tensor these restrictions over j, and use $(mT)(m'T') \cong mm'(TT')$, $\sum_j N_{ij}k_{ij} = W$ and (2.1). With $X_i = \prod_j E_{ij}(2M_{ij}-1)^{6k_{ij}}$ and $Y_i = \prod_j M_{ij}^{6k_{ij}}$, this gives $X_i Q_{r,b}^{\otimes W} \mathsf{M}_{U_i} \succeq Y_i Q_{c_i,r}^{\otimes W} \mathsf{M}_{V_i}$. Here $Q_{r_i,b_i} \cong Q_{r,b}$ by (2.2). If $c' \le c$ and $r' \le r$, then $C(c,r) \succeq C(c',r')$: set to zero x_α for $\alpha \ge c'$, y_η for $\eta \ge r'$ and z_ζ for $\zeta \ge c' + r' - 1$; the surviving terms are those with $\alpha < c'$ and $\eta < r'$. Doing this in all six oriented factors gives $Q_{c,r} \succeq Q_{c',r'}$. Hence a crop and possibly an exchange give $Q_{c_i,r} \succeq Q_{r_{i+1},b_{i+1}}$, and

$$X_i\, Q_{r_i,b_i}^{\otimes W} \mathsf{M}_{U_i} \succeq Y_i\, Q_{r_{i+1},b_{i+1}}^{\otimes W} \mathsf{M}_{V_i}. \tag{A.1}$$

Induction. We show, for $i = 0, \dots, J$, that

$$r_* \mathsf{M}_A \succeq \Big(\prod_{l \le i} Y_l\Big)\Big(\prod_{l > i} X_l\Big) Q_{r_{i+1},b_{i+1}}^{\otimes W} \mathsf{M}_{\mathfrak{b}_i}. \tag{A.2}$$

Since $r_* = s^{6W} \prod_l X_l$, and $s^{6W} \mathsf{M}_1 \succeq Q_{r_1,b_1}^{\otimes W}$ over $\mathbb{Q}$ by (2.2), tensoring with M_A gives the case $i = 0$. Suppose (A.2) holds for $i-1$. Its right side is a direct sum of $(\prod_{l<i} Y_l)(\prod_{l>i} X_l)$ groups, each of X_i copies of $Q_{r_i,b_i}^{\otimes W} \mathsf{M}_{\mathfrak{b}_{i-1}}$. Setting to zero the variables of $\mathsf{M}_{\mathfrak{b}_{i-1}}$ with an index at least $U_i f_i$ restricts it to $\mathsf{M}_{U_i f_i} \cong \mathsf{M}_{U_i} \mathsf{M}_{f_i}$, so each group restricts to $(X_i Q_{r_i,b_i}^{\otimes W} \mathsf{M}_{U_i}) \mathsf{M}_{f_i}$. Apply (A.1), tensored with M_{f_i}, and use $\mathsf{M}_{V_i} \mathsf{M}_{f_i} \cong \mathsf{M}_{\mathfrak{b}_i}$. Each group becomes $Y_i Q_{r_{i+1},b_{i+1}}^{\otimes W} \mathsf{M}_{\mathfrak{b}_i}$, which proves (A.2) for i. Finally, a crop and possibly an exchange give $Q_{r_{J+1},b_{J+1}} \succeq Q_{1,t} \cong \mathsf{M}_{t^2}$ by (2.2), so the case $i = J$ gives $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_{t^{2W} \mathfrak{b}_J} = m_* \mathsf{M}_B$. All maps are restrictions with rational coefficients. No degeneration is composed with another, since each stage relation is already exact.

(a) Since $\mathfrak{b}_i = f_i V_i \le \mathfrak{b}_{i-1} V_i / U_i$, we get $\mathfrak{b}_J \le A \prod_i V_i/U_i$, that is, $B/A \le \bar{B}/\bar{A}$. For the source supply, induction on i gives $\mathfrak{b}_{i-1} = \prod_{l \ge i} U_l \prod_{l<i} V_l = U_i f_i$, so every inequality is an equality.

(b) The first stage is legal, so $b \ge c_1 + 1 \ge 2$ in Definition 7.1(b), and $s \ge 2$. Hence $r_*/m_* = s^{6W} \prod_{i,j} E_{ij} \big((2M_{ij}-1)/M_{ij}\big)^{6k_{ij}} \ge s^{6W} > 1$. If $B > A$, then $\bar{B}/\bar{A} \ge B/A > 1$ by (a), so both restrictions are chain schemes. Their pooled exponents have the same positive numerator $\log(r_*/m_*)$, and $\log(B/A) \le \log(\bar{B}/\bar{A})$.

(c) Let $\tilde{U}$ be the maximum in the hypothesis. We show by induction on $i = 1, \dots, J+1$ that $\mathfrak{b}_{i-1} \ge \frac{A}{2} \prod_{l<i} (V_l/U_l)$, and that $\mathfrak{b}_{i-1} \ge 2JU_i$ if $i \le J$. Suppose this holds for all smaller i. Then $f_l \ge 1$ for $l < i$, and, with $y = \mathfrak{b}_{l-1}/U_l$, the bound $\lfloor y \rfloor \ge y - 1$ gives $\mathfrak{b}_l \ge (\mathfrak{b}_{l-1} V_l/U_l)(1 - U_l/\mathfrak{b}_{l-1}) \ge (1 - \frac{1}{2J}) \mathfrak{b}_{l-1} V_l/U_l$ for $l < i$. By Bernoulli's inequality, $(1 - \frac{1}{2J})^{i-1} \ge 1 - \frac{i-1}{2J} \ge \frac{1}{2}$, so $\mathfrak{b}_{i-1} \ge \frac{A}{2} \prod_{l<i} (V_l/U_l)$. If $i \le J$, this is at least $2J\tilde{U} \prod_{l<i} (V_l/U_l) \ge 2JU_i$, by the definition of $\tilde{U}$. At $i = J+1$ it gives $B/A \ge \bar{B}/(2\bar{A})$. □

The floors in Remark 7.3. Let $n_\infty = AB/(B-A)$, as in Section 5, and let $\mathfrak{b}_0 = n \ge 2n_\infty$ and $\mathfrak{b}_{l+1} = \lfloor \mathfrak{b}_l/A \rfloor B$. Since $\lfloor y \rfloor \ge y-1$, we have $\mathfrak{b}_{l+1} \ge (B/A)\mathfrak{b}_l - B$, that is, $\mathfrak{b}_{l+1} - n_\infty \ge (B/A)(\mathfrak{b}_l - n_\infty)$, so $\mathfrak{b}_l \ge n_\infty + (B/A)^l(n-n_\infty) \ge n_\infty(1+(B/A)^l)$. As $B > A$, we have $n_\infty > A$, so every step can be applied. Moreover $\mathfrak{b}_{l+1} \ge (B/A)\mathfrak{b}_l(1-u_l)$ with $u_l = A/\mathfrak{b}_l \le (1-A/B)/(1+(B/A)^l)$, so $u_l \le \frac12$ and $u_l \le (1-A/B)(A/B)^l$. Since $-\log(1-u) \le 2u$ for $0 \le u \le \frac12$ and $\sum_{l\ge0}(1-A/B)(A/B)^l = 1$, we get $\log \mathfrak{b}_j \ge \log n + j\log(B/A) - 2$.

A.2 The Hankel lemmas

Proof of Lemma 7.4. For $C(r,b)$, the entry at row (ζ,a) and column (η,e) is $[\zeta = a+e+\eta] = \sum_{m=0}^{b+v-2}[\zeta = a+m][m = e+\eta]$. So $\mathcal{L}_u(C(r,b))$ is a product of two matrices with inner dimension $b+v-1$. Conversely, the rows $(m,0)$ and the columns $(\min(m,b-1), m-\min(m,b-1))$, for $0 \le m \le b+v-2$, form an identity submatrix; the rows exist because $m \le b+r-2$, as $v \le r$.

For $C^{\mathrm{mid}}(r,c)$, put $a' = u-1-a$ and $e' = v-1-e$. As $r-1 = (u-1)+(v-1)$, the entry is $[\eta = r-1-a-e+\zeta] = [\eta = \zeta + a' + e'] = \sum_{m=0}^{c+u-2}[m = \zeta + a'][\eta = m + e']$, a product with inner dimension $c+u-1$. The rows with $(\zeta,a') = (\min(m,c-1), m-\min(m,c-1))$ and the columns with $(\eta,e') = (m,0)$, for $0 \le m \le c+u-2$, form an identity submatrix; the columns exist because $u \le r$. All matrices here have entries 0 and 1, so the ranks are the same over every field. □

Proof of Lemma 7.6. If r is odd, take $u = v = (r+1)/2$. Then $c+u-1 = c+v-1 = (r+2c-1)/2$, and (7.2) gives $M/q \le (\hat b/\hat c)^N$ with $\hat b = r+2b-1 \ge \hat c = r+2c-1 \ge 2$. Since $\hat b^2(\hat c^2-1) \le \hat c^2(\hat b^2-1)$, we get $\hat b^2/\hat c^2 \le \varphi_r(b)/\varphi_r(c)$. If r is even, write (7.2) for $(u,v) = (\frac r2, \frac r2+1)$ and for $(u,v) = (\frac r2+1, \frac r2)$, and put $c_+ = c+\frac r2$ and $c_- = c+\frac r2 - 1 \ge 1$. The right sides are $\sum_g c_+^{N-n_{\mathrm{mid}}(g)} c_-^{n_{\mathrm{mid}}(g)}$ and $\sum_g c_-^{N-n_{\mathrm{mid}}(g)} c_+^{n_{\mathrm{mid}}(g)}$, and by the Cauchy–Schwarz inequality their product is at least $M^2(c_+c_-)^N$. The product of the left sides is $q^2\big((b+\frac r2)(b+\frac r2-1)\big)^N$. Finally $4(y+\frac r2)(y+\frac r2-1) = \varphi_r(y)$. □

Proof of Lemma 7.7. (a) $\partial_r\Psi = 2r+4b-2 > 0$ and $\partial_b\Psi = 4r+2b-2 > 0$.

(b) Fix r and put $\psi(y) = \frac23 \log\Psi(r,y) - \frac12\log\varphi_r(y)$ for real $y \ge 1$. With $\hat y = r+2y-1 \ge 2$ and $\delta = r-y$, we have $\varphi_r(y) = \hat y^2 - 1$ and $\varphi_r'(y) = 4\hat y$, and expanding both sides shows

$$\tfrac23(4r+2y-2)\,\varphi_r(y) - 2\hat y\,\Psi(r,y) = \tfrac13(2\hat y\delta^2 - 4\delta + \hat y).$$

The left side is $\Psi(r,y)\varphi_r(y)\psi'(y)$, and $\hat y(2\hat y\delta^2 - 4\delta + \hat y) = 2(\hat y\delta - 1)^2 + \hat y^2 - 2 > 0$. As Ψ and φ_r are positive, ψ is increasing, so $\psi(b) > \psi(c)$.

(c) With $s = r+b-1$, we have $\Psi = s^2 - \frac76 + 2rb$ and $rb \le (s+1)^2/4$, so $\Psi \le \frac32 s^2 + s - \frac23$, and $\frac{11}{6}s^2 - (\frac32 s^2 + s - \frac23) = \frac13(s-1)(s-2) \ge 0$ for $s \ge 2$. By (a), $\Psi(r,b) \ge \Psi(1,b)$, and $(6\Psi(1,b))^2 - 121b^3 = 36(b-1)^4 + 167(b-1)^3 + 345(b-1)^2 + 165(b-1) \ge 0$ as $b \ge 1$. □

A.3 Contraction and endpoints

Lemma A.1 (Contraction)**.** *Every legal stage $(r,b) \to (c,r)$ of arity d satisfies $\hat\Psi(c,r) \le \mu_d^{-2}\,\hat\Psi(r,b)$.*

Proof. Write $\lambda = \lambda_d$ and $\nu = \lambda + \frac{1}{d+2}$, so that $\lambda\nu = \lambda^2 + \frac{\lambda}{d+2} = \frac{d}{d+2}$. Put $\tilde r = r - \frac13$, $\tilde b = b - \frac13$ and $c^* = (db - r + 1)/(d+2)$. Legality gives $c \le c^*$, and $\tilde c^* = c^* - \frac13 = (d\tilde b - \tilde r)/(d+2)$. Comparing the coefficients of $\tilde r^2$, $\tilde r\tilde b$ and $\tilde b^2$, and using $\lambda\nu = \frac{d}{d+2}$, shows

$$\lambda^2\hat\Psi(r,b) - \hat\Psi(c^*,r) = \lambda^2(\tilde r^2 + 4\tilde r\tilde b + \tilde b^2) - (\tilde c^{*2} + 4\tilde c^*\tilde r + \tilde r^2) = (1-\nu^2)(\tilde r - \lambda\tilde b)^2.$$

For instance, the coefficients of $\tilde b^2$ are $\lambda^2 - \frac{d^2}{(d+2)^2} = \lambda^2(1-\nu^2)$. Since f_d increases on $[0,\infty)$, $f_d(\lambda) = 0$ and $f_d(\frac{d+1}{d+2}) = 1$, we have $\lambda < \frac{d+1}{d+2}$ and $0 < \nu < 1$. Hence $\hat\Psi(c,r) \le \hat\Psi(c^*,r) \le \lambda^2\hat\Psi(r,b)$. □

Lemma A.2 (Endpoints). *All chain data satisfy* $\log s \ge \log \kappa_0 + \sum_{i=1}^{J} \log \mu_{d_i} + \frac{4}{5} \log t$.

Proof. Let $(r, b) \to (c_i, r)$ be the stage of Definition 7.1(b). By Lemma A.1, symmetry and monotonicity, $\hat{\Psi}(r_{i+1}, b_{i+1}) \le \hat{\Psi}(c_i, r) \le \mu_{d_i}^{-2} \hat{\Psi}(r_i, b_i)$. Also $\hat{\Psi}(1, t) \le \hat{\Psi}(r_{J+1}, b_{J+1})$. Hence $\hat{\Psi}(1, t) \le \hat{\Psi}(r_1, b_1) \prod_i \mu_{d_i}^{-2}$.

Start. Since $2xy \le \frac{1}{2}(x + y)^2$, we have $x^2 + 4xy + y^2 = (x + y)^2 + 2xy \le \frac{3}{2}(x + y)^2$ for real x, y. With $x + y = r_1 + b_1 - \frac{2}{3} = s + \frac{1}{3}$, this gives $\hat{\Psi}(r_1, b_1) \le \frac{3}{2}(s + \frac{1}{3})^2 \le \frac{9}{4} s^2$, because $12(\frac{9}{4} s^2 - \frac{3}{2}(s + \frac{1}{3})^2) = 9s^2 - 12s - 2 = (s - 2)(9s + 6) + 10 > 0$ for $s \ge 2$ (see the proof of Lemma 7.2(b)).

End. $\hat{\Psi}(1, t) = t^2 + 2t - \frac{1}{3}$. We show $\hat{\Psi}(1, t) \ge \frac{9}{4} \kappa_0^2 t^{8/5}$, where $\frac{9}{4} \kappa_0^2 = \frac{3969}{1600}$. For $t = 1$ this is $\frac{8}{3} \ge \frac{3969}{1600}$. For $t \ge 2$, we have $2t - \frac{1}{3} \ge \frac{11}{6} t$, and the weighted arithmetic–geometric mean inequality gives

$$t^2 + \tfrac{11}{6} t = \tfrac{3}{5} \cdot \tfrac{5}{3} t^2 + \tfrac{2}{5} \cdot \tfrac{55}{12} t \ge (\tfrac{5}{3})^{3/5} (\tfrac{55}{12})^{2/5} t^{8/5}.$$

Here $(\frac{5}{3})^3 (\frac{55}{12})^2 = \frac{378125}{3888} > 97 > 94 > (\frac{3969}{1600})^5$.

Combining, $\frac{9}{4} \kappa_0^2 t^{8/5} \le \frac{9}{4} s^2 \prod_i \mu_{d_i}^{-2}$. Take half the logarithm. □

Lemma A.3 (Source and terminal). *All chain data satisfy:*

(a) $t \le s$;

(b) if $d_i = d$ *for some* i, *then* $s \ge (1 + \frac{1}{d}) t + \frac{1}{d}$;

(c) $s \ge \frac{40}{49} Z t + \frac{76}{147} Z - \frac{1}{3} \ge \frac{40}{49} Z t$, *where* $Z = \prod_i \mu_{d_i}$.

Proof. (a) A legal stage $(r, b) \to (c, r)$ has $c < b$, so the pair (c, r) has a smaller sum of entries than (r, b) and no larger entry; crops and exchanges increase neither. Hence $t \le \max(r_{J+1}, b_{J+1}) \le \max(r_1, b_1) \le s$.

(b) Let $(r, b) \to (c, r)$ be the stage of arity d. By the proof of (a), $s \ge r + b - 1$ and $t \le \max(c, r)$. If $t \le r$, then $db \ge (d+2)c + r - 1 \ge r + d + 1$, so $s \ge (1 + \frac{1}{d}) r + \frac{1}{d}$. If $t \le c$, then $s \ge b \ge (1 + \frac{2}{d}) c \ge (1 + \frac{1}{d}) c + \frac{1}{d}$.

(c) The proof of Lemma A.2 gives $\hat{\Psi}(r_1, b_1) \ge Z^2 \hat{\Psi}(1, t)$ and $\hat{\Psi}(r_1, b_1) \le \frac{3}{2}(s + \frac{1}{3})^2$, so $s + \frac{1}{3} \ge Z \sqrt{\frac{2}{3} \hat{\Psi}(1, t)}$. Also $\frac{2}{3}(t^2 + 2t - \frac{1}{3}) - (\frac{40}{49} t + \frac{76}{147})^2 = \frac{2(t-1)(t+1763)}{7203} \ge 0$. This gives the first inequality, and the second holds because $Z \ge 1$ and $\frac{76}{147} > \frac{1}{3}$. □

A.4 The gap and the type counts

Lemma A.4 (Gap). *For* $d \ge 2$, $g_d \le 2/d$ *and* $\omega_d - \frac{9}{4} \ge \frac{1}{24(d+2)^2} \ge \frac{1}{96 d^2}$. *In particular,* $\log \mu_d > \frac{3}{4} g_d$.

Proof. First, $g_d = \log(1 + \frac{2}{d}) \le \frac{2}{d}$. Put $z = (d/(d+2))^{1/4} \in (0, 1)$, so that $d + 2 = 2/(1 - z^4)$ and $g_d = -4 \log z$. Substituting $d = 2z^4/(1 - z^4)$ gives

$$(1 - z^4) f_d(z^3) = 2z^6 + z^3(1 - z^4) - 2z^4 = z^3 (1 - z)^2 (1 - z^2),$$

so $f_d(z^3) = z^3 (1 - z)^2 / (1 + z^2) > 0$. Since f_d increases on $[0, \infty)$ and $f_d(\lambda_d) = 0$, we get $\lambda_d < z^3$, and

$$\omega_d - \tfrac{9}{4} = \frac{-3 \log \lambda_d + 9 \log z}{g_d} = \frac{3}{g_d} \log \frac{z^3}{\lambda_d}.$$

By the mean value theorem, $f_d(z^3) = f_d(z^3) - f_d(\lambda_d) \le (2d+5)(z^3-\lambda_d)$, as $f_d'(y) = 2(d+2)y+1 \le 2d+5$ on $[0,1]$. With $\log y \ge 1 - 1/y$,

$$\log \frac{z^3}{\lambda_d} \ge \frac{z^3-\lambda_d}{z^3} \ge \frac{(1-z)^2}{(1+z^2)(2d+5)}.$$

Now $1-z^4 = (1-z)(1+z+z^2+z^3) \le 4(1-z)$ gives $1-z \ge \frac{1}{2(d+2)}$. With $1+z^2 \le 2$, $2d+5 \le \frac{9}{2}d$ and $g_d \le \frac{2}{d}$, we get $\omega_d - \frac{9}{4} \ge 3 \cdot \frac{1}{4(d+2)^2} \cdot \frac{1}{2 \cdot \frac{9}{2} d \cdot \frac{2}{d}} = \frac{1}{24(d+2)^2}$. Finally $d+2 \le 2d$, and $\omega_d > \frac{9}{4}$ is $\log \mu_d > \frac{3}{4} g_d$. □

Lemma A.5 (Type counts)**.** *Let $d \ge 2$ and $N \ge 1$, and let M be the sector count of a type of width N at arity d. Then:*

(a) $1 \le M < (d+2)^N$;

(b) if $M > d^N$, *then* $N \ge 3$ *and* $d \le 2N$;

(c) if $N \ge 3$ *and* $d \le 2N$, *then* $M \le (d+2)^N/2$;

(d) if $N \ge 2(d+2)$, *then* $M < 12(d+2)^{N+1}/N$.

Proof. (a) is Lemma 4.6.

(b) For $N=1$ the counts are $d-1$, 2 and 1, all at most d. For $N=2$ they are $(d-1)^2$, $4(d-1)$, $2(d-1)$, 4, 4 and 1, all at most d^2, since $d^2 - 4(d-1) = (d-2)^2$. If $d \ge 2N+1$, then for the type $(n_{\mathrm{K}}, n_{\mathrm{O}}, n_{\mathrm{mid}})$, using $N!/n_{\mathrm{K}}! \le N^{n_{\mathrm{O}}+n_{\mathrm{mid}}}$,

$$\frac{M}{(d-1)^N} \le \frac{1}{n_{\mathrm{O}}!\, n_{\mathrm{mid}}!} \Big(\frac{2N}{d-1}\Big)^{n_{\mathrm{O}}} \Big(\frac{N}{d-1}\Big)^{n_{\mathrm{mid}}} \le 1,$$

so $M \le (d-1)^N < d^N$.

(c) Put $p_{\mathrm{K}} = \frac{d-1}{d+2}$, $p_{\mathrm{O}} = \frac{2}{d+2}$ and $p_{\mathrm{M}} = \frac{1}{d+2}$. Then $M/(d+2)^N$ is the probability that N independent trials with outcome probabilities $p_{\mathrm{K}}, p_{\mathrm{O}}, p_{\mathrm{M}}$ produce the type. This is at most $\pi(n_{\mathrm{K}})$, where $\pi(a) = \binom{N}{a} p_{\mathrm{K}}^a (1-p_{\mathrm{K}})^{N-a}$ is the probability of a kernel outcomes. We have $\pi(a+1) \ge \pi(a)$ if and only if $a+1 \le (N+1)p_{\mathrm{K}}$, and $1 \le (N+1)p_{\mathrm{K}} \le N$, because $N(d-1) \ge 3$ and $d-1 \le 3N$. So π has a maximum at some a with $1 \le a \le N-1$ (if $(N+1)p_{\mathrm{K}} = N$, then $\pi(N-1) = \pi(N)$). At such an a, the ratios $\pi(a-1)/\pi(a)$ and $\pi(a+1)/\pi(a)$ have product $\frac{a}{a+1} \cdot \frac{N-a}{N-a+1} \ge \frac{1}{4}$, hence sum at least 1. So $2\pi(a) \le \pi(a-1) + \pi(a) + \pi(a+1) \le 1$.

(d) Let $\tau = (\tau_{\mathrm{K}}, \tau_{\mathrm{O}}, \tau_{\mathrm{M}})$ be a type that maximizes the probability of (c), and let $P_{\max}$ be this probability; then $M/(d+2)^N \le P_{\max}$. Moving one trial from class j to class i multiplies the probability by $\tau_j p_i / ((\tau_i+1)p_j) \le 1$, and $\tau_j p_i \le (\tau_i+1)p_j$ holds trivially if $\tau_j = 0$. Summing over $j \ne i$ gives $p_i(N-\tau_i) \le (\tau_i+1)(1-p_i)$, that is, $\tau_i \ge (N+1)p_i - 1$. Since $N \ge 2d+4$, this gives $\tau_{\mathrm{K}} \ge \frac{N+1}{4} - 1 \ge \frac{N}{8}$, $\tau_{\mathrm{O}} \ge \frac{N}{d+2}$ and $\tau_{\mathrm{M}} \ge \frac{N}{2(d+2)} \ge 1$. For integers $y \ge 1$,

$$\sqrt{y}\,(y/e)^y \le y! \le e\sqrt{y}\,(y/e)^y.$$

Indeed, log is concave, so the midpoint rule gives $\log y! \ge \int_{1/2}^{y+1/2} \log u\, du \ge y \log y - y + \frac{1}{2}\log y$, and the trapezoid rule gives $\log y! - \frac{1}{2}\log y \le \int_1^y \log u\, du = y \log y - y + 1$. Since $\sum_i \tau_i = N$, these bounds and $\log u \le u - 1$ give

$$P_{\max} \le e\sqrt{\frac{N}{\tau_{\mathrm{K}}\tau_{\mathrm{O}}\tau_{\mathrm{M}}}} \prod_i \Big(\frac{Np_i}{\tau_i}\Big)^{\tau_i} \le e\sqrt{\frac{N}{\tau_{\mathrm{K}}\tau_{\mathrm{O}}\tau_{\mathrm{M}}}} \le \frac{4e(d+2)}{N} < \frac{12(d+2)}{N}.$$ □

A.5 The resource lower bound

Proof of Theorem 7.12. Put $\Lambda_1 = \log(1/\varepsilon)$, $d_\varepsilon = (96\varepsilon)^{-1/2}$ and $p_0 = \frac{9}{4} + \varepsilon$. The hypothesis $\varepsilon < 1/9216$ enters through four conditions that it implies: $\varepsilon \le \frac{3}{20}$, $\varepsilon \le \frac{9}{64}$, $\varepsilon \le \frac{9}{16384}$ and $\varepsilon < \frac{1}{9216}$ itself.

Step 0: reduction. Let $p' \le p_0$ be the pooled exponent of the given chain scheme. By Lemma 7.2(b), the same chain data with the source supply give a chain scheme $r_*\mathsf{M}_{\bar{A}} \succeq m_*\mathsf{M}_{\bar{B}}$ with pooled exponent $p \le p' \le p_0$. As $\varepsilon \le \frac{3}{20}$, $p_0 \le \frac{12}{5} < 3$. The quantities r_*, m_*, W, s, t and d_i do not depend on the supply.

Step 1: normalization. Put $w_{ij} = N_{ij}k_{ij}/W$, so that $\sum_j w_{ij} = 1$, and $\delta_{ij} = N_{ij}\log(d_i+2) - \log M_{ij}$, which is positive by Lemma A.5(a). Taking logarithms in (7.1) and dividing by $2W$ gives

$$\frac{\log(\bar{B}/\bar{A})}{2W} = \log t + \sum_{i,j} w_{ij}\Big(g_{d_i} - \frac{\delta_{ij}}{N_{ij}}\Big), \qquad \frac{\log(r_*/m_*)}{2W} \ge 3\log s + \sum_{i,j} w_{ij}\frac{\log E_{ij}}{2N_{ij}k_{ij}},$$

where the factors $(2M_{ij}-1)/M_{ij} \ge 1$ were dropped.

Step 2: the profit inequality. Since $p \le p_0$, $\log(r_*/m_*) \le p_0\log(\bar{B}/\bar{A})$. By Lemma A.2, $3\log\mu_d = \omega_d g_d$ and $\sum_j w_{ij} = 1$, we have $3\log s \ge 3\log\kappa_0 + \sum_{i,j} w_{ij}\omega_{d_i}g_{d_i} + \frac{12}{5}\log t$. Inserting both into Step 1 gives

$$\sum_{i,j} w_{ij}\vartheta_{ij} \ge 3\log\kappa_0 + (\tfrac{12}{5} - p_0)\log t \ge 3\log\kappa_0, \qquad \vartheta_{ij} = (p_0 - \omega_{d_i})g_{d_i} - \frac{p_0\delta_{ij}}{N_{ij}} - \frac{\log E_{ij}}{2N_{ij}k_{ij}}.$$

Since $\omega_{d_i} > \frac{9}{4}$ (Lemma A.4), $\delta_{ij} > 0$ and $E_{ij} \ge 1$, every block has

$$\vartheta_{ij} < (p_0 - \omega_{d_i})g_{d_i} < \varepsilon g_{d_i}. \tag{A.3}$$

Let I be the set of blocks with $\vartheta_{ij} > 0$. Then $\sum_{(i,j)\in I} w_{ij}\vartheta_{ij} \ge 3\log\kappa_0 > 0$, so I is not empty.

Step 3: a profitable block. Fix $(i,j) \in I$, and write d, N, k, M, δ, E, g and ω for its quantities.

- By (A.3), $\omega < p_0$. By Lemma A.4, $\frac{1}{96d^2} \le \omega - \frac{9}{4} < \varepsilon$, so $d > d_\varepsilon$.
- As $\vartheta > 0$ and $p_0 - \omega < \varepsilon$,

$$\frac{p_0\delta}{N} < \varepsilon g \qquad \text{and} \qquad \frac{\log E}{2Nk} < \varepsilon g. \tag{A.4}$$

- Hence $\frac{1}{N}\log M = \log(d+2) - \frac{\delta}{N} > \log(d+2) - g = \log d$, since $\varepsilon < p_0$. So $M > d^N$.
- By Lemma A.5(b) and (c), $\delta \ge \log 2$. By (A.4) and $g \le \frac{2}{d}$, $N > \frac{p_0\delta}{\varepsilon g} \ge \frac{9}{4}\log 2 \cdot \frac{d}{2\varepsilon} > \frac{9d}{16\varepsilon}$.
- As $\varepsilon \le \frac{9}{64}$, $N > 4d \ge 2(d+2)$, so Lemma A.5(d) applies. With $d+2 \le 2d$, it gives $\delta > \log\frac{N}{24d} > \log\frac{3}{128\varepsilon} \ge \frac{\Lambda_1}{2}$, the last step because $\varepsilon \le \frac{9}{16384}$.
- By (A.4) again, $N > \frac{p_0\delta}{\varepsilon g} > \frac{9}{4}\cdot\frac{\Lambda_1}{2}\cdot\frac{d}{2\varepsilon} = \frac{9d\Lambda_1}{16\varepsilon}$.
- $E = M + (6k-1)(M-1) \ge M > d^N$, so (A.4) gives $\frac{\log d}{2k} < \varepsilon g \le \frac{2\varepsilon}{d}$, that is, $k > \frac{d\log d}{4\varepsilon}$.
- As $\varepsilon < \frac{1}{9216}$, $\log d_\varepsilon = \frac{1}{2}(\Lambda_1 - \log 96) > \frac{\Lambda_1}{4}$. Since $d > d_\varepsilon > 1$ and $y\log y$ increases for $y \ge 1$, $d\log d > d_\varepsilon\log d_\varepsilon > \frac{d_\varepsilon\Lambda_1}{4}$.

These are the bounds on d_i, N_{ij} and k_{ij} in the theorem. Since $W \geq Nk$ and $d_\varepsilon^2 = \frac{1}{96\varepsilon}$,

$$W > \frac{9d_\varepsilon\Lambda_1}{16\varepsilon} \cdot \frac{d_\varepsilon\Lambda_1}{16\varepsilon} = \frac{3\Lambda_1^2}{8192\,\varepsilon^3}.$$

Step 4: the profitable sum. For $(i,j) \in I$, (A.3) and Lemma A.4 give $\vartheta_{ij} < \varepsilon g_{d_i} \leq \frac{2\varepsilon}{d_i} < \frac{2\varepsilon \log d_i}{d_\varepsilon \log d_\varepsilon}$, as $d_i \log d_i > d_\varepsilon \log d_\varepsilon$. Weight by w_{ij} and sum over I:

$$\Sigma_I = \sum_{(i,j)\in I} w_{ij} \log d_i > \frac{3 \log \kappa_0\, d_\varepsilon \log d_\varepsilon}{2\varepsilon} > \frac{3 \log \kappa_0\, d_\varepsilon \Lambda_1}{8\varepsilon}.$$

Step 5: conclusion. Since $\sum_j w_{ij} = 1$, Steps 3 and 4 give

$$\log \bar{A} = 2W \sum_{i,j} w_{ij} \log d_i \geq 2W\,\Sigma_I > 2 \cdot \frac{3\Lambda_1^2}{8192\,\varepsilon^3} \cdot \frac{3 \log \kappa_0\, d_\varepsilon \Lambda_1}{8\varepsilon} = \frac{9 \log \kappa_0}{32768\sqrt{96}}\, \varepsilon^{-9/2} \Lambda_1^3.$$

As $\log(1+y) \geq y/(1+y)$, $\log \kappa_0 \geq \frac{1}{21}$, and $\sqrt{96} < 10$, so the coefficient exceeds $\frac{9}{21 \cdot 32768 \cdot 10} > 1.3 \cdot 10^{-6}$. Under the source supply, $A = \bar{A}$, which proves (c).

For (a), the proof of Lemma 7.2(b) gives $r_*/m_* \geq s^{6W} > 1$. Since $p \leq 3$, $\log(\bar{B}/\bar{A}) \geq \frac{1}{3}\log(r_*/m_*) \geq 2W \log s$. By (7.1), $\bar{B} = t^{2W} m_*^{1/3}$, so $\frac{1}{3}\log m_* \geq \log \bar{A} + 2W \log(s/t)$. By Lemma A.3(a), $t \leq s$, so $\log m_* \geq 3 \log \bar{A}$. Finally $3 \cdot \frac{9}{21 \cdot 32768 \cdot 10} > 3.9 \cdot 10^{-6}$.

For (b), Step 2 and (A.3) give $3 \log \kappa_0 \leq \sum_{i,j} w_{ij} \vartheta_{ij} < \varepsilon \sum_i g_{d_i}$. By Lemmas A.2 and A.4, with $\log \kappa_0 > 0$ and $t \geq 1$, $\log s \geq \sum_i \log \mu_{d_i} > \frac{3}{4} \sum_i g_{d_i} > \frac{9 \log \kappa_0}{4\varepsilon}$. For the given scheme, $p' \leq p_0 \leq \frac{12}{5}$ gives

$$\log(B/A) = \frac{\log(r_*/m_*)}{p'} \geq \frac{\log(r_*/m_*)}{p_0} \geq \frac{6W \log s}{12/5} > \frac{45 \log \kappa_0}{8} \cdot \frac{W}{\varepsilon} > \frac{45}{8 \cdot 21} \cdot \frac{3\Lambda_1^2}{8192\,\varepsilon^4},$$

and $\frac{135}{8 \cdot 21 \cdot 8192} = \frac{45}{458752} > 9.8 \cdot 10^{-5}$. □

Remark A.6 (Joint exactification of one stage). Suppose that the blocks of each stage are exactified together, so that the factors E_{ij} of r_* are replaced by $E_i = 1 + \sum_j 6k_{ij}(M_{ij} - 1)$. Then Theorem 7.12 still holds, except for the bound on k_{ij}. In the proof above, Step 1 holds with $\sum_i \log E_i/(2W)$ in place of $\sum_{i,j} w_{ij} \log E_{ij}/(2N_{ij}k_{ij})$, and Step 2 gives $\sum_i \vartheta_i \geq 3 \log \kappa_0$ with one profit term per stage,

$$\vartheta_i = \sum_j w_{ij} \Big((p_0 - \omega_{d_i}) g_{d_i} - \frac{p_0 \delta_{ij}}{N_{ij}} \Big) - \frac{\log E_i}{2W} < \varepsilon g_{d_i}.$$

If $\vartheta_i > 0$, then some block j of stage i has $(p_0 - \omega_{d_i}) g_{d_i} > p_0 \delta_{ij}/N_{ij}$, and Step 3 gives the bounds on d_i and N_{ij} for this block. Moreover $\log E_i/(2W) < \varepsilon g_{d_i} \leq 2\varepsilon/d_i$ and $E_i \geq M_{ij} > d_i^{N_{ij}}$ give $W > N_{ij} d_i \log d_i/(4\varepsilon)$, which replaces $W \geq N_{ij}k_{ij}$ in the bound on W. Steps 4 and 5 run over the stages with $\vartheta_i > 0$, with Σ_I replaced by the sum of $\log d_i$ over these stages.

A.6 The recycled main construction

Proof of Corollary 7.13. We use the notation of Theorem 4.8 and its proof. Here A and B are the sides of Theorem 4.8, that is, of the source supply, $W = Nk$, $t = 1$, every stage has one block with the largest sector count M_i, and, as there, $a_N = 2\log(N+1)/N$ and $\Lambda = \log(4096/\varepsilon)$. Read forwards, the chain has $L \geq 1$ bulk stages of arity D, followed by ramp stages, and every

arity satisfies $d_i \le 24D-3$ (Lemma 4.5). We use these facts from the proof of Theorem 4.8: $N = k \ge 256DH/\varepsilon \ge 4608/\varepsilon$ (as $D \ge 2$ and $H \ge 9$), $a_N < 3\varepsilon/(128D)$ and $Lg_D \ge 64/\varepsilon$ (Steps 0 and 1); $\log(B/A) \ge \frac{497}{512}\cdot 2NkG$ with $G \ge 64/\varepsilon$ (Steps 1 and 4; see also (5.2)); and $D \le 3\varepsilon^{-1/2}$, $J \le 240\,\varepsilon^{-3/2}$ and $N \le 2305\,\varepsilon^{-3/2}\Lambda$ (Step 5).

Funding. By Lemma 4.6, $\log M_i \ge N\log(d_i+2) - 2\log(N+1)$, so $\gamma_i = \log(V_i/U_i) = 2k\log M_i - 2Nk\log d_i \ge 2Nk(g_{d_i} - a_N)$.

- Every stage gains: $g_{d_i} \ge \frac{2}{d_i+2} > \frac{1}{12D}$, as $d_i + 2 < 24D$, and $a_N < \frac{3\varepsilon}{128D} \le \frac{3}{128D} < \frac{1}{12D}$. So $\gamma_i > 0$.
- The bulk pays for the larger arities of the ramp: $La_N < (\frac{64D}{\varepsilon}+1)\frac{3\varepsilon}{128D} \le \frac{3}{2} + \frac{3}{256} < 2$, so $\sum_{i\le L}\gamma_i \ge 2Nk(Lg_D - La_N) > 2Nk(\frac{64}{\varepsilon} - 2) \ge 124Nk$.

For $i \le L$, $U_i = D^{2Nk}$ and $\prod_{l<i} U_l/V_l \le 1$, with equality for $i = 1$. For $i > L$, $U_i \le (24D)^{2Nk}$ and $\prod_{l<i} U_l/V_l < e^{-124Nk} < 24^{-2Nk}$. Hence $\max_i U_i \prod_{l<i} U_l/V_l = D^{2Nk}$, and Lemma 7.2(c) applies to $A' = 4JD^{2Nk}$: the recycling supply is a supply, and $B'/A' \ge B/(2A)$. By Lemma 7.2(a), $B'/A' \le B/A$.

Exponent. We have $\log(B/A) \ge \frac{497}{512}\cdot\frac{128}{\varepsilon}Nk > \frac{124Nk}{\varepsilon} \ge \frac{124\cdot 4608^2}{\varepsilon}$, so $\log(B'/A') \ge \log(B/A) - \log 2 > 0$, and $B' > A'$. The two pooled exponents have the same numerator $p\log(B/A)$, so

$$p' - p \le \frac{p\log 2}{\log(B/A) - \log 2} < \frac{\frac{11}{4}\log 2}{124\cdot 4608^2 - 1}\,\varepsilon < 10^{-9}\varepsilon,$$

using $p < \frac{11}{4}$ and $\varepsilon \le 1$. With $p < \frac{9}{4} + \frac{719}{1988}\varepsilon$ (Theorem 4.8) and $\frac{719}{1988} + 10^{-9} < \frac{1}{2}$, this gives $p' < \frac{9}{4} + \frac{\varepsilon}{2}$. Also $p' \ge p > \frac{9}{4}$, by Lemma 7.2(b) and Theorem 4.8.

Source side. $\log A' = \log(4J) + 2Nk\log D$. Here $4J \le 960\,\varepsilon^{-3/2} \le (4096/\varepsilon)^{3/2}$, so $\log(4J) \le \frac{3}{2}\Lambda$; $2\log D \le \log(9/\varepsilon) \le \Lambda$; and $Nk = N^2 \le 2305^2\varepsilon^{-3}\Lambda^2$. As $\Lambda \ge \log 4096 > 8.3$, $\frac{3}{2}\Lambda < 0.022\,\Lambda^3$. Hence

$$\log A' < (2305^2 + 0.022)\,\varepsilon^{-3}\Lambda^3 < 5.32\cdot 10^6\,\varepsilon^{-3}\Lambda^3.$$

Target side. Since $M_i < (d_i+2)^N$ and $t = 1$, $\log(B/A) < 2NkG$, where $G = \log F(R,R) + Lg_D$ (Lemma 4.5). Here $Lg_D \le (\frac{64D}{\varepsilon}+1)\frac{2}{D} \le \frac{128}{\varepsilon} + 1$. By [2, (3.6)], $\log F(R,R) \le \frac{4}{3}\log(2R-1)$, and $2R - 1 < 16D \le 48\,\varepsilon^{-1/2}$; since $48^4 < 4096^2$, this gives $\log F(R,R) < \frac{2}{3}\Lambda$. The function $\varepsilon\log(4096/\varepsilon)$ increases on $(0,1]$, so $\varepsilon\Lambda \le \log 4096 < 8.32$. Hence

$$\log(B/A) < Nk(\tfrac{256}{\varepsilon} + 2 + \tfrac{4}{3}\Lambda) \le 2305^2\varepsilon^{-4}\Lambda^2(256 + 2\varepsilon + \tfrac{4}{3}\varepsilon\Lambda) < 1.43\cdot 10^9\,\varepsilon^{-4}\Lambda^2.$$

By Lemma 7.2(a), $\log B' \le \log A' + \log(B/A)$, and $\log A' < 5.32\cdot 10^6\,\varepsilon^{-3}\Lambda^3 < 4.5\cdot 10^7\,\varepsilon^{-4}\Lambda^2$, as $\varepsilon\Lambda < 8.32$. So $\log B' < 1.5\cdot 10^9\,\varepsilon^{-4}\Lambda^2$. The restriction holds over $\mathbb{Q}$ by Lemma 7.2, with the r_* and m_* of Theorem 4.8. □

A.7 Small batches

Proof of Proposition 7.15. (a) For integers $N, k \ge 1$, $d \ge 2$ and $M \ge 1$, put $E = 1 + 6k(M-1)$ and $\upsilon(N,d,k,M) = \frac{1}{2Nk}\log(d^{6Nk}(2M-1)^{6k}E/M^{12k})$. Let $\upsilon_{ij} = \upsilon(N_{ij}, d_i, k_{ij}, M_{ij})$ and let $\bar\upsilon_i = \sum_j (N_{ij}k_{ij}/W)\upsilon_{ij}$, a weighted average. Under the source supply, $\log A = \sum_{i,j} 2N_{ij}k_{ij}\log d_i$, so (7.1) gives

$$\frac{\log(r_*/m_*) - 3\log(B/A)}{2W} = 3\log\frac{s}{t} + \sum_i \bar\upsilon_i, \tag{A.5}$$

and $p < 3$ if and only if this is negative. If $M \le d^N$, then $\upsilon(N,d,k,M) \ge 0$, because $d^{6Nk} \ge M^{6k}$, $2M-1 \ge M$ and $E \ge 1$. If a type of width N has sector count $M > d^N$, then $N \ge 3$ and $d \le 2N$ (Lemma A.5(b)).

Suppose that $p < 3$ and $m_* < M_0^{12}$, and let $\mathcal{K}$ be the set of stages with $\bar\upsilon_i < 0$. For $i \in \mathcal{K}$, let N, k, M be the width, power and sector count of a block of stage i with the least υ_{ij}, and υ_i° its υ_{ij}. Then $\upsilon_i^\circ \le \bar\upsilon_i < 0$, so $M > d_i^N$, $N \ge 3$, $d_i \le 2N$, and $M^{6k} \le m_* < M_0^{12}$. By (A.5), $0 > 3\log(s/t) + \sum_{i\in\mathcal{K}} \upsilon_i^\circ$. If $\mathcal{K} = \emptyset$, this contradicts Lemma A.3(a). If $\mathcal{K} = \{i\}$ and $d = d_i$, Lemma A.3(b) gives $s > (1+\frac{1}{d})t$, so

$$0 > 3\log\frac{d+1}{d} + \upsilon_i^\circ = \frac{1}{2Nk}\log\frac{(d+1)^{6Nk}(2M-1)^{6k}E}{M^{12k}},$$

which needs $M > (d+1)^N$ (as above) and contradicts check (i) below. If $|\mathcal{K}| \ge 2$, Lemma A.3(c) and $\mu_d > 1$ give $3\log(s/t) \ge 3\log\frac{40}{49} + 3\sum_{i\in\mathcal{K}} \log\mu_{d_i}$, and check (ii) gives $3\log\mu_{d_i} + \upsilon_i^\circ \ge \log\frac{34}{25}$. So $0 > 3\log\frac{40}{49} + 2\log\frac{34}{25}$, which is false, as $34^2 \cdot 40^3 = 73\,984\,000 > 73\,530\,625 = 25^2 \cdot 49^3$.

The two checks concern all (N,d,k,M) with $N \ge 3$, $2 \le d \le 2N$, $k \ge 1$, M a sector count of width N at arity d, and $M^{6k} < M_0^{12}$: (i) if $M > (d+1)^N$, then $(d+1)^{6Nk}(2M-1)^{6k}E \ge M^{12k}$; (ii) if $M > d^N$, then $25^{2Nk}(vd)^{6Nk}(2M-1)^{6k}E \ge 34^{2Nk}u^{6Nk}M^{12k}$, where $u/v \ge \lambda_d$ is a rational number fixed for each d. Since $\log\mu_d \ge \log(v/u)$, (ii) gives the bound used above. Both lists are finite, since $d^{6Nk} < M^{6k} < 2^{495}$ gives $N \le 82$, and $d \ge 33$ would give $N \ge 17$ and $d^{6N} \ge 33^{102} > 2^{495}$. The supplementary script `shifted_chains_check_small_sources.py` (Part C) runs both checks in exact integer arithmetic, with u/v the upper end of an enclosure of λ_d from an integer square root, certified by $f_d(u/v) \ge 0$. All 5 792 cases of (i) and 31 677 cases of (ii) pass.

(b) Every restriction $r\mathsf{M}_A \succeq m\mathsf{M}_B$ in the construction has $r \ge m$ (for chain schemes since $2M_{ij} - 1 \ge M_{ij}$), and the operations preserve this. By induction on the construction, $B > A$ and $r/m < (B/A)^3$ imply $m \ge M_0^{12}$. For chain schemes this is (a); classical restrictions have $r/m = (B/A)^3$; and (i) keeps r and m and does not increase B/A. Under (ii) and (iii), $m = m^{(1)}m^{(2)}$, $r/m = (r^{(1)}/m^{(1)})(r^{(2)}/m^{(2)})$ and $B/A \le (B^{(1)}/A^{(1)})(B^{(2)}/A^{(2)})$, since $f \le B^{(1)}/A^{(2)}$. So $r/m < (B/A)^3$ gives $r^{(l)}/m^{(l)} < (B^{(l)}/A^{(l)})^3$, hence $B^{(l)} > A^{(l)}$ as $r^{(l)} \ge m^{(l)}$, for some l, and $m \ge m^{(l)} \ge M_0^{12}$. The composition is a restriction by the construction of Remark 7.3, which does not use $B^{(l)} > A^{(l)}$. Finally, the maps of $r\mathsf{M}_A \succeq m\mathsf{M}_B$ replace each variable of $r\mathsf{M}_A$ by a linear form in the mB^2 variables of the same leg of $m\mathsf{M}_B$, and each of these occurs in $m\mathsf{M}_B$, hence in some form. As $B \ge 2$, this gives at least $mB^2 \ge 4M_0^{12} > 2^{496}$ nonzero coefficients per leg.

(c) The stage is legal, since $4 \le 2\cdot 182 - 361 + 1$, and $s = 542$. So $A = 2^{92}$, $B = 361^{92}M_0^4$, $r_* = 542^{276}(2M_0-1)^{12}(12M_0-11)$ and $m_* = M_0^{12}$. The script checks $r_*A^3 < m_*B^3$, that is $p < 3$, in exact integer arithmetic, and its enclosures give $p \in [2.999998425472, 2.999998425473]$. □

B The maps of the scheme and their coefficients

This appendix writes each of the three maps of the scheme $\mathcal{S}_\varepsilon$ of Section 5 as a product of explicit matrices, in the order in which the proofs of Proposition 3.5, Corollary 3.6 and Proposition 3.8 apply them. From this list we read off a common denominator and bounds on the numerators, and we bound the cost of a simple generator. This proves parts (a) and (c) of Proposition 5.4. We keep the notation of Proposition 3.8 and Section 5; in particular $N = k$, the terminal pair is $(1,1)$, and $\mathcal{D}$ and β_ε are as in Proposition 5.4.

B.1 Substitution matrices

A restriction $T \succeq T'$ replaces each variable of leg λ of T by a linear form in the leg-λ variables of T'. Its *leg-λ matrix* has rows indexed by the leg-λ variables of T, columns indexed by those of T', and, in row x and column y, the coefficient of y in the form that replaces x. Composition multiplies leg matrices: if $T \succeq T'$ and $T' \succeq T''$ have leg-λ matrices Γ and Γ', then $T \succeq T''$ has $\Gamma\Gamma'$. The restrictions $mT \succeq mT'$ and $TT'' \succeq T'T''$, with the identity on T'', have leg matrices $I_m \otimes \Gamma$ and $\Gamma \otimes I$; tensor products of restrictions have Kronecker products of leg matrices; and isomorphisms have permutation matrices. A degeneration has leg matrices $\mathsf{S}(t)$ with Laurent polynomial entries, and $\mathsf{S}(u)$ is the matrix at $t = u \neq 0$. The restriction $E\,T^{\otimes k} \succeq \widetilde{T}_0^{\otimes k}$ of [2, Lemma 9.1(c)] has, on leg λ, rows indexed by pairs (u, x) with $1 \le u \le E$; row (u, x) is row x of $\mathsf{S}(u)^{\otimes k}$, multiplied by $w_u = (-1)^{u-1}\binom{E}{u}$ on one fixed leg.

For a rational matrix Γ, let $\|\Gamma\|$ be its largest absolute row sum. It is submultiplicative, multiplicative under $\otimes$, equal to 1 for permutation matrices, and at least the absolute value of every entry. A *common denominator* of Γ is a positive integer $\mathfrak{d}$ with $\mathfrak{d}\Gamma$ integral; the product of common denominators of Γ and Γ' is one of $\Gamma\Gamma'$ and of $\Gamma \otimes \Gamma'$.

B.2 The maps

For a positive integer n and $1 \le \ell \le n$, let

$$\varpi_{n,\ell}(\phi) = \prod_{\substack{1 \le \ell' \le n \\ \ell' \neq \ell}} \frac{\phi - \ell'}{\ell - \ell'} = \sum_{\gamma=0}^{n-1} \varpi_{n,\ell,\gamma}\, \phi^{\gamma}$$

be the ℓ-th Lagrange basis polynomial of the nodes $1, \dots, n$, in an indeterminate ϕ, and write $[\phi^e]$ for the coefficient of ϕ^e; [2, Section 9.1] writes ξ for ϕ.

Fix a leg $\lambda \in \{1, 2, 3\}$. In the notation of the proof of Proposition 3.8, that proof passes through the tensors $\mathcal{I}_0 = r_* \mathsf{M}_A$ and

$$\mathcal{I}_j = \Big(\prod_{i<j} \rho_i'\Big)\Big(\prod_{i \ge j} \rho_i\Big)\mathcal{T}_j\, \mathsf{M}_{A_{\ge j} B_{<j}} \qquad (1 \le j \le J+1), \tag{B.1}$$

where $\rho_i = E_i(2M_i - 1)^{6k}$, $\rho_i' = M_i^{6k}$, $\mathcal{T}_j = Q_{r_j,b_j}^{\otimes Nk}$, and $\mathcal{I}_{J+1} \cong m_* \mathsf{M}_B$. Let Γ_0 be the leg-λ matrix of the source decomposition $\mathcal{I}_0 \succeq \mathcal{I}_1$, and Γ_i that of stage i, $\mathcal{I}_i \succeq \mathcal{I}_{i+1}$. Up to the permutation of columns given by $\mathcal{I}_{J+1} \cong m_* \mathsf{M}_B$, the leg-$\lambda$ matrix of $\mathcal{S}_\varepsilon$ is

$$\Gamma = \Gamma_0 \Gamma_1 \cdots \Gamma_J. \tag{B.2}$$

Leg dimensions. Leg λ of $Q_{r,b}$ has dimension $(rb(r+b-1))^2 \le (r+b-1)^6$, since each leg of $C(r,b)$ lies on leg λ in two of the six orientations. Along the chain, $r+b-1$ decreases, because $c_i + r_i - 1 < r_i + b_i - 1$. Since $\rho_i' \le \rho_i$, every leg of every $\mathcal{I}_j$, $0 \le j \le J+1$, has dimension at most

$$\mathsf{n} = r_* A^2 B^2 \le A^{59/4},$$

by Lemma 5.1.

The source. Proposition 3.8 decomposes each of the $6Nk$ oriented factors $C(r_1, b_1)^\pi$ of $\mathcal{T}_1$ by [2, Lemma 9.1(a)], with the nodes $1, \dots, s$. So $\Gamma_0 = I \otimes \Gamma_{\mathrm{src}} \otimes I$ (with the identity on the copies and on M_A), where the rows of Γ_{src} are indexed by tuples of nodes $\ell_f \in \{1, \dots, s\}$, one for each factor f, and the entry of Γ_{src} is the product over f of ℓ_f^α, ℓ_f^β or $\varpi_{s,\ell_f,\gamma}$, according as factor f contributes its first, second or third leg (variable x_α, y_β or z_γ) to leg λ. Each case occurs for $2Nk$ factors.

One stage in one orientation. Fix a stage i, and write $(r,b)=(r_i,b_i)$, $c=c_i$, $h=b-c$, $d=d_i$, $M=M_i$, $E=E_i$, and let $U=C(r,b)B_X(d)$ be the tensor (3.2), with variables x_α, $y_{j,\beta}$ and $z_{j,\gamma}$. Fix $\pi\in S_3$, and let $\iota=\pi^{-1}(\lambda)$ be the leg of $U^{\otimes N}$ that lies on leg λ of $(U^{\otimes N})^\pi$. In the proof of Proposition 3.5, the oriented degeneration $(2M-1)(U^{\otimes N})^\pi \trianglerighteq M(\mathcal{V}B_X(M))^\pi$ has, on leg λ, the matrix

$$\mathsf{S}_\pi(t)=(I_{2M-1}\otimes K_\iota)\,\mathsf{P}_\iota(t)\,\Sigma_\iota,$$

where the factors are as follows.

- K_1 is the identity. For $\iota=2,3$, K_ι is the change to the shifted bases of Lemma 3.1 at all N positions, followed by the crop and the type restriction of Lemma 3.4, which set coordinates to zero; as a matrix, the latter is a coordinate selection, with entries $0,1$ and at most one nonzero entry in each row and column. Let Π_ℓ be the matrix whose columns are the shifted basis of Lemma 3.1(a), written in the monomial basis $\xi^m e_j$. As in the proof of Lemma 3.1(c), leg 2 changes by $\Pi_b^{\otimes N}$, and leg 3 contragrediently, by $(\Pi_{b+r-1}^{-\top})^{\otimes N}$.
- $\mathsf{P}_\iota(t)$ is the separator (2.4), in the form of [2, Lemma 9.1(b)], with M sectors $g=0,\dots,M-1$, the words of the type, and shared leg $\pi(1)$. In source copy $\ell\in\{1,\dots,2M-1\}$ it maps

$$x_\alpha\mapsto\sum_{g'<M}\varpi_{M,g'+1}(\ell)X_{\alpha,g'},\quad y_{g,\beta}\mapsto\sum_{e<M}\frac{(\ell-g-1)^e}{t^e}Y_{g,\beta,e},\quad z_{g,\gamma}\mapsto\sum_{e'<M}\zeta_{\ell,g,e'}\,t^{e'}Z_{g,\gamma,e'},$$

  for $\iota=1,2,3$ respectively, where $\zeta_{\ell,g,e'}=[\phi^{e'}]\,\varpi_{2M-1,\ell}(g+1+\phi)$.
- Σ_ι is the word-dependent reindexing, a permutation matrix.

One stage. On leg λ, the six-orientation degeneration of Proposition 3.5 has the matrix $\mathsf{S}(t)=\Sigma_{\mathrm{in}}\big(\bigotimes_{\pi\in S_3}\mathsf{S}_\pi(t)\big)\Sigma_{\mathrm{out}}$, where Σ_{in} and Σ_{out} are the permutation matrices of the identifications of $Q_{r,b}^{\otimes N}\mathsf{M}_{d^{2N}}$ with $\bigotimes_\pi(U^{\otimes N})^\pi$ and of $\bigotimes_\pi(\mathcal{V}B_X(M))^\pi$ with $Q_{c,r}^{\otimes N}\mathsf{M}_{M^2}$. As π runs through S_3, the leg $\iota=\pi^{-1}(\lambda)$ takes each value twice. Corollary 3.6 exactifies this degeneration at power k by [2, Lemma 9.1(c)]; let Θ_i be the resulting leg-λ matrix, with rows (u,x) as in Appendix B.1. Finally, Γ_i is $I\otimes\Theta_i\otimes I$ (identity on the other copies and on $\mathsf{M}_{A_{\ge i+1}B_{<i}}$), up to permutation matrices on both sides.

Primitive bounds. The primitive matrices have the following entries, row-sum bounds and common denominators. Crops, the type restriction, the reindexing and the identifications are coordinate selections or permutations, with row sums at most 1 and denominator 1; the exactification weights $w_u=(-1)^{u-1}\binom{E}{u}$ are integers with absolute value below 2^E.

- The shifted basis consists of the lifts, which are monomials, and the kernel vectors $\kappa_{j'}(\xi^i)=\xi^i e_{j'+1}-\xi^{i+h}e_{j'}$. A monomial $\xi^m e_j$ occurs in at most one lift and in at most two kernel vectors, those with $(j',i)=(j,m-h)$ or $(j-1,m)$; so a row of Π_b has at most three nonzero entries, all ±1, and row sum at most 3. A row of $\Pi_{b+r-1}^{-\top}$ lists the coefficients of one monomial in the shifted basis; by (3.3), it has at most d nonzero entries, all 1. Both matrices are integral.
- For $1\le\ell\le 2M-1$ and $1\le j\le M$ we have $|\ell-j|\le 2M-2$, so $|\varpi_{M,j}(\ell)|\le(2M-2)^{M-1}$ and $\sum_{g'<M}|\varpi_{M,g'+1}(\ell)|\le(2M)^M$; the denominator $(j-1)!\,(M-j)!$ of $\varpi_{M,j}$ divides $(M-1)!$. Likewise $|\ell-g-1|\le 2M-2$ gives $\sum_{e<M}|\ell-g-1|^e u^{-e}\le(2M)^M$, with denominator u^{M-1}.

- The polynomial $\varpi_{2M-1,\ell}(g+1+\phi)$ is $\prod_{\ell'\neq\ell}(\phi+g+1-\ell')$ divided by $\prod_{\ell'\neq\ell}(\ell-\ell') = \pm(\ell-1)!\,(2M-1-\ell)!$, a divisor of $(2M-2)!$. The absolute values of the coefficients of a product of linear factors $\phi+a$ sum to at most $\prod(1+|a|)$, and $1+|g+1-\ell'|\le 2M$. Hence $\sum_{e'<M}|\zeta_{\ell,g,e'}|u^{e'} \le E^{M-1}(2M)^{2M}$ for $u\le E$, with denominator $(2M-2)!$.
- Similarly $\sum_{\alpha<r_1}\ell^\alpha \le s^s$, $\sum_{\beta<b_1}\ell^\beta \le s^s$, and $\sum_\gamma|\varpi_{s,\ell,\gamma}| \le \prod_{\ell'\neq\ell}(1+\ell') \le (s+1)^s$ for $1\le\ell\le s$; the denominators of the $\varpi_{s,\ell,\gamma}$ divide $(s-1)!$.

Bounds for the factors. For $\iota=1,2,3$, these bounds give $\|\mathsf{S}_\pi(u)\|\le\mathfrak{s}_\iota$ with $\mathfrak{s}_1=(2M)^M$, $\mathfrak{s}_2=3^N(2M)^M$ and $\mathfrak{s}_3=d^N(2M)^{2M}E^{M-1}$, and the common denominators $(M-1)!$, u^{M-1} and $(2M-2)!$. Since each ι occurs twice, $\|\mathsf{S}(u)\|\le(\mathfrak{s}_1\mathfrak{s}_2\mathfrak{s}_3)^2 \le (3d)^{2N}(2M)^{8M}E^{2M}$, with common denominator $((M-1)!\,(2M-2)!)^2u^{2M-2}$. For $u\le E$, the number $u^{2k(M-1)}$ divides $(E!)^{2k(M-1)}$. Hence

$$\|\Gamma_i\| \le \mathfrak{n}_i = 2^{E_i}\big((3d_i)^{2N}(2M_i)^{8M_i}E_i^{2M_i}\big)^k, \qquad \mathfrak{d}_i = \big((M_i-1)!\,(2M_i-2)!\big)^{2k}(E_i!)^{2k(M_i-1)}, \tag{B.3}$$

where $\mathfrak{d}_i$ is a common denominator of Γ_i. For the source,

$$\|\Gamma_0\| \le \mathfrak{n}_0 = (s+1)^{6Nks}, \qquad \mathfrak{d}_0 = ((s-1)!)^{2Nk}. \tag{B.4}$$

So $\mathcal{D}=\mathfrak{d}_0\mathfrak{d}_1\cdots\mathfrak{d}_J$, for each of the three legs.

B.3 Denominators and bit lengths

Proof of Proposition 5.4(a). Fix a leg. By (B.2) and the rules of Appendix B.1, every partial product $\Gamma_0\cdots\Gamma_j$ has the common denominator $\mathfrak{d}_0\cdots\mathfrak{d}_j$ and norm at most $\mathfrak{n}=\mathfrak{n}_0\mathfrak{n}_1\cdots\mathfrak{n}_J$ (all $\mathfrak{n}_i\ge1$). As $\mathcal{D}=\mathfrak{d}_0\cdots\mathfrak{d}_J$, every coefficient of the map is $\mathfrak{c}/\mathcal{D}$ with an integer $\mathfrak{c}$ and $|\mathfrak{c}|\le\mathfrak{n}\mathcal{D}$. It remains to show $\log_2\mathfrak{n}+\log_2\mathcal{D}<\beta_\varepsilon$.

Put $\mathcal{Y}=(24D)^N$, so that $\beta_\varepsilon=\mathcal{Y}^3$. We use: $N=k$; $2\le M_i\le(d_i+2)^N<\mathcal{Y}$; $2M_i\le E_i\le 6kM_i<6N\mathcal{Y}\le\mathcal{Y}^2$, so $\log_2E_i<2\log_2\mathcal{Y}$; $3d_i<(24D)^2$; $J\le N/28$, since $J=16D-2+L\le 80D/\varepsilon$ and $N\ge 256DH/\varepsilon\ge 2304D/\varepsilon$, as $H\ge9$; and $s<\mathcal{Y}^{1/9}$, since $6N^2\log_2 s\le\log_2 r_*\le\frac{35}{4}\log_2A\le\frac{35}{4}\cdot\frac{N^3}{14}\log_2(24D)$ by Lemma 5.1. With $\log_2(m!)\le m\log_2 m$,

$$\log_2\mathcal{D}\le 2N^2s\log_2 s+\sum_{i=1}^{J}(6NM_i\log_2(2M_i)+2NM_iE_i\log_2E_i),$$

$$\log_2\mathfrak{n}\le 6N^2s\log_2(s+1)+\sum_{i=1}^{J}(E_i+2N^2\log_2(3d_i)+8NM_i\log_2(2M_i)+2NM_i\log_2E_i).$$

The source terms add up to at most $8N^2\mathcal{Y}\log_2\mathcal{Y}$. In the i-th summands, $2NM_iE_i\log_2E_i<24N^2\mathcal{Y}^2\log_2\mathcal{Y}$, the terms with $\log_2(2M_i)$ or $\log_2E_i$ and a factor NM_i add up to less than $32N\mathcal{Y}\log_2\mathcal{Y}$, $E_i<6N\mathcal{Y}$, and $2N^2\log_2(3d_i)<4N\log_2\mathcal{Y}$. So each stage contributes less than $25N^2\mathcal{Y}^2\log_2\mathcal{Y}$, and, as $\log_2\mathcal{Y}=N\log_2(24D)$,

$$\log_2\mathfrak{n}+\log_2\mathcal{D}<8N^2\mathcal{Y}\log_2\mathcal{Y}+\frac{N}{28}\cdot25N^2\mathcal{Y}^2\log_2\mathcal{Y}\le N^4\log_2(24D)\,\mathcal{Y}^2.$$

Finally $\mathcal{Y}\ge48^{N-1}\cdot24D>N^4\log_2(24D)$, since $N\ge4608$. Hence $\log_2\mathfrak{n}+\log_2\mathcal{D}<\mathcal{Y}^3=\beta_\varepsilon$. □

The proof also shows $\|\Gamma\|<2^{\beta_\varepsilon}$ for each of the three leg matrices Γ, which Remark 5.5 uses.

B.4 The generator

The *generator* takes a rational $\varepsilon \in (0,1]$, given by its numerator and denominator, and works with integers only.

1. Compute the parameters of Theorem 4.8 and J; build the chain backwards from $(1,1)$ by Lemma 4.1; for each arity, compute all $(N+1)(N+2)/2$ sector counts and take the first maximizer in the lexicographic order of $(n_{\mathrm{K}}, n_{\mathrm{O}})$; compute M_i, E_i, s, A, B, r_*, m_*, the denominators $\mathfrak{d}_i$ and $\mathcal{D}$.

2. For every stage, compute the primitive data of Appendix B.2 as integer numerators over their denominators: the nonzero entries of the rows of Π_b and of $\Pi_{b+r-1}^{-\top}$, from the definition of the shifted basis and (3.3); the words of the type in lexicographic order; the values $\varpi_{M,g'+1}(\ell)$; the coefficients of $\prod_{\ell' \neq \ell}(\phi + g + 1 - \ell')$, by multiplying linear factors; the powers u^e; and the binomial coefficients $\binom{E}{u}$. For the source, compute the powers ℓ^α and the coefficients of $\prod_{\ell' \neq \ell}(\phi - \ell')$.

3. For each leg, put $\Gamma_{\leq 0} = \mathfrak{d}_0 \Gamma_0$ as a dense integer matrix, and for $i = 1, \dots, J$ form $\mathfrak{d}_i \Gamma_i$ as a dense integer matrix and put $\Gamma_{\leq i} = \Gamma_{\leq i-1} \cdot \mathfrak{d}_i \Gamma_i$.

4. Output $\mathcal{D}$ and, for each leg, the matrix $\Gamma_{\leq J}$ with its columns permuted by the identification $\mathcal{I}_{J+1} \cong m_* \mathsf{M}_B$.

Proof of Proposition 5.4(c). By Step 3, $\Gamma_{\leq J} = \mathfrak{d}_0 \cdots \mathfrak{d}_J \, \Gamma_0 \cdots \Gamma_J$, so by (B.2) Step 4 outputs $\mathcal{D}\Gamma$ for each leg, which is correct by part (a).

Entries of Γ_i. Each entry of Γ_i is zero or an entry of Θ_i, because the other factors of Γ_i are identities and permutations. An entry of Θ_i is w_u (on the weighted leg) times a product of $6k$ entries of matrices $\mathsf{S}_\pi(u)$, one for each orientation and each of the k tensor factors. An entry of $\mathsf{S}_\pi(u)$ is a single product of an entry of K_ι and an entry of $\mathsf{P}_\iota(u)$: for a given column of $\mathsf{P}_\iota(u)\Sigma_\iota$, only one row of $\mathsf{P}_\iota(u)$ in the given source copy has a nonzero entry, because the separator keeps the sector, the variable and (on the shared leg) the index α. An entry of K_ι is a product of N entries of Π_b or $\Pi_{b+r-1}^{-\top}$, or 1. So an entry of $\mathfrak{d}_i \Gamma_i$ is obtained from at most $6k(N+1)+1$ numerators of the primitive matrices and their denominators: multiply the numerators, multiply the denominators (their product divides $\mathfrak{d}_i$), divide $\mathfrak{d}_i$ exactly by this product, and multiply. This takes at most $12k(N+2)$ integer operations, and the same holds for the entries of $\mathfrak{d}_0 \Gamma_0$.

Index work. To form one entry of $\mathfrak{d}_i \Gamma_i$, the generator also decodes its row and column indices and, for each of the $6k$ orientations and tensor factors, finds the words of the sectors in the list of Step 2 by linear search, checks the crop and the type, locates the needed entries of Π_b or $\Pi_{b+r-1}^{-\top}$ in their stored rows, and applies the reindexing position by position. For each orientation and tensor factor this takes at most $4(N+3)$ divisions, $2M_iN$ comparisons for the words, $4N$ operations for the crop, the type and the reindexing, and $(d_i+3)N$ comparisons for the entries of Π_b or $\Pi_{b+r-1}^{-\top}$; in total at most $3M_iN(d_i+1)$, as $M_i, d_i \geq 2$. So one entry of $\mathfrak{d}_i \Gamma_i$, with its loop counter, takes at most $12k(N+2) + 20kNM_i(d_i+1) < \mathcal{Y}^2 < \mathfrak{n}$ operations, in the notation of the proof of part (a), and likewise one entry of $\mathfrak{d}_0 \Gamma_0$.

Sizes of the integers. Every numerator of a primitive entry is at most its row-sum bound times its denominator, so every product of such numerators within one entry, and every coefficient formed while multiplying out the linear factors of Step 2, is at most $\mathfrak{n}_i \mathfrak{d}_i \leq \mathfrak{n}\mathcal{D}$ in absolute value. The entries and the partial sums of the products in Step 3 are at most $\mathfrak{d}_0 \cdots \mathfrak{d}_i \, \|\Gamma_0 \cdots \Gamma_{i-1}\| \max |\Gamma_i| \leq \mathcal{D}\mathfrak{n}$. The other integers of Steps 1, 2 and 4 are at most $\mathcal{D}\mathfrak{n}$ or at most r_* (this covers the numerator and denominator of ε, which are at most A by hypothesis, the parameters, the sector counts, A, B and

m_*), and the indices and loop counters are at most $\mathsf{n} \le A^{59/4}$, where $\log_2 \mathsf{n} \le \frac{59}{4} \cdot \frac{N^3}{14} \log_2(24D) < \beta_\varepsilon$ by the proof of part (a). By part (a), every integer formed has absolute value below 2^{β_ε}.

Counting. The matrices of Step 3 have at most n rows and columns. As each entry takes at most n operations, forming $\mathfrak{d}_i \Gamma_i$ takes at most n^3, and the product with $\Gamma_{\le i-1}$, with its loop counters, at most $5\mathsf{n}^3$. Step 4 takes at most $4\mathsf{n}^2$ operations for each leg. Steps 1 and 2 take at most $16J\mathfrak{m}^4$ operations, where $\mathfrak{m} = J + N + s + \max_i E_i$: for one stage, the sector counts of its arity take at most $2(N+1)^3$ operations, $\mathfrak{d}_i$ at most $2E_i + 2M_i + 2k + 2$, the rows of Π_b and $\Pi_{b+r-1}^{-\top}$ at most $5(d_i+3)d_i s$, the words $3M_i N$, the values $\varpi_{M_i,g'+1}(\ell)$ at most $4M_i^3$, the coefficients $\zeta_{\ell,g,e'}$ at most $8M_i^4$ (each of the $(2M_i-1)M_i$ polynomials is a product of $2M_i-2$ linear factors), the powers u^e and the weights at most $E_iM_i + 3E_i$, and its factors of A, B, r_* and m_* at most $2N^2 + 15k$; the parameters, the chain, the source data, $\mathfrak{d}_0$ and $\mathcal{D}$ take at most $3D + H + 11J + 2s^3 + 8N^2 + s + 21$. In the notation of the proof of part (a), $\mathfrak{m} < \mathcal{Y}^3$, so $16J\mathfrak{m}^4 < 16J\mathcal{Y}^{12} \le 16JA^{4/k} < A^2 \le \mathsf{n}$. For three legs, the generator therefore performs at most $3((6J+1)\mathsf{n}^3 + 4\mathsf{n}^2) + \mathsf{n} \le 18(J+1)\mathsf{n}^3$ operations on integers with at most β_ε bits, hence at most $288(J+1)\mathsf{n}^3\beta_\varepsilon^2$ bit operations. Finally $J + 1 \le \log_2 A$, $\mathsf{n}^3 \le A^{177/4}$ and $\beta_\varepsilon^2 \le A^{2/k}$, and $288 \log_2 A \cdot A^{2/k} \le A^{7/4}$ because $A \ge 2^{2NkJ}$ and $k \ge 4608$. So the generator uses at most A^{46} bit operations. □

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